Seed, Envelope, Detail: Coarse-to-Fine Diffusion Attention for Training-Free Segmentation

Supplementary Material

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1. Method details

Notation. t indexes UNet calls (0–30 for CA, 10–20 for SA), h heads (eight per module) and m modules. $s \in \{16, 32, 64\}$ is the side of an SA grid, $A^s_{t,h} \in \mathbb{R}^{s^2 \times s^2}$ a row-stochastic SA matrix and $\bar{A}^s_t$ its head mean. U is bilinear $2\times$ upsampling (U_{64}: to 64×64), D area downsampling, and norm min–max normalization. The SA modules are the final transformer block of the decoder at each resolution: `up_blocks.1.attentions.2` (16 px), `up_blocks.2.attentions.2` (32 px) and `up_blocks.3.attentions.2` (64 px). The eleven CA modules are, in sorted `attn2` order, indices 0, 1 (down, 64 px), 4, 5 (down, 16 px), 6 (mid, 8 px), 7, 8, 9 (up, 16 px) and 13, 14, 15 (up, 64 px). The five 32-pixel modules (2, 3, 10, 11, 12) are excluded.

Cell (+R2)	VOC forced OOF	folds vs. V0	COCO
historical A3 → fine only (V0)	.8228	–	.6623
GoCA-oriented A3 → fine only (G0)	.8228	3/5	.6617
historical A3 → hierarchy + cap (V3)	.8292	5/5	.6967
GoCA-oriented A3 → hierarchy + cap (G3)	.8304	5/5	.6961

Table 1. **A3 orientation (paired rerun).** Both seeds come from the same diffusion run. G0 − V0 = −0.0000 (CI $[-0.0001, +0.0001]$); G3 − G0 = +0.0075 (5/5, CI $[+0.0020, +0.0128]$); G3 − V3 = +0.0012 (4/5).

A3 weights. At each call, F is the conditional query of `up_blocks.2.attentions.1`'s CA module, with heads concatenated, and $P_t = \mathrm{softmax}(FF^\top/\sqrt{d})$. For CA module m, $\tilde{S}_{t,m}$ is the head-mean SA affinity of the same transformer block, resized to P_t's grid. Following GoCA [5], the key axis is resized bilinearly and the query axis by nearest neighbour. The score $\langle \tilde{S}_{t,m}, P_t \rangle$ is clipped at zero and normalized over $\mathcal{M}$ within each call (Eq. 2 of the main paper). Figure 1 shows the resulting weights. The 8-pixel module receives a quarter of the total weight and the five 16-pixel modules 60%, while the 64-pixel modules together receive 16%. The 8-pixel module dominates the first calls, when the image is still coarse.

Orientation of the A3 score. In our implementation, the resized SA affinity is indexed [key, query], so the score is $\langle \tilde{S}^\top, P \rangle$ rather than GoCA's $\langle \tilde{S}, P \rangle$. A paired rerun accumulated both seeds in the same diffusion run (Table 1). The two seeds agree to Pearson 0.99999 (mean absolute difference 0.0008). The control after R2 changes by −0.0000, inside the pre-registered ±0.002 tie band, and the hierarchy's gain is unchanged on the corrected seed. All results use the historical orientation.

The cap as a projection. Let $q \geq 0$ be the 32-pixel proposal and $r \geq 0$ the 16-pixel parent. Consider the problem

$$\min_{z \geq 0} \sum_k \left(z_k \log \frac{z_k}{q_k} - z_k + q_k \right) \quad \text{s.t.} \quad \sum_{k \in g} z_k \leq 4 r_g \;\; \forall g. \tag{1}$$

The objective and the constraints separate over groups. Within a group, the stationarity condition gives z_k =

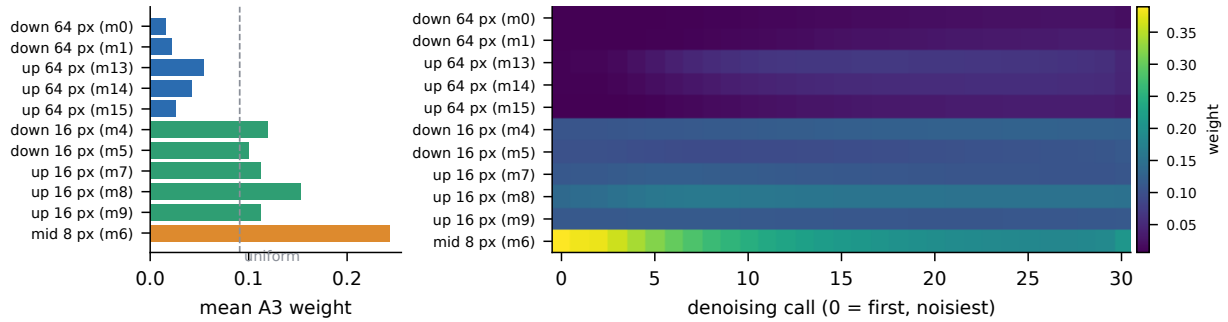


Figure 1. **A3 CA module weights** (VOC-Sim, all images). Left: mean weight per module, against uniform weighting (dashed). Right: mean weight per denoising call. Weighting by agreement with SA moves 84% of the weight to the 8- and 16-pixel modules.

$q_k e^{-\lambda_g}$ with multiplier $\lambda_g \geq 0$. If $\sum_{k \in g} q_k \leq 4r_g$, then $\lambda_g = 0$ and $z = q$; otherwise the constraint is active and $e^{-\lambda_g} = 4r_g / \sum_{k \in g} q_k$. This is Eq. 4 of the main paper. Replacing $\leq$ by $=$ gives *exact conservation*, $z_k = q_k \, 4r_g / \sum_{k \in g} q_k$ for every group. That forces under-budget groups to grow as well, so children must fill a parent's budget even where the parent was wrong. The relaxed variant (ρ=0.5) moves halfway from q towards the conserved solution. The variants with conservation in Table 3 of the main paper operate on per-call distributions (each call normalized to unit mass), whereas the cap operates on the raw maps. This is why the plain hierarchy, rather than the conservation rows, is the cap's direct control: if no group binds, the capped map equals the plain hierarchy exactly.

DenseCRF. R2 calls the SimpleCRF wrapper used by OVAM. The unary term is the hard binary mask (p(fg)=1 inside, 0 outside), and the parameters are $(w_1, \theta_\alpha, \theta_\beta, w_2, \theta_\gamma, \text{iterations}) = (10, 80, 13, 3, 3, 5)$ on the full-resolution RGB image. Neither these parameters nor the threshold are tuned on COCO-Cap.

Cost. All attention is captured during the 30-step generation; nothing is optimized afterwards. The seed reads eleven CA modules at every call, and the hierarchy reads three SA modules at eleven calls. Its largest operation is applying the eight 4096×4096 head matrices of the 64-pixel module to one map per call, which is the same operation as fine-only propagation. DenseCRF runs once per image on the CPU.

2. Building the fine-only baseline

The hierarchy of the main paper replaces the propagation of an earlier, fully registered pipeline: CA seed → one 64-pixel SA module, two observation-wise passes → DenseCRF, with SA restricted to calls 10–20. This section records how that baseline was selected; it is the "fine only" reference throughout the main paper.

One SA module. One pass through all sixteen SA modules improves the frozen seed, but the last 64-pixel module (`up_blocks.3.attentions.2`) is better. OOF mIoU rises from 0.756 to 0.789 in the combined SA confirmation, all five folds select it, and the paired mean-image gain is $+0.042$ (CI $[+0.037, +0.047]$). Separate screens favoured call 30, resolutions 8+64 and heads $\{1, 4, 6\}$, but none survived the combination. A Cartesian audit of six CA seeds and twelve SA settings found module 15 the best SA partner for every seed.

Two observation-wise passes. For each call–head observation i, the fine-only map is

$$p_2 = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} A_i^2 c. \tag{2}$$

The square is taken within each observation before averaging, which is a different operator from $(\frac{1}{|\mathcal{I}|} \sum_i A_i)^2 c$. A second pass raises OOF mIoU from 0.788 to 0.800, especially for chair ($+0.078$) and person ($+0.030$); a third pass is not promoted. Figure 2 shows the trade-off between completion and leakage. In a later diagnostic (A3 seed, calls 10–20, threshold 0.425), pixels first accepted at depth one are 86.5% true positives on VOC-Sim and 70.8% on COCO-Cap, but pixels first accepted at depth three are only 39.3% and 14.0%.

Refinement absorbs propagation gains. DenseCRF was selected in all five folds and improved all 20 classes in its screen (0.788→0.805); with two passes the combined system reached 0.813 OOF. Twenty-four families of graph operators were then registered against one pass (Sec. 7). Observation-reliability operators (consensus edges, sign consensus, robust aggregation, keeping the 50% most consistent observations) raised one-pass mIoU to 0.797–0.798, which is the level of a second pass. Their per-class gains also correlate with those of the second pass (0.88 and 0.82), and none survived on top of two passes plus DenseCRF. Graph-restricting operators (absorbing walks, confidence locks, reciprocal links, bottlenecks, region labelling) lost. From then on, every candidate was selected on the refined mask.

Setting	P2	R2	Mid	Forced OOF	COCO
One pass	–	–	–	.7873	.6172
Two passes	✓	–	–	.7997	.6246
+ DenseCRF	✓	✓	–	.8132	.6372
+ calls 10–20	✓	✓	✓	.8168	.6438

Table 2. **The fine-only baseline in one run** (SG22, uniform seed). The registered selector chose MID-P2+R2 in four of five folds (selection path 0.8158 OOF).

The SA call window. Selecting on the refined mask found one change: SA calls 10–20 instead of 0–30 (Table 2). Late calls recover some object pixels but also add distant false positives (Fig. 3), and their utility varies by class.

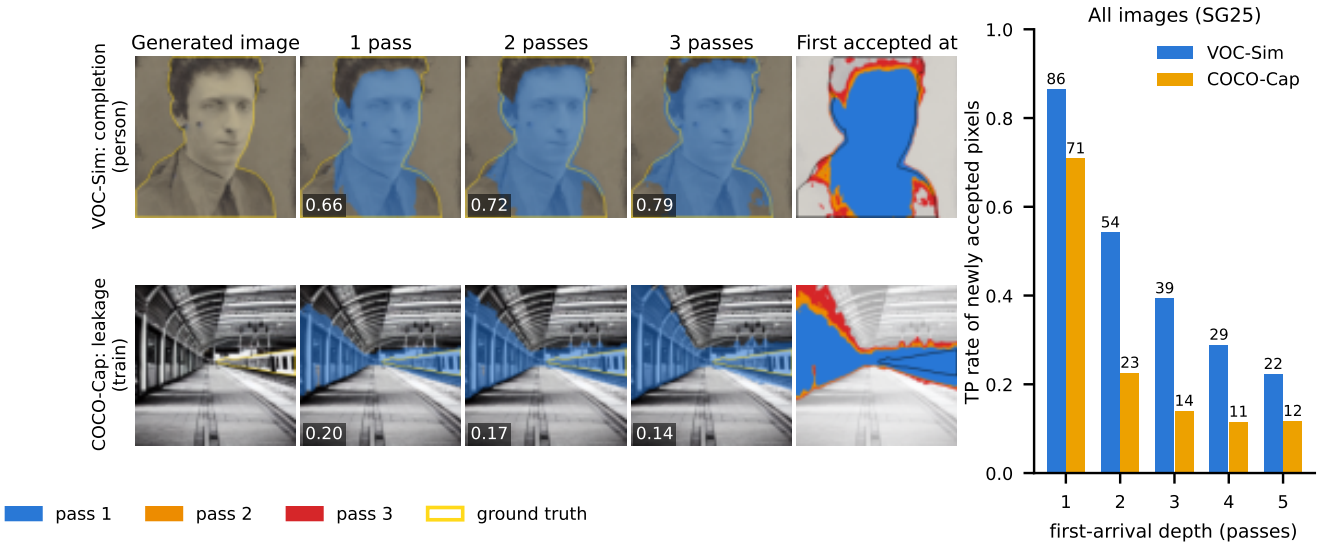


Figure 2. **How far fine SA should propagate.** Left: one, two and three observation-wise passes on a completion example (VOC-Sim, 90th percentile of the two-pass gain) and a leakage example (COCO-Cap, 10th percentile of the three-pass gain), with the pass at which each pixel was first accepted. Right: true-positive rate of newly accepted pixels by depth, pooled over all images. The supplementary gallery (Fig. 14) applies both rules to both datasets.

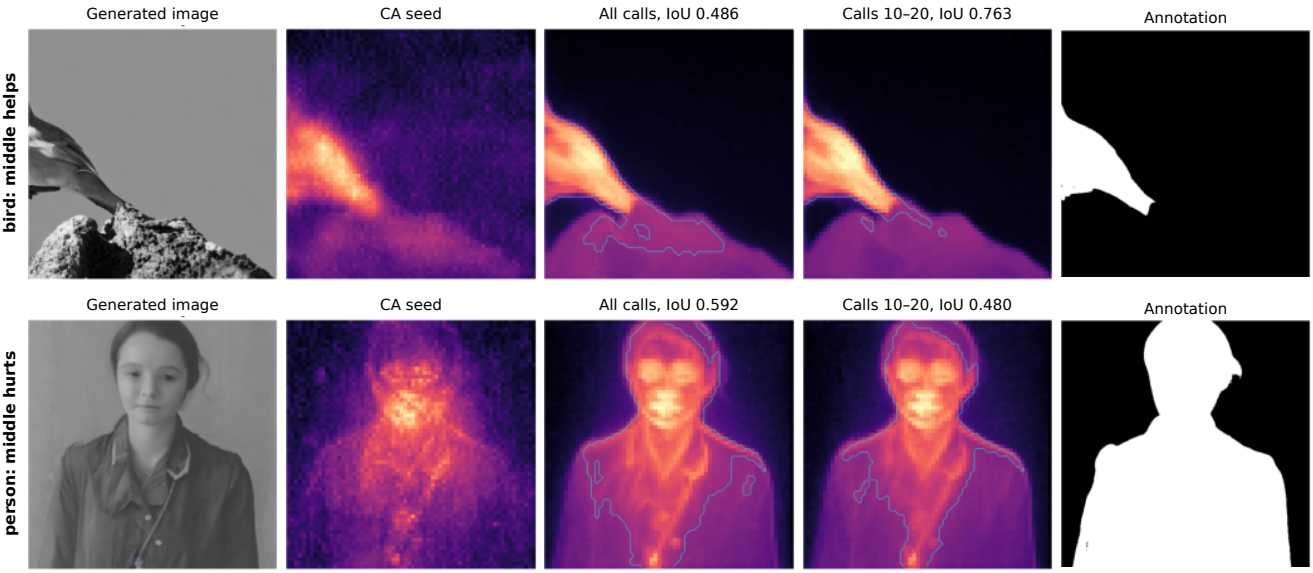


Figure 3. **A temporal window can remove leakage or remove object pixels.** Two VOC-Sim examples from the one-pass, raw-map diagnostic, propagated with SA calls 0–30 or 10–20. Bird gains when a false region is suppressed; person loses when useful extent is suppressed. These are deliberately opposing examples, not images from the final evaluation.

3. Evidence for each stage of the final pipeline

Figure 4 repeats Fig. 2 of the main paper with every intermediate map.

3.1. Seed

Figure 5 shows the per-resolution CA maps. 8-pixel CA is blob-like and 16-pixel CA follows object parts, while 64-pixel CA fires on texture and edges throughout the image. A3 down-weights the 64-pixel modules, so its seed is smoother than the uniform one. Table 3 lists every weighting scheme of the same run. Per-pixel content-token rescaling (A1, A4) collapses on COCO-Cap, where the attribution prompt has a single content token, and projected-output head weighting (A2) loses on both benchmarks.

Weighting	CA only	final (MID-P2+R2)	COCO final
A0 uniform	.7135	.8169	.6437
A1 content-token mass	.6406	.8005	.0000
A2 projected head output	.6897	.8033	.6366
A3 SA agreement	.7049	**.8230**	**.6620**
A4 all three	.6288	.8004	.0000

Table 3. **CA weighting schemes** (one run, VOC forced OOF, COCO-Cap frozen). A3 won 4/5 folds against A0 (paired mean-image interval positive). The registered multi-candidate screen chose A3 in only 3/5 folds and kept A0. A3 was adopted afterwards by owner decision.

Propagation (two passes)	VOC	VOC + R2	COCO	COCO + R2
64 px only (fine)	.8097	.8228	.6599	.6621
16 px only (coarse)	.7225	.7538	.6282	.6652
32 px only	.7924	.8138	.6693	.6948
16 → 64	.8020	.8170	.6869	.6975
32 → 64	.8089	.8217	.6695	.6717
64 → 16	.7505	.7825	.6403	.6705

Table 4. **SA scale and order** (one run, A3 seed, calls 10–20; VOC forced OOF, COCO-Cap frozen).

3.2. Coarse and fine SA

Table 4 gives every scale cell of the run behind Fig. 3 of the main paper. Coarse 16-pixel SA alone is far below fine SA on VOC-Sim but competitive on COCO-Cap after R2 (0.665 vs. 0.662). Going from 16 to 64 pixels gains on COCO-Cap but not on VOC-Sim, the gap that the cap closes. The reverse order (64→16) fails, so the order of the scales matters.

3.3. The hierarchy: every variant

Table 5 adds the cells omitted from Table 3 of the main paper: budgeted completion (v1 C7–C9), gated completion (v2 V6), and three single changes to the capped hierarchy. The three changes are a cap applied to each head before averaging (W1), a native 16-pixel seed instead of the downsampled one (W2), and a temporal cap of each call by the running mean of earlier calls (W3). None improves on the one-sided cap. The cap binds in 58% (VOC-Sim) and 65% (COCO-Cap) of 2×2 groups but removes only 7.6% and 8.3% of the proposal mass (mean scaling factor of binding groups 0.84).

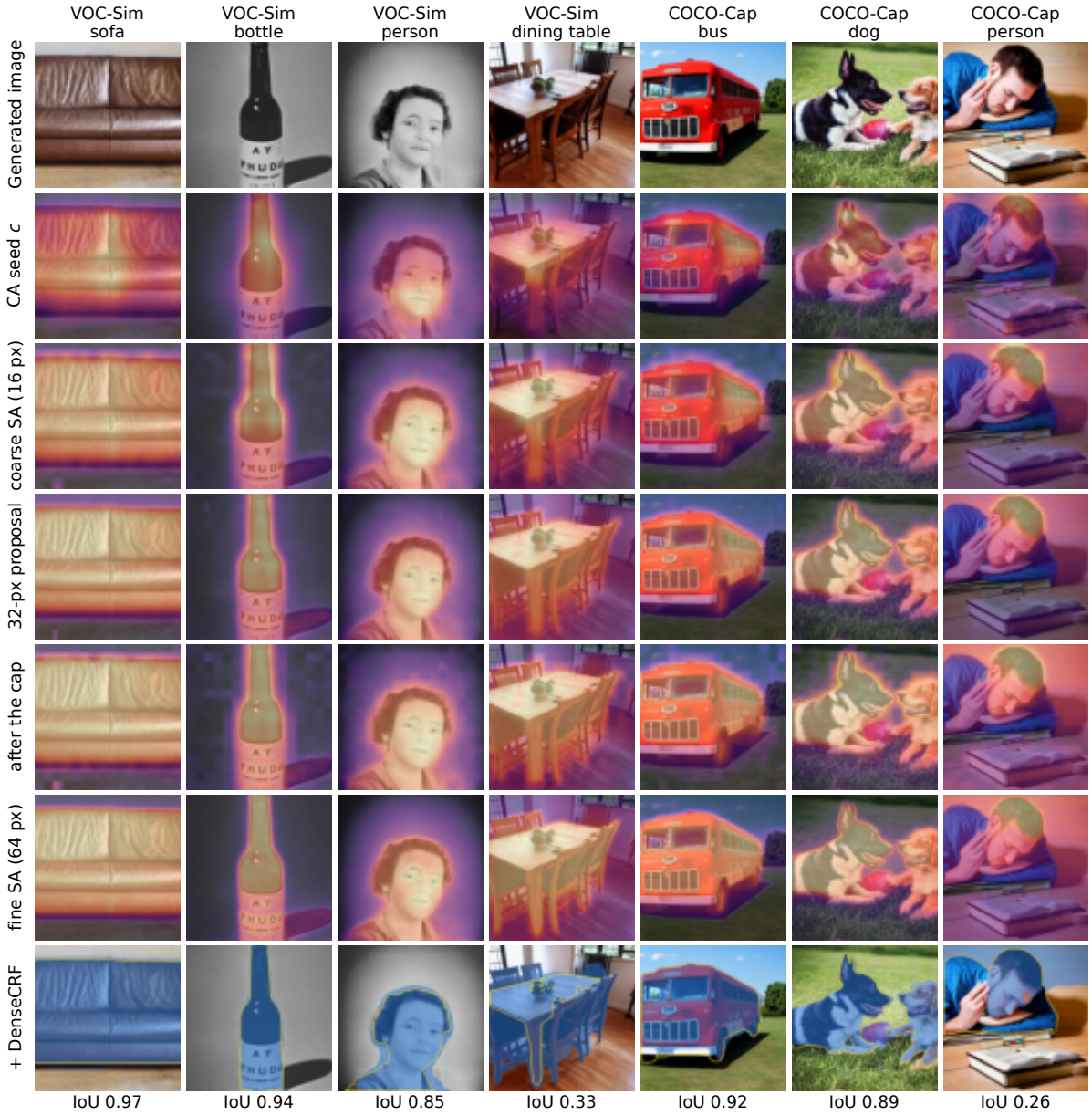


Figure 4. **Every stage, same fixed-rule images as Fig. 2 of the main paper.** Rows: generated image; A3 seed; coarse 16-pixel SA r; 32-pixel proposal q; capped map z; fine 64-pixel SA y; final mask after DenseCRF with the annotation outline. Regenerated maps reproduce the committed per-image scores (largest image-IoU difference 0.005, one COCO-Cap image).

3.4. DenseCRF

Across the 490 VOC-Sim images, DenseCRF changes the capped pipeline's image IoU by more than 0.005 in 375 images, improving 278. On COCO-Cap it improves 427 of 707 images and 18 of 20 classes (Table 4 of the main paper). Figure 6 shows the per-class change.

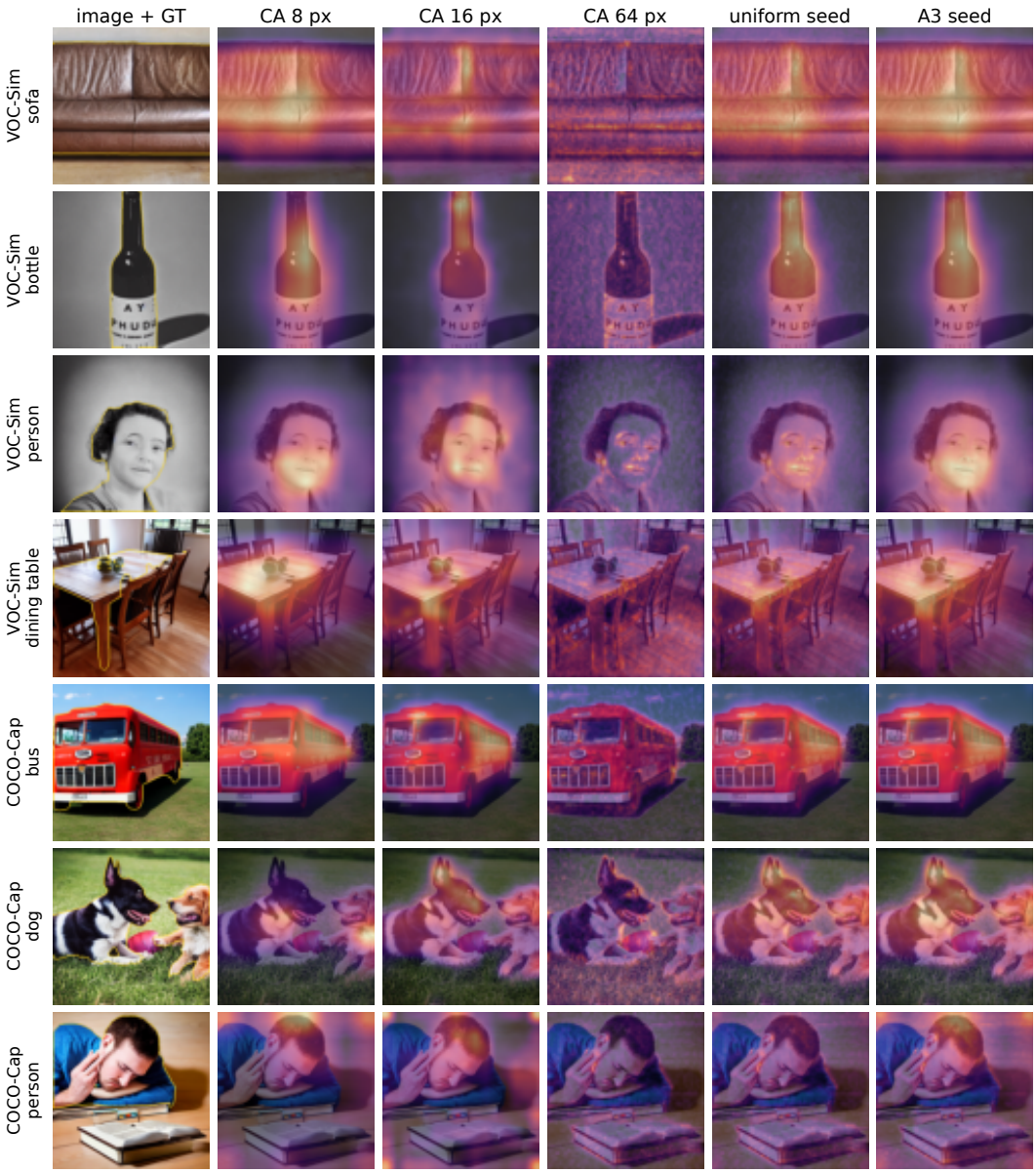


Figure 5. **CA by resolution.** Unweighted mean CA of the selected modules at 8, 16 and 64 pixels, the uniform seed and the A3 seed, for the images of Fig. 2 of the main paper.

Run	Variant	VOC raw	VOC + R2	COCO + R2
v1	exact conservation (both) + background-only completion budget (C7)	.7785	.8047	.7054
v1	exact conservation (both) + semantic/background odds budget (C8)	.7774	.8031	.7061
v1	relaxed conservation + odds budget (C9)	.7951	.8140	.7057
v2	one-sided cap + completion only where $P_1 \geq B$ (V6)	.8143	.8256	.6792
v3	one-sided cap, reference (R1, = v2 V3)	.8157	.8293	.6964
v3	head-wise cap: $z = \frac{1}{8}\sum_h \mathrm{cap}(r, A_h^{32}Ur)$ (W1)	.8161	.8294	.6972
v3	native 16-px CA seed c_{16}^* instead of $D(c)$ (W2)	.8155	.8281	.6975
v3	temporal cap by the mean of earlier calls (W3)	.8116	.8261	.6828

Table 5. **Remaining hierarchy variants** (VOC forced OOF, COCO-Cap frozen; each run has its own same-generation references, which agree with Table 3 of the main paper to 0.0003). Against R1 after R2: W1 $+0.0001$ (CI spans 0), W2 -0.0012 (2/5), W3 -0.0032 (1/5). The head-wise cap removes almost the same mass (7.7% vs. 7.6%). The native seed correlates 0.996 with $D(c)$. The temporal cap hurts most on small objects (-0.016 in the smallest GT-area bin).

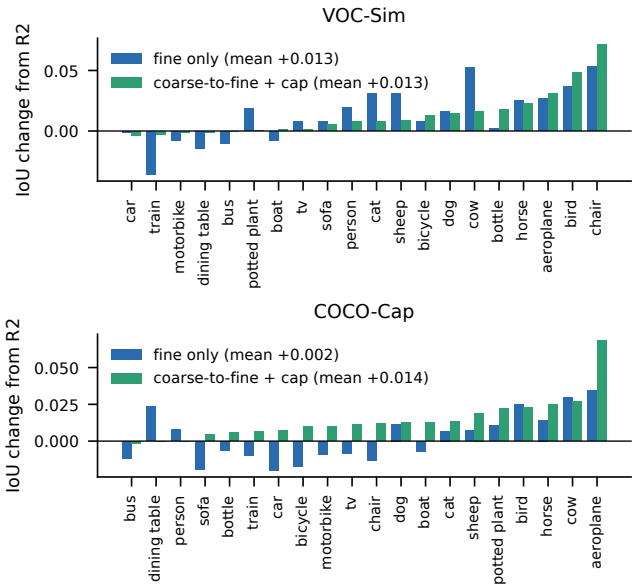


Figure 6. **Per-class effect of DenseCRF** on fine-only and capped coarse-to-fine maps (same run; thresholds from VOC-Sim).

Map (+R2)	Rule	VOC OOF	vs. same map T0	sheep Δ	COCO
fine only	T0 global	.8230	–	–	.6623
fine only	T1 Otsu	.8283	+.0054 (4/5)	−.041	.6411
fine only	T4 stable	.8255	+.0026 (2/5)	−.042	.6809
fine only	T6 Otsu+d	.8276	+.0046 (3/5)	−.041	.6411
C2F + cap	T0 global	**.8293**	–	−.092	**.6964**
C2F + cap	T1 Otsu	.8238	−.0055 (1/5)	−.079	.6359
C2F + cap	T2 two-Gauss.	.8031	−.0262 (0/5)	−.080	.6258
C2F + cap	T3 $\mu+k\sigma$	.7864	−.0429 (0/5)	−.254	.6498
C2F + cap	T4 stable	.8239	−.0054 (1/5)	−.108	.6722
C2F + cap	T5 fine-only area	.8246	−.0047 (0/5)	−.007	.6659
C2F + cap	T6 Otsu+d	.8297	+.0004 (2/5)	−.090	.6586

Table 6. **Per-image threshold rules** (after R2; sheep Δ vs. fine-only T0). Both registered selectors kept the fine-only global threshold. Per-image oracle (best grid threshold per image, using ground truth): 0.896 fine-only vs. 0.895 C2F + cap on VOC-Sim.

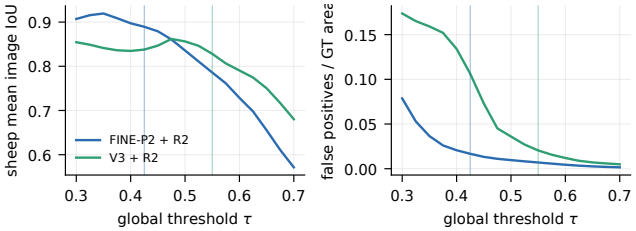


Figure 7. **Sheep across thresholds** (25 VOC-Sim images, after R2). Vertical lines: the global thresholds of the two pipelines.

4. Calibration, thresholds and the sheep failure

Per-image thresholds. If the hierarchy's gain is calibration, could a per-image threshold computed from each map recover it, or repair sheep? We tested seven rules on the counts and stored maps of the main run. Each rule's τ_I is snapped to the 17-point grid, so the existing counts score it exactly, raw and with R2 (the stored maps reproduce the counted mask areas with Pearson $\geq$ 0.9998). The rules are: T0, the global threshold; T1, Otsu; T2, the crossing of a two-Gaussian fit; T3, $\mu_I + k\sigma_I$; T4, the most stable threshold (smallest area change); T5, the capped map's ranking at the fine-only map's area; and T6, Otsu plus an offset. One-parameter rules choose their parameter inside the training folds. The rule set was registered before the run. Table 6 gives the outcome. No rule improves the capped pipeline. Matching the fine-only map's area (T5) repairs sheep (-0.007 instead of -0.092) but gives back most of the gain. Otsu helps the fine-only map in four folds but loses 0.021 on COCO-Cap. Two-Gaussian and $\mu+k\sigma$ thresholds often fall outside the grid (20–87% of images).

Sheep. The sheep loss is a loss of recall: 294,665 fewer true-positive pixels, against only 12,173 more false positives. It is concentrated in close-ups where the sheep covers 79–96% of the frame. Figure 7 shows that the capped map is better on sheep at every threshold above 0.45. Below that, its false positives are twice those of the fine-only map, so sheep cannot use the low threshold it prefers. Across all classes, objects that fill more than 75% of the frame gain from the cap at every threshold. Sheep is the exception because its surroundings are grouped with it at low thresholds. Figure 8 shows the median sheep image.

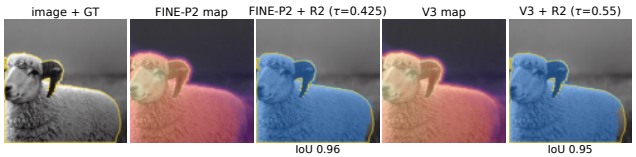


Figure 8. **The median sheep image** (median change of the capped pipeline against fine-only, both after R2, at their global thresholds).

5. Selection status of every component

Table 7 lists how each component of the final pipeline entered it. Two components were adopted after the registered rule kept a simpler reference; their registered outcomes are reported here in full.

The negative record at a glance. Figure 9 places the 45 registered comparisons of the fine-only phase (studies up to SG28) against the reference of its own run; Sec. 7 lists them, and Sec. 12 gives every cell.

Component	How it entered	Registered evidence
CA without 32-px modules	promoted	4/5 folds, OOF 0.665→0.691, paired CI $[+0.015, +0.029]$
All CA heads and calls, conditional query, mean of word pieces	promoted / kept	head and call subsets inside the 0.002 tie band once crossed with scale
SA module 15 (64 px)	promoted	5/5 folds, 0.756→0.789
Two observation-wise passes	promoted	0.788→0.800; confirmed with R2 in 5/5 folds (0.813)
DenseCRF (R2), fixed parameters	promoted	5/5 folds, all 20 classes improved in its screen
SA calls 10–20	promoted	selected after R2 in 4/5 folds (selection path 0.8158 vs. 0.8132)
A3 module weights	**owner decision**	forced 4/5 folds, +0.0061 OOF, COCO +0.018; multi-candidate screen chose A3 in 3/5 folds and kept uniform
Coarse-to-fine hierarchy with one-sided cap	**owner decision**	forced 5/5 folds vs. fine-only and plain hierarchy; selectors kept fine-only because the worst class lost 0.057–0.104 in the training folds (guard: 0.050); training-fold gain +0.0039 to +0.0071

Table 7. **Selection status.** "Owner decision" marks components adopted after the pre-registered selector retained its reference. Their numbers in the main paper are forced out-of-fold estimates, never in-sample, and their COCO-Cap numbers are frozen transfer.

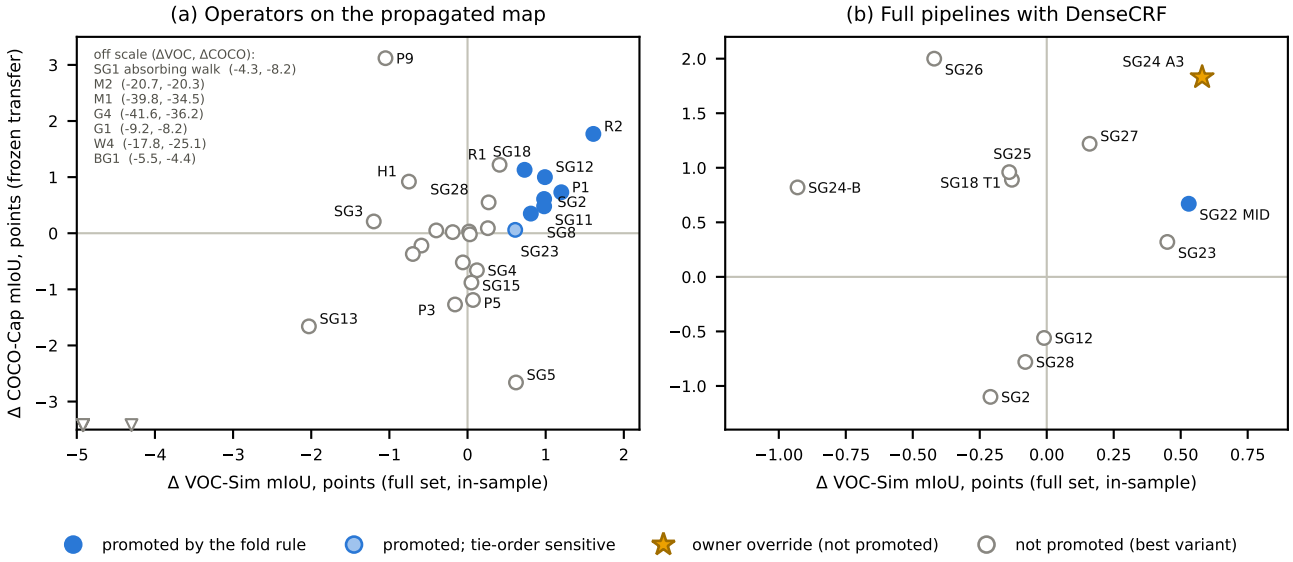


Figure 9. **The search at a glance.** Each point is one registered experiment family, compared with the reference cell of the same run. x: change in VOC-Sim mIoU at each cell's full-VOC threshold (in-sample, descriptive); y: change in frozen COCO-Cap transfer. (a) Operators on the propagated map, mostly against one SA pass. The promoted cluster (blue) gains about one point on both datasets, but apart from the second pass and DenseCRF these operators do not improve on two passes plus DenseCRF. (b) Complete pipelines with DenseCRF. Only the middle-call window is promoted, and the A3 weighting (star) was adopted by owner decision. Several multi-scale studies raise COCO-Cap transfer but are not selected on VOC-Sim. The map covers the studies up to SG28; later studies, including the hierarchy, are in Secs. 3 and 12.

6. Protocol

The full protocol, with every registration, result and review in the order they happened, is `docs/ATTENTION_SEGMENTATION_EXPERIMENT_PROTOCOL.md` (135 sections); the registry of experiment families is `docs/ATTENTION_FUSION_AND_TOKEN_IDEAS.md`. This section summarizes the rules that every reported number obeys.

Fixed conditions. Stable Diffusion 1.5 generates every image from its released prompt and seed, with 30 scheduler steps, classifier-free guidance 7.5, a float32 model and float16 storage of compact attention. "Calls 0–30" are the 31 recorded UNet call indices, inclusive. CA uses the conditional branch for query and key, a softmax over all 77 positions of the padded attribution prompt, and the mean over every tokenizer piece of the class name within each head. All maps are combined on a 64×64 grid. Generated images and attention tensors are discarded after scoring; per-image TP, FP and FN at every threshold of the grid are kept, so that folds, bootstrap intervals and later analyses need no regeneration.

Folds. `configs/voc_sim_folds.csv` assigns the 490 VOC-Sim images to five class-stratified folds (about five images per class per fold), with a fixed seed, committed before any run. In each outer fold, the method parameter and the mask threshold are chosen on the other four folds and scored on the held-out fold; the reference is treated the same way.

Decision rule (v1.2). A candidate is promoted if it improves training-fold mIoU by at least 0.010, or by 0.005–0.0099 with at least 12 of 20 classes improved and no class losing more than 0.05; it must win at least four of five fold votes and give a positive pooled OOF gain. Within 0.002, the simpler setting wins (the reference, then fewer added operations, then fewer free parameters). Each family is screened against its own reference; family winners then go to a combined confirmation.

Registry rules. R1: only VOC-Sim selects; COCO-Cap is scored once per frozen pipeline. R2: the frozen CA/SA pair is the reference input. R3: folds and thresholds as above, on the 17-point grid 0.30–0.70. R4–R5: the decision and tie rules above. R6: per-image records, commit and environment for every shard. R7: each package runs in one environment and compares only with its own reference. R8: grids are committed before the run; later changes are logged and labelled post hoc. R9: every package reports fold choices, OOF scores, paired bootstrap intervals, per-class changes and the number of configurations searched. R10: training-free, few-shot and external-segmenter tracks are reported separately. R11: no token is trained on an image of the fold it is scored on.

Deviations. Five departures from these rules affect reported numbers, and each is labelled where it appears. (i) The A3 CA weighting was retained by owner decision after the registered selector kept its reference (SG24-A); all later studies (SG25–SG33, the SG-combined studies and SG-threshold) build on it. The paired SG-A3G rerun showed that GoCA's orientation of the A3 score leaves the results unchanged (Sec. 1). (v) The coarse-to-fine hierarchy with the one-sided cap, the method of the main paper, was adopted by owner decision (protocol Section 135) after every registered selector kept fine-only propagation because sheep failed the class guard (Sec. 5). (ii) SG13's candidate cells use a 0–1 threshold grid by 0.005, registered before the run; its controls keep the 17-point grid. (iii) SG22–SG24 shards ran on a mix of V100 and A100 nodes, against R7 (Sec. 10). (iv) SG23's raw promotion depends on the order of tied candidates, which the registration did not fix (Sec. 10).

7. Every registered family

Table 8 summarizes the 45 comparisons of Fig. 9; Sec. 12 gives every registered decision and every cell. The table lists them with their reference, plotted cell, number of candidates and both changes. Every value is recomputed from the committed per-image shards of the named package (`scripts/paper/figure_decision_map.py`; values in `outputs/segmentation/paper_figures/decision_map.csv`). Three choices keep the comparison uniform. Each point compares two cells of one run, so no environment difference enters. Each cell is scored at its own full-VOC threshold on the 17-point grid, which places the point but is in-sample; the promotion decision is the out-of-fold rule of the protocol section given. Cells that repeat another package's operator by design (SG26 S1/S2, SG27 S1/S2, which reproduce SG25) are excluded from the best-variant choice.

What each family tested. *Graph restriction.* SG1 estimates, for each uncertain pixel, the probability that a walk reaches confident foreground before confident background. SG3 preserves pixels on which the call–head observations agree and propagates mainly elsewhere. SG9 replaces each affinity by a reciprocal form of A and $A^\top$ (geometric mean, minimum or soft gate). SG10 scores each pixel by its widest-path (bottleneck) connectivity to the confident seed. SG4 and SG13 partition the SA graph into regions and label whole regions from CA evidence. *Backlog* (Sec. 83): seeded progressive expansion (P9), a propagated

Family	Sec.	Reference	Plotted cell	Cand.	ΔVOC	ΔCOCO	Decision
(a) Operators on the propagated map							
P1 two passes	43	W/REF	W/P1	1	+1.20	+0.73	promoted
R2 DenseCRF	44	W/REF	W/REF+R2	1	+1.61	+1.77	promoted
SG1 absorbing walk	71	W/REF	W/SG1/f90_b10_r1	12	-4.30	-8.22	–
SG3 confidence lock	71	W/REF	W/SG3/f95_b05_r0_n1	24	-1.20	+0.21	–
SG5 contrastive seed	70	W/REF	WN/REF	5	+0.62	-2.66	–
SG8 observation consensus	78	W/REF	W/C/rho0p80	15	+0.81	+0.35	promoted
SG9 reciprocal links	78	W/REF	W/R1_geometric	3	+0.26	+0.09	–
SG10 bottleneck	79	W/REF	W/B3_G2_q95	16	-0.59	-0.22	–
SG2 consensus graph	85	W/REF	W/G_cv_g20_n1	12	+0.98	+0.61	promoted
SG4 SA regions	85	W/REF	W/SPEC32_rmean	18	+0.12	-0.66	–
SG11 robust consensus	89	W/REF	W/SG11/trim20/rho0p80	6	+0.98	+0.48	promoted
SG12 observation filtering	89	W/REF	W/SG12/top50	4	+0.99	+1.00	promoted
SG13 region inference	90	W/REF	W/SG13/R1	3	-2.03	-1.66	–
R1 (backlog)	83	REF/one_pass	R1/w100	3	+0.41	+1.22	–
M2 (backlog)	83	REF/one_pass	M2/k2	3	-20.66	-20.32	–
M1 (backlog)	83	REF/one_pass	M1/ca_prototype	1	-39.75	-34.48	–
G4 (backlog)	83	REF/one_pass	G4/s070	3	-41.59	-36.24	–
G1 (backlog)	83	REF/one_pass	G1/tau090	2	-9.25	-8.22	–
H1 (backlog)	83	REF/one_pass	H1/proportional	1	-0.75	+0.92	–
W4 (backlog)	83	REF/one_pass	W4/converged	1	-17.75	-25.08	–
W2 (backlog)	83	REF/one_pass	W2/k6	3	-0.40	+0.05	–
W1 (backlog)	83	REF/one_pass	W1/T050	3	+0.02	+0.03	–
BG1 (backlog)	83	REF/one_pass	BG1/g050	5	-5.49	-4.38	–
F2 (backlog)	83	REF/one_pass	F2/a075	4	-0.70	-0.37	–
P9 (backlog)	83	REF/one_pass	P9/s050_bg0	6	-1.05	+3.12	–
P5 (backlog)	83	REF/one_pass	P5/n2	4	+0.07	-1.19	–
P3 (backlog)	83	REF/one_pass	P3/t400	4	-0.16	-1.27	–
SG14 combined consensus (vs SG11)	92	W/SG11/trim20/rho0p80	W/SG14	1	+0.03	-0.02	–
SG15 refinement (vs SG11)	92	W/SG11/trim20/rho0p80	W/SG11/trim20/rho0p80/SG15-B	4	+0.05	-0.88	–
SG17 filtered 2nd pass (vs P1)	97	W/P1	W/SG17-Q70	3	-0.06	-0.52	–
SG18 middle calls	97	W/REF	W/T1	3	+0.73	+1.13	promoted
SG21 head weighting (vs T1)	101	SG18/T1	SG21-DIST	4	-0.19	+0.02	–
SG23 middle-call SG2 (vs MID one-pass)	102	MID-R0	MID-SG2	3	+0.61	+0.06	promoted[a]
SG28 PEARL-inspired (vs A3 P2)	112	P0	B2	11	+0.27	+0.55	–
(b) Complete pipelines with DenseCRF							
SG2 + CRF (vs FTF)	89	W/P1+R2	W/SG2+R2	1	-0.21	-1.10	–
SG12 + CRF (vs FTF)	92	W/P1+R2	W/SG12/top50+R2	1	-0.01	-0.56	–
T1 + CRF (vs FTF)	98	W/P1+R2	W/T1+R2	1	-0.13	+0.89	–
SG22 middle calls + CRF (vs FTF)	102	ALL-P2+R2	MID-P2+R2	3	+0.53	+0.67	promoted
SG23 middle-call SG2/11/12 + CRF	102	ALL-P2+R2	MID-SG11+R2	6	+0.45	+0.32	–
SG24-A CA weighting (A3)	103	A0/FINAL	A3/FINAL	4	+0.58	+1.83	owner[b]
SG24-B 16 CA modules	103	B0-A3/FINAL	B1-A3/FINAL	1	-0.93	+0.82	–
SG25 multi-resolution SA	104	FINE-P2+R2	32-to-64+R2	5	-0.14	+0.96	–
SG26 coarse-guided SA	105	S0+R2	S5+R2	4	-0.42	+2.00	–
SG28 PEARL-inspired + CRF	112	P0+R2	P3+R2	11	-0.08	-0.78	–
SG27 late fusion	109	S0+R2	S8+R2	39	+0.16	+1.22	–

Table 8. **Every registered family in Fig. 2.** Changes in mIoU points against the reference cell of the same run; VOC at each cell's full-VOC threshold (in-sample), COCO at that threshold (frozen transfer). The plotted cell is the promoted one, or else the best of the family's candidates (Cand.) on full VOC. Sec.: protocol section with the registration and result. [a]Promotion depends on the unregistered tie order (Sec. 10). [b]Not promoted by the registered selector; retained by owner decision.

background map (BG1), iteration to convergence (W4), multi-resolution fusion (H1), DiffSeg-style attention merging (G1), seeded region growing (G4), CA-weighted single and multiple prototypes (M1, M2), DenseCRF with SA edges (R1), a heat kernel over pass counts (P3), a symmetric normalized affinity (P5), a column-mass gate (F2), entropy-weighted observations (W1) and per-image head selection by agreement with the seed (W2). *Observation reliability.* SG2 weights each SA edge by its consistency across observations; SG8 combines the 248 propagated maps by per-pixel trimmed means, variance gating or sign agreement (promoted: sign agreement of a fraction $\rho = 0.8$); SG11 aggregates observations robustly (trimmed mean, median) where they agree; SG12 discards the least consistent observations. SG14 and SG15 combine or refine SG11. SG17 filters the second pass's additions by their reliability. *Time, heads and scale.* SG18 restricts SA to call windows; SG21 weights SA heads by structural roles; SG22 and SG23 cross the window with the number of passes and with the reliability operators, before and after DenseCRF. SG24 weights CA modules by their agreement with SA (A) or restores the 32-pixel CA modules (B). SG25–SG27 propagate through coarser SA modules alone, as guidance, or as late fusion. SG28 adds PEARL-inspired components (Procrustes alignment, boundary-aware Laplacians, cross-scale trust) [7]. Five further paper-inspired studies follow: entropy head weights and stopped random walks (SG29, NERVE [4]), nonlinear affinity sharpening by expansion, inflation and

pruning (SG30, SPARK [3]), coarse-to-fine re-seeding with background propagation (SG31, SeeDiff [6]), a second SA pass sharpened only where cross-scale reliability is low (SG32), and hierarchical residuals with a class-seeded Markov coarse start (SG33, FA-Seg and M2N2 [1, 2]). The SG-combined studies (v1, its C10 add-on, v2, v3) constrain mass transport through the 16→32→64 hierarchy: exact and relaxed conservation and completion budgets (v1), the one-sided cap (v2) and three single changes to it (v3). SG-A3G reruns the A3 seed with GoCA's orientation, and SG-threshold tests label-free per-image thresholds on the v3 maps. Only two raw-stage promotions result, SG33's Markov start M2 and C10a, which replays it; after DenseCRF every registered screen keeps the reference (Sec. 12.4). The one-sided cap (v2 V3, v3 R1) wins every fold against both references but fails the class guard on sheep; it was adopted by owner decision (Sec. 5).

8. Per-class effects

Figure 10 shows the per-class change of fourteen within-run comparisons, from the committed per-class tables (`scripts/narmyn/attention/cross_package_classes.py`). Three observations support the main text. First, the promoted raw-stage operators (top block) resemble the second pass on VOC-Sim: over the 20 classes, the change from SG2 correlates with the change from a second pass at 0.88 and that from SG12 at 0.82 (SG11: 0.44; SG8: 0.34). This is consistent with their acting through the same mechanism, object completion. Second, after DenseCRF (bottom block) the pattern weakens, and only the middle-call window and A3 keep a positive VOC-Sim mean. Third, a change's per-class effect on VOC-Sim predicts its effect on COCO-Cap poorly: the correlation over classes is between -0.01 and $+0.46$ for every comparison. Person is the most volatile class on VOC-Sim, with seven gains and seven losses beyond ± 0.005; potted plant and chair lose most often (eight comparisons each), and bicycle loses twice as often as it gains; on COCO-Cap, dining table loses in nine of fourteen comparisons. Figure 11 gives the class detail of the SG22 middle-call decision.

9. Exploratory record

Before the fold protocol existed, seventeen exploratory experiments (B1–B17) chose settings on the same images they reported. They motivated the controlled screens but are never used as evidence in the paper: CA time windows and resolutions (B1), single and leave-out CA layers (B2), joint time and resolution (B3), the exhaustive 65,535 CA layer subsets (B4), threshold and normalization (B5), heads, time–layer, fusion and cleanup (B6), SA screens and progressive SA (B7–B10), component ablation and COCO-Cap transfer of an earlier final method (B11–B13), CA call windows (B14), CA-only screens (B15), OVAM's released masks (B16), and OVAM's optimized token with DenseCRF (B17). Their maxima are optimistic by construction. Their tables and figures are in Sec. 12.7, and Table 9 lists each with the controlled screen that later tested the same idea.

10. Reproducibility and audits

Environments. Every shard records its commit and software environment. Packages ran in one of two conda environments, `gen` (PyTorch 2.4.1) and `trdp` (PyTorch 2.6 and NumPy 1.26 at the time of the runs), on V100 or A100 GPUs. Rule R7 requires one environment per package, and every comparison in the paper is within one package. Two exceptions are known: SG22–SG24 shards ran on a mix of V100 and A100 nodes, and the SG17, SG18 and SG21 shards do not record their environment. Regenerating images in another environment changes per-image counts slightly through floating-point differences in attention; the same two-pass-plus-DenseCRF pipeline, for example, scores 0.8132–0.8135 forced OOF across packages. We therefore never compare cells across packages.

Independent replication. Each package was reviewed by a second analyst, who re-derived its decisions from the per-image shards before the result entered the protocol; selected packages were also regenerated independently. The B2 token packages were independently regenerated (protocol Section 65), and the token optimizer was audited against OVAM's released code, with one recipe deviation recorded (Section 69). For SG17–SG27, every report was regenerated from the committed per-image shards: the regenerated tables match the committed ones byte for byte apart from line endings and last-digit floating-point differences, and every number quoted in the protocol matches. One dependence was found. SG23's raw promotion of MID-SG2 depends on the order in which tied candidates are visited, which the registration did not fix: the finalizer's order gives 5/5 fold votes, a neutral order 4/5, and the reverse order 3/5, in which case the reference is retained. SG23 is therefore marked as tie-order sensitive in Fig. 2 and Tab. 8. The same check found no effect of tie order on SG22. SG28 was audited the same way, and its code was also checked against the registration. Every reported number matches, and its baseline equals the SG27 baseline counts on all 490 VOC-Sim and 707 COCO-Cap images. The run enforced that per-image replay only for its two smoke images; the full replay was confirmed afterwards from the shards (protocol Section 130).

The A3 affinity orientation. SG24's A3 score compares each CA module's query–key affinity with a resized SA

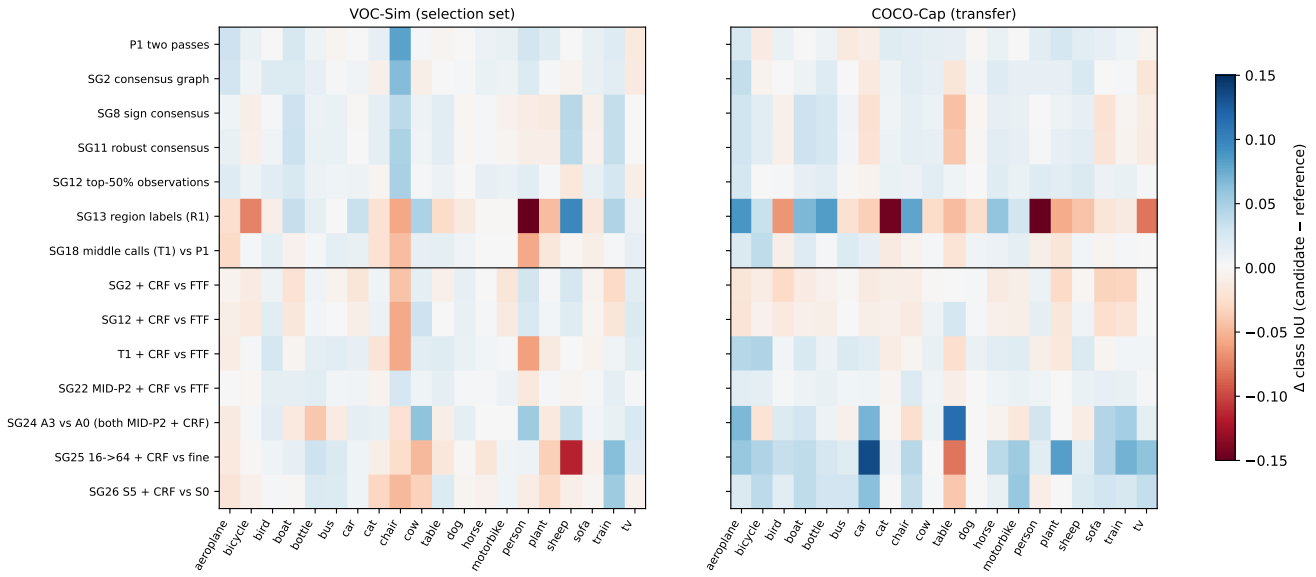


Figure 10. **Per-class IoU change, candidate minus same-run reference.** Rows: fourteen comparisons, raw operators above the line and complete pipelines with DenseCRF below; VOC-Sim at each cell's full-VOC threshold, COCO-Cap at that threshold (frozen). Blue: gain; red: loss; colour saturates at ± 0.15.

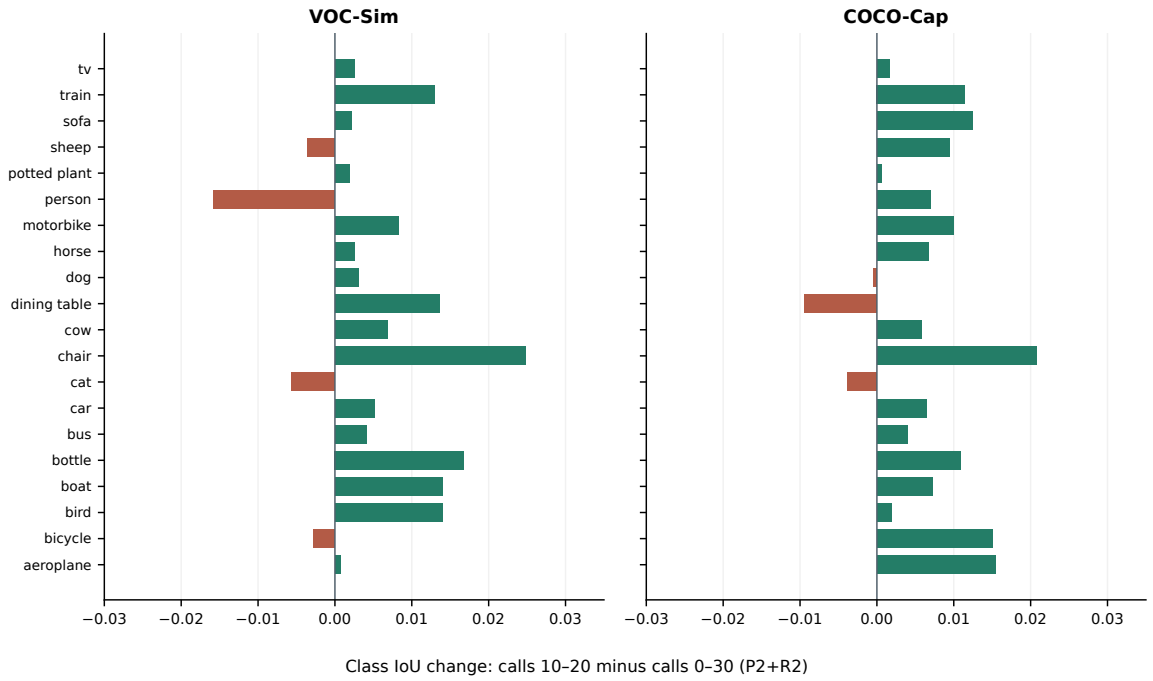


Figure 11. **The middle-call gain is heterogeneous.** Per-class IoU change for MID-P2+R2 versus ALL-P2+R2 in the same SG22 evaluation, with the threshold chosen on VOC only. Chair improves on both datasets; VOC person loses 0.016.

affinity. In the implementation (`src/segmentation/sg24_ca.py`), the resized SA affinity is not transposed back, so the score is $\langle M^\top, P \rangle$ rather than GoCA's $\langle M, P \rangle$ [5]. A paired rerun (SG-A3G) accumulated both seeds in one diffusion run. They agree to Pearson 0.99999, the control changes by -0.0000 after R2, and the hierarchy's gain holds on the corrected seed (Table 1). All A3 numbers are those of the implemented operator, and the orientation is immaterial to them.

Regenerated figure maps. The qualitative figures of the coarse-to-fine pipeline regenerate fixed-rule images on the GPU (`scripts/paper/select_story_images.py`, `figure_story_maps.py`). Each image's raw IoU is checked against the committed SG-combined-v3 counts: 13 of 14 images agree to 10^{-4}, and one COCO-Cap image differs by 0.005 (GPU nondeterminism across node types).

Sources of figures and tables. Earlier-pipeline gallery (Sec. 11): `scripts/narmyn/attention/select_figure_images.py` (rule), `figure_maps.py` (regeneration; each final-mask IoU is checked against the SG5 shards) and `scripts/paper/figure_pipeline_stages.py`. Fig. 9 and Tab. 8: `scripts/paper/figure_decision_map.py`. Propagation-depth figure: `scripts/narmyn/attention/figure_passes_maps.py` (rule and regeneration, checked against the SG22 shards) and `scripts/paper/figure_passes.py`. Per-class figure: `scripts/narmyn/attention/cross_package_classes.py`. Earlier-pipeline figures (temporal examples, CA resolution example): `scripts/paper/build_story_figures.py`. Coarse-to-fine figures of the main paper and this supplement: `scripts/paper/figure_story_qualitative.py` (qualitative, from the regenerated maps) and `figure_coarse_to_fine.py` (quantitative, from committed count shards and result files). The main-paper tables are hand-written from the result files named in `paper/README.md`. No figure script selects a method or changes a result.

11. Qualitative figures of the fine-only baseline

These figures show the earlier fine-only pipeline of Sec. 2 (uniform seed, two passes through 64-pixel SA, DenseCRF at 0.45). Figure 12 shows, by a fixed rule, the four classes with the lowest class IoU of that pipeline on each dataset, each shown by its median-IoU image. No image was inspected before the choice. The COCO-Cap columns show three different failures. For the chair, propagation leaks into the wooden wall behind it. For the dining table, which fills the frame, the mask covers the plate and bread on it but not its surface. For the bottle, the mask covers both bottles, but only one is annotated, so part of the error is in the label rather than the mask.

CA resolution. Figure 13 illustrates the CA resolution decision (Table 2 of the main paper) on one image.

Propagation depth. Figure 14 applies both selection rules of Fig. 2 to both datasets.

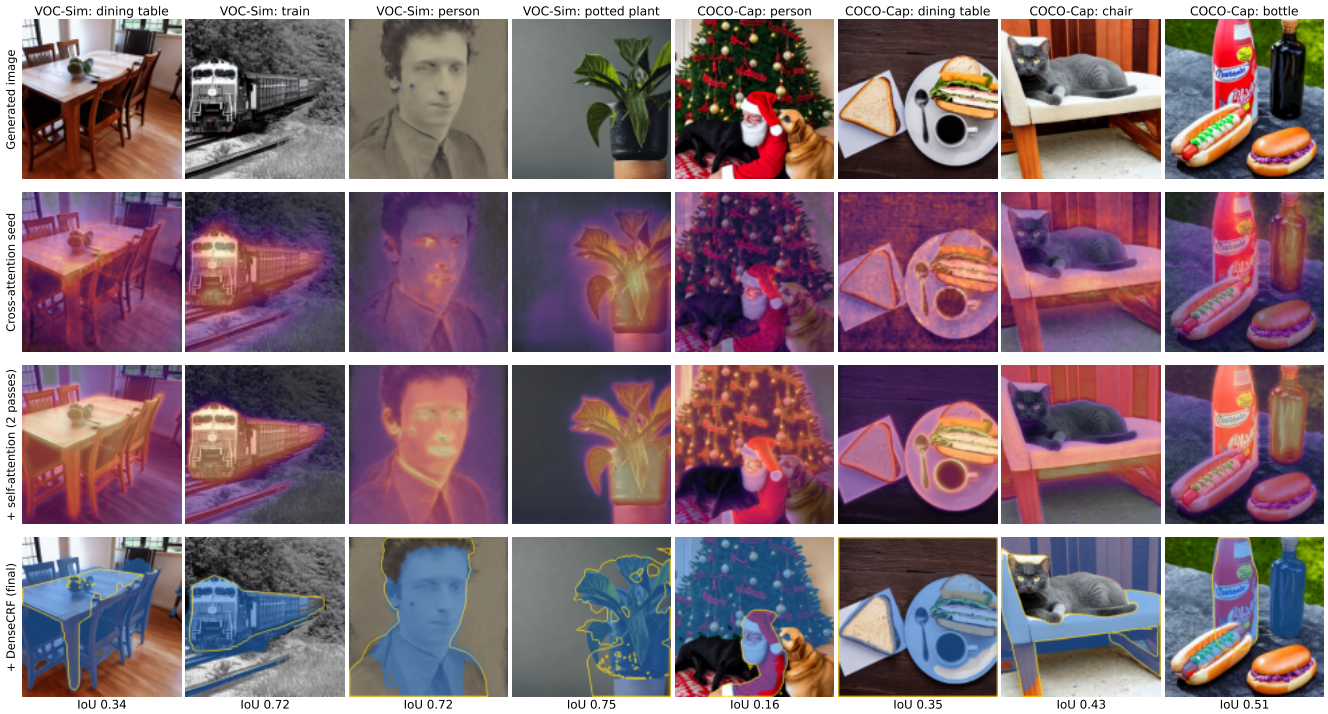


Figure 12. **The weakest classes, one median image each.** Rows: generated image; CA seed; two SA passes; final mask after DenseCRF at 0.45 (blue) with the ground-truth outline (yellow) and the image IoU. Columns: the four lowest class IoUs on VOC-Sim, then on COCO-Cap, from the lowest.

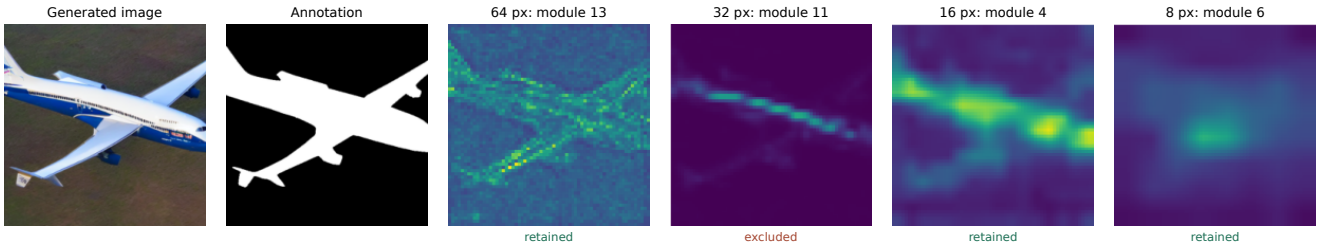


Figure 13. **Why CA resolution is a real design choice.** One VOC-Sim airplane generated image and annotation (left), followed by representative head-averaged CA modules from the same image and all calls. All heatmaps use one color scale. The selected seed retains 8, 16 and 64-pixel modules and omits the 32-pixel group. This visual example motivates the controlled resolution test; the fold-level evidence is in Table 2 of the main paper.

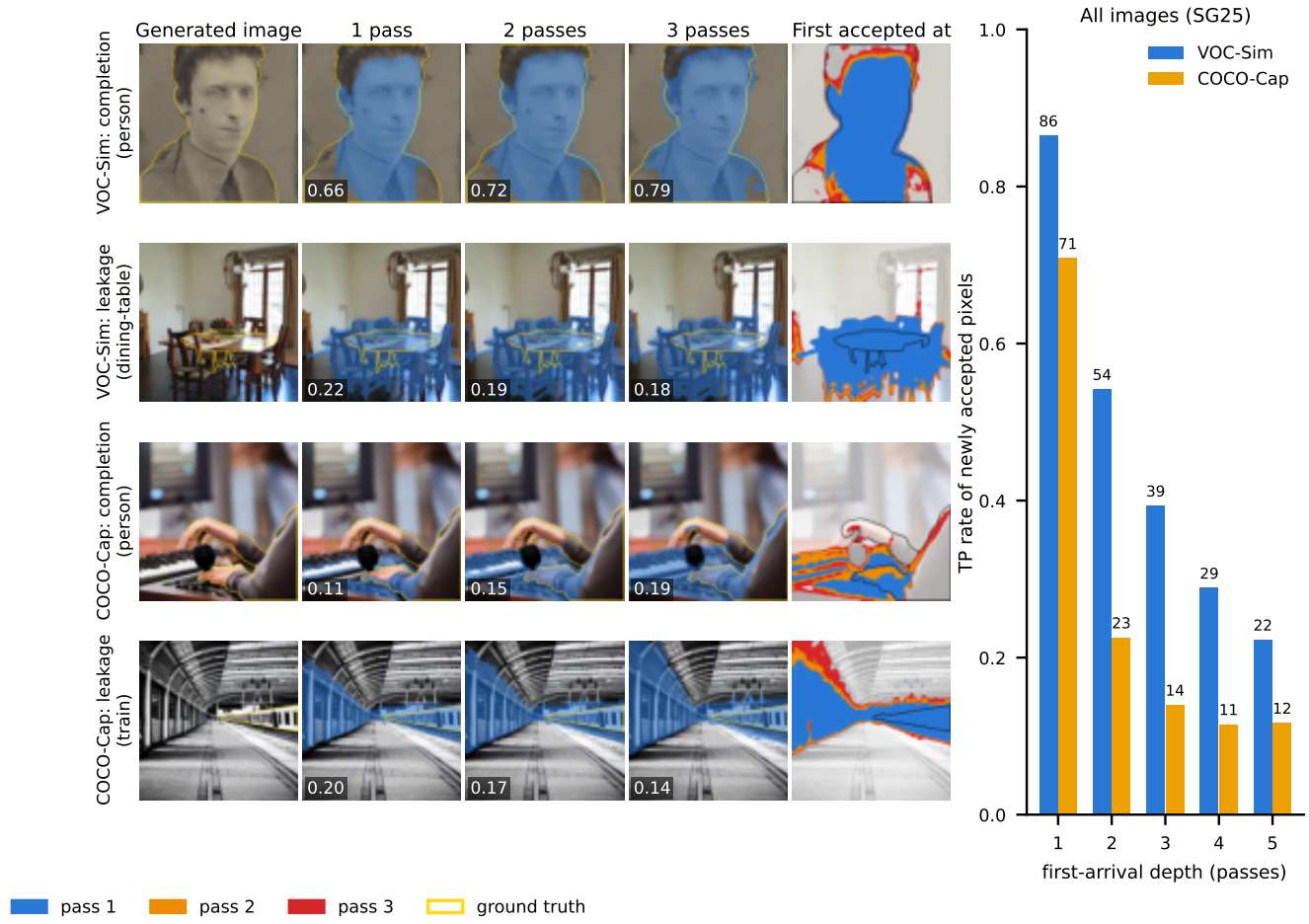


Figure 14. **One, two and three SA passes, both rules on both datasets.** Rows: on each dataset, the image at the 90th percentile of the two-pass gain (completion) and at the 10th percentile of the three-pass gain (leakage), from the all-call raw maps of SG22 at threshold 0.45. Columns and right panel as in Fig. 2.

12. Complete experiment record

This section lists every experiment of the study, including those the paper mentions only in passing. Nothing here is new evidence: every number is read from a committed result file. Three kinds of number appear, and they should not be mixed.

- **Out-of-fold (OOF)** VOC-Sim mIoU: a choice made on four folds and scored on the fifth, pooled over the five rotations. This is the number every decision uses.
- **Full-VOC** mIoU: a cell at its best threshold on all 490 images. It is in-sample and only describes the cell.
- **COCO-Cap** mIoU: frozen transfer at the threshold chosen on VOC-Sim. Nothing is ever chosen on COCO-Cap.

Experiments come in three groups: exploratory runs before the fold protocol (B1–B17, Sec. 12.7); the controlled screens of the first three phases (A1–A12, P3-C, T-C, Sec. 12.3); and the later registered packages (SG1–SG33, the SG-combined studies and the training-free backlog, Sec. 12.4). Table 9 lists all of them; Table 10 lists every registered fold decision.

12.1. Ledger and registered decisions

Table 9. **Ledger of every experiment.** B: exploratory, before the fold protocol (in-sample, never used as evidence). A, P3-C, T-C: controlled screens of the first three phases. SG, BL: later registered packages. Sec.: protocol section with the result. Best/decision: for B rows, the best in-sample configuration and mIoU; for controlled rows, the promoted cell(s) or "none".

ID	Experiment	Status	Cells	Best / decision
B1	CA time windows and resolutions	exploratory, in-sample	11	first look at time and resolution
B2	Single and leave-out CA layers	exploratory, in-sample	24	layer screening
B3	Joint time x resolution	exploratory, in-sample	21	interaction check
B4	Exhaustive CA layer subsets	exploratory, in-sample	17	chose BEST_12 = layers 0, 2, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15
B5	Threshold x normalization	exploratory, in-sample	153	chose percentile 1-99 and threshold 0.45
B6	Stage-1 suite (heads, time-layer, fusion, cleanup)	exploratory, in-sample	590	chose the fixed top-4 CA heads per layer
B7	Self-attention screens and progressive SA	exploratory, in-sample	5320	first evidence that SA propagation helps
B8	Full streaming ablations	exploratory, in-sample (choices from B1-B7 frozen)	6261	SA at U3.2 (64x64), 4 passes: 0.7777
B9	Focused SA follow-up	exploratory, in-sample	1632	top-4-head seed, 2 passes, t=0.45: 0.7922
B10	Joint SA head x time x passes x threshold	exploratory, in-sample	4680	final method: 4 passes, heads 0,1,2,4,5,6, calls 15-30, t=0.45, 1% cleanup: 0.7997
B11	Component ablation of the final method	exploratory (COCO untuned)	18	SA propagation is the largest component
B12	COCO-Cap generalization of the final method	untuned transfer	6	0.5719 → 0.6080
B13	Revised system (1 pass, all 16 CA layers) and its ablation	exploratory; informed by COCO-Cap	14	revised system: VOC 0.7824, COCO 0.6322
B14	CA call windows with all SA heads	exploratory	4	no gain over all calls
B15	Clean CA-only screens (pre-sync)	exploratory, in-sample; audited, superseded by A1	1573	ratified as correct; re-run with folds as A1
B16	OVAM released masks	reference (authors' masks)	3	0.7036 / 0.8247 / 0.6616
B17	OVAM optimized token and DenseCRF on the final method	exploratory; nothing selected	32	token + DenseCRF: VOC 0.8525, COCO 0.6612
A1	Controlled CA package B: CA-L, CA-H, CA-A	controlled, 5-fold cross-validated	65961	CA-L drop modules 10-11; CA-H drop head 1; CA-A keep reference (combined in A3)
A2	Controlled CA package A: CA-T, CA-R (Codex)	controlled, 5-fold cross-validated	1279	CA-T keep all calls; CA-R keep 8+16+64 (drop 32x32) (combined in A3)
A3	Controlled CA confirmation (CA-C) and the frozen CA	controlled, 5-fold cross-validated	6	frozen CA: CA-R alone (drop the 32x32 modules), all heads
A4	Controlled SA package B: SA-L, SA-H, operator check	controlled, 5-fold cross-validated	65951	SA-L only module 15 (U3.2, 64x64); SA-H heads 1, 4, 6; A x kept (combined in A6)
A5	Controlled SA package A: SA-T, SA-R (Codex)	controlled, 5-fold cross-validated	1279	SA-T call 30 only; SA-R resolutions 8+64; both calibration-dependent (combined in A6)
A6	Controlled SA confirmation (SA-C, Codex) and the frozen SA	controlled, 5-fold cross-validated	13	frozen SA: module 15 (U3.2) only, all calls, all heads
A7	Joint CA x SA interaction audit (Codex)	controlled, 5-fold cross-validated	78	sequential pair kept 5/5; Section 11 unchanged
A8	Phase 3 propagation package (Codex): P1, W3, P2, P6	controlled, 5-fold cross-validated; threshold chosen in the folds	40	two passes mean_i(A_i c) promoted to P3-C
A9	Phase 3 fusion and calibration package: P4, F3, F4, N2-N4, R2	controlled, 5-fold cross-validated; threshold chosen in the folds	12	DenseCRF promoted to P3-C; the rest kept at REF
A10	Released baselines scored with OVAM's procedures	reproduction of published values	19	every printed OVAM-notebook value reproduced
AN1	Why a second SA pass helps (analysis, nothing selected)	analysis of committed counts		second pass fills under-covered objects; the token seed already covers them; COCO-Cap leakage cancels the gain
A11	Token track B1/B1b: OVAM's optimized tokens on the frozen pipeline	controlled, 5-fold cross-validated; few-shot track (2-3 annotated images per class)	27	token promoted on every pipeline: REF 0.787 → 0.851 (conditional query)
A12	Token learning B2a/S4/B2b (Codex): retrained tokens, shots, fold-safe token	controlled, fixed T-C pipeline and threshold 0.475; linear-objective variant of OVAM's recipe (audit)	9	retrained all-support token = released on VOC-Sim (0.868 vs 0.865), +0.0075 on COCO-Cap; second support gives the gain; released token kept

ID	Experiment	Status	Cells	Best / decision
P3-C	P3-C (Sec. 51)	controlled, 5-fold	30	promoted: `P1/obs_n2|N0/global|R2/crf`, `P1/obs_n2|N4/otsu|R2/crf`
SG5	SG5 contrastive cross-attention seed (Sec. 66, 70)	controlled, 5-fold	40	promoted: none
SG1/SG3	SG1/SG3 anchored graph refinement (Sec. 68, 71)	controlled, 5-fold	76	promoted: none
SG3-T	Locked token-SG3 confirmation (Sec. 72, 78, 80)	controlled, 5-fold	6	promoted: none
SG8	SG8 observation-consensus propagation (Sec. 73, 78, 80)	controlled, 5-fold	32	promoted: `T/A/median`, `W/C/rho0p80`
SG9	SG9 reciprocal SA propagation (Sec. 74, 78, 80)	controlled, 5-fold	8	promoted: none
SG10	SG10 seed-connected bottleneck propagation (Sec. 75, 77, 79)	controlled, 5-fold	54	promoted: none
SG2	SG2 consensus-edge SA graph (Sec. 81, 85)	controlled, 5-fold	56	promoted: `T/G_cv_g20_n1`, `W/G_cv_g20_n1`
SG4	SG4 SA regions labelled by CA (Sec. 81, 85)	controlled, 5-fold	40	promoted: none
BL	Training-free Track-A backlog (14 families) (Sec. 82, 83, 86)	controlled, 5-fold	44	promoted: none
SG11/SG12	SG11 robust consensus and SG12 reliable observations (Sec. 84, 89)	controlled, 5-fold	14	promoted: `W/SG11/trim20/rho0p80`, `W/SG12/top50`
SG2+R2	Locked SG2 + DenseCRF confirmation (Sec. 87, 89)	controlled, 5-fold	6	promoted: none
SG13	SG13 target-conditioned region labelling (Sec. 88, 90)	controlled, 5-fold	7	promoted: none
SG12+R2	Locked SG12 + DenseCRF confirmation (Sec. 91, 92)	controlled, 5-fold	5	promoted: none
SG14	SG14 SG11 x SG12 complementarity (Sec. 91, 92)	controlled, 5-fold	4	promoted: `W/SG11/trim20/rho0p80`, `W/SG14`
SG15	SG15 prior-driven SPEC32 region refinement (Sec. 91, 92, 93)	controlled, 5-fold	16	promoted: none
SG17	SG17 reliability-filtered second-pass additions (Sec. 94, 95)	controlled, 5-fold	4	promoted: none
SG18	SG18 temporal-windowed one-pass propagation (Sec. 96, 97)	controlled, 5-fold	5	promoted: `W/T1`
SG18-T1+R2	SG18 follow-up: T1 window + DenseCRF (Sec. 98)	controlled, 5-fold	4	promoted: none
SG21	SG21 structural SA-head weighting (Sec. 101)	controlled, 5-fold	5	promoted: none
SG22	SG22 call window x propagation depth factorial (Sec. 102)	controlled, 5-fold	12	promoted: `MID-P2+R2`
SG23	SG23 call window x reliability rule factorial (Sec. 102)	controlled, 5-fold	18	promoted: `MID-SG2`
SG24-A	SG24-A automatic CA seed construction (Sec. 103)	controlled, 5-fold	10	promoted: none
SG24-B	SG24-B CA module set: 11 vs all 16 (Sec. 103)	controlled, 5-fold	4	promoted: none
SG25	SG25 multi-resolution SA propagation (+ Pre-SG26 first-arrival diagnostic) (Sec. 104)	controlled, 5-fold	12	promoted: none
SG26	SG26 coarse-gated fine SA (Sec. 105)	controlled, 5-fold	14	promoted: none
SG27	SG27 late fusion of three SA output maps (Sec. 106, 109)	controlled, 5-fold	84	promoted: none
SG28	SG28 PEARL-inspired components (Procrustes alignment, Laplacian smoothing) (Sec. 110, 112)	controlled, 5-fold	24	promoted: none
SG29	SG29 NERVE-inspired head weighting and stopped random walk (Sec. 113, 115)	controlled, 5-fold	20	promoted: none
SG30	SG30 SPARK-inspired nonlinear affinity sharpening (Sec. 116, 118)	controlled, 5-fold	18	promoted: none
SG31	SG31 SeeDiff-inspired re-seeding and background propagation (Sec. 117, 119, 120)	controlled, 5-fold	22	promoted: none
SG32	SG32 reliability-adaptive second SA pass (Sec. 121)	controlled, 5-fold	24	promoted: none
SG33	SG33 hierarchical residuals and coarse Markov reachability (Sec. 122)	controlled, 5-fold	24	promoted: `M2`
SG-combined-v1	SG-combined-v1 constrained transport and budgeted completion (Sec. 123)	controlled, 5-fold	20	promoted: none
SG-combined C10	SG-combined-v1 C10 add-on: SG33 M2 coarse source (Sec. 124, 127)	controlled, 5-fold	8	promoted: `C10a`
SG-combined-v2	SG-combined-v2 one-sided coarse caps and route diagnostic (Sec. 126, 127)	controlled, 5-fold	14	promoted: none
SG-A3G	SG-A3G GoCA-oriented A3 seed and paired rerun (Sec. 128, 131)	controlled, 5-fold	8	promoted: none
SG-combined-v3	SG-combined-v3 single changes to the V3 capped hierarchy (Sec. 129, 132)	controlled, 5-fold	10	promoted: none
SG-threshold	SG-threshold label-free per-image thresholds (CPU, on SG-combined-v3 counts) (Sec. 133, 134)	controlled, 5-fold		promoted: none

Table 10. **Every registered fold decision.** One row per screen, in protocol order. Ref.: the screen's reference cell; Cand.: number of candidates; Fold picks: the cell chosen on each set of four training folds (ref = reference); OOF: pooled out-of-fold VOC-Sim mIoU of the reference and of the selection procedure. A change is promoted only if one candidate wins at least four of five folds and the OOF guard passes. Rows without OOF values come from screens whose record keeps only the votes.

Pkg. (Sec.)	Screen	Ref.	Cand.	Fold picks	OOF ref	OOF sel.	Gain	Outcome
Cross-attention seed								
A1 (21)	CA-L	L/all_16	158	L/leave2_10_11 5	0.6661	0.6923	+2.62	promoted
A1 (21)	CA-H	H/0_1_2_3_4_5_6_7	254	H/0_2_3_4_5_6_7 4; ref 1	0.6661	0.6741	+0.80	promoted
A1 (21)	CA-A	A/cond_padded_all	11	ref 5	0.6661	0.6661	+0.00	kept
A2 (20)	CA-T	time/all_00_30	78	ref 5	0.6661	0.6661	+0.00	kept
A2 (20)	CA-R	resolution/r8_r16_r32_r64	14	resolution/r8_r16_r64 5	0.6661	0.6957	+2.96	promoted
A3 (27)	CA-C	C/reference	5	C/R 4; C/L 1	0.6653	0.6914	+2.60	promoted
Self-attention module								
A4 (30)	SA-L	SL/all_16	158	SL/single_15 5	0.7553	0.7884	+3.31	promoted
A4 (30)	SA-H	SH/0_1_2_3_4_5_6_7	254	SH/1_4_6 5	0.7553	0.7610	+0.57	promoted
A5 (31)	SA-T	time/all_00_30	78	time/step_30 5	0.7556	0.7710	+1.53	promoted
A5 (31)	SA-R	resolution/r8_r16_r32_r64	14	resolution/r8_r64 5	0.7556	0.7676	+1.19	promoted
A6 (33)	SA-C	C/reference	11	C/L 5	0.7556	0.7887	+3.30	promoted
A7 (36)	Joint CA-SA	CA=C/R|SA=C/L	77	ref 5	0.7887	0.7887	+0.00	kept
Propagation, refinement and tokens								
A8 (43)	P1 observation-wise powers	REF/one_pass	3	P1/obs_n2 4; P1/obs_n3 1	0.7875	0.7991	+1.16	promoted
A8 (43)	W3 average-before powers	REF/one_pass	4	ref 4; W3/mean_n2 1	0.7875	0.7875	-0.00	kept
A8 (43)	P2 observation-wise restart	REF/one_pass	12	ref 4; P2/obs_b10_n4 1	0.7875	0.7877	+0.02	kept
A8 (43)	P2 average-before restart	REF/one_pass	12	ref 5	0.7875	0.7875	+0.00	kept
A8 (43)	P6 sparse affinity	REF/one_pass	7	ref 5	0.7875	0.7875	+0.00	kept
A9 (44)	P4 residual mix	REF/one_pass	4	ref 5	0.7875	0.7875	+0.00	kept
A9 (44)	F3/F4 gating	REF/one_pass	2	ref 5	0.7875	0.7875	+0.00	kept
A9 (44)	N2-N4 calibration	REF/one_pass	3	ref 5	0.7875	0.7875	+0.00	kept
A9 (44)	R2 DenseCRF	REF/one_pass	1	R2/crf 5	0.7875	0.8049	+1.75	promoted
P3-C (51)	P3-C mandatory confirmation	REF/one_pass|N0/global|R0/none	4	P1/obs_n2|N0/global|R2/crf 5	0.7875	0.8135	+2.60	promoted
P3-C (51)	P3-C expanded confirmation	REF/one_pass|N0/global|R0/none	30	P1/obs_n2|N4/otsu|R2/crf 5	0.7875	0.8207	+3.32	promoted
A11 (50)	B1 released token, OVAM form (unconditional query)	W/REF	1	TU/REF 5	0.7873	0.8428	+5.55	promoted
A11 (50)	B1 released token, conditional query	W/REF	1	TC/REF 5	0.7873	0.8507	+6.34	promoted
T-C (52)	T-C mandatory confirmation (token on REF, P1, R2, P1+R2)	TC/REF	3	TC/R2 4; TC/P1+R2 1	0.8507	0.8635	+1.28	promoted
Graph, seed and observation operators								
SG5 (70)	SG5-W seed under REF	W/REF	5	ref 5	0.7873	0.7873	+0.00	kept
SG5 (70)	SG5-W in the training-free pipeline (FTF)	W/P1+R2	5	ref 5	0.8134	0.8134	+0.00	kept
SG5 (70)	SG5-T seed under REF	TC/REF	1	ref 5	0.8496	0.8496	+0.00	kept
SG5 (70)	SG5-T in the few-shot pipeline (T-C)	TC/R2	1	ref 5	0.8647	0.8647	+0.00	kept
SG1/SG3 (71)	W-SG1	W/REF	12	ref 5	0.7873	0.7873	+0.00	kept
SG1/SG3 (71)	W-SG3	W/REF	24	ref 5	0.7873	0.7873	+0.00	kept
SG1/SG3 (71)	T-SG1	T/REF	12	ref 5	0.8507	0.8507	+0.00	kept
SG1/SG3 (71)	T-SG3	T/REF	24	ref 3; T/SG3/f95_b05_r0_n1 2	0.8507	0.8534	+0.27	kept
SG8 (78)	W-A	W/REF	3	W/A/trim20 3; W/A/median 2	0.7875	0.7942	+0.67	kept (votes)
SG8 (78)	W-B	W/REF	4	ref 5	0.7875	0.7875	+0.00	kept
SG8 (78)	W-C	W/REF	4	W/C/rho0p80 4; W/C/rho0p70 1	0.7875	0.7970	+0.95	promoted
SG8 (78)	W-D	W/REF	4	ref 5	0.7875	0.7875	+0.00	kept
SG8 (78)	T-A	T/REF	3	T/A/median 4; ref 1	0.8508	0.8544	+0.36	promoted
SG8 (78)	T-B	T/REF	4	ref 5	0.8508	0.8508	+0.00	kept
SG8 (78)	T-C	T/REF	4	ref 5	0.8508	0.8508	+0.00	kept
SG8 (78)	T-D	T/REF	4	ref 5	0.8508	0.8508	+0.00	kept
SG9 (78)	W	W/REF	3	ref 5	0.7875	0.7875	+0.00	kept
SG9 (78)	T	T/REF	3	ref 5	0.8508	0.8508	+0.00	kept
SG10 (79)	SG10-W	W/REF	16	ref 5	0.7873	0.7873	+0.00	kept
SG10 (79)	SG10-T	T/REF	16	ref 5	0.8496	0.8496	+0.00	kept
SG2 (85)	SG2-W-1pass	W/REF	12	W/G_cv_g20_n1 5	0.7873	0.7984	+1.11	promoted
SG2 (85)	SG2-W-2pass	W/P1	12	ref 5	0.7997	0.7997	+0.00	kept
SG2 (85)	SG2-T-1pass	T/REF	12	T/G_cv_g20_n1 5	0.8496	0.8568	+0.72	promoted
SG2 (85)	SG2-T-2pass	T/P1	12	ref 5	0.8534	0.8534	+0.00	kept

Pkg. (Sec.)	Screen	Ref.	Cand.	Fold picks	OOF ref	OOF sel.	Gain	Outcome
SG4 (85)	SG4-W	W/REF	18	ref 5	0.7873	0.7873	+0.00	kept
SG4 (85)	SG4-T	T/REF	18	ref 5	0.8496	0.8496	+0.00	kept
BL (83)	P3	REF/one_pass	4	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	P5	REF/one_pass	4	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	P9	REF/one_pass	6	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	F2	REF/one_pass	4	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	BG1	REF/one_pass	5	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	W1	REF/one_pass	3	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	W2	REF/one_pass	3	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	W4	REF/one_pass	1	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	H1	REF/one_pass	1	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	G1	REF/one_pass	2	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	G4	REF/one_pass	3	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	M1	REF/one_pass	1	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	M2	REF/one_pass	3	ref 5	0.7875	0.7875	+0.00	kept
BL (83)	R1	REF/one_pass	3	ref 5	0.7875	0.7875	+0.00	kept
SG11/SG12 (89)	SG11	W/REF	6	W/SG11/trim20/rho0p80 5	0.7873	0.7984	+1.11	promoted
SG11/SG12 (89)	SG12	W/REF	4	W/SG12/top50 5	0.7873	0.7967	+0.94	promoted
SG2+R2 (89)	primary_vs_FTF	W/P1+R2	1	ref 5	0.8132	0.8132	+0.00	kept
SG2+R2 (89)	secondary_vs_one_pass_CRF	W/REF+R2	1	ref 3; W/SG2+R2 2	0.8047	0.8057	+0.10	kept
SG13 (90)	R1	W/REF	1	ref 5	0.7875	0.7875	+0.00	kept
SG13 (90)	R2	W/REF	1	ref 5	0.7875	0.7875	+0.00	kept
SG13 (90)	R3	W/REF	1	ref 5	0.7875	0.7875	+0.00	kept
SG13 (90)	joint	W/REF	3	ref 5	0.7875	0.7875	+0.00	kept
SG12+R2 (92)	primary	W/P1+R2	1	ref 5	0.8132	0.8132	+0.00	kept
SG14 (92)	versus_W/REF	W/REF	1	W/SG14 5	0.7873	0.7987	+1.14	promoted
SG14 (92)	versus_W/SG11/trim20/rho0p80	W/SG11/trim20/rho0p80	1	ref 5	0.7984	0.7984	+0.00	kept
SG14 (92)	versus_W/SG12/top50	W/SG12/top50	1	ref 5	0.7967	0.7967	+0.00	kept
SG14 (92)	joint	W/REF	3	W/SG11/trim20/rho0p80 5	0.7873	0.7984	+1.11	promoted
SG15 (92)	W/P1_SG15-A	W/P1	3	ref 5	0.7997	0.7997	+0.00	kept
SG15 (92)	W/P1_SG15-B	W/P1	1	ref 5	0.7997	0.7997	+0.00	kept
SG15 (92)	W/SG11/trim20/rho0p80_SG15-A	W/SG11/trim20/rho0p80	3	ref 5	0.7984	0.7984	+0.00	kept
SG15 (92)	W/SG11/trim20/rho0p80_SG15-B	W/SG11/trim20/rho0p80	1	ref 5	0.7984	0.7984	+0.00	kept
SG15 (92)	W/SG12/top50_SG15-A	W/SG12/top50	3	ref 5	0.7967	0.7967	+0.00	kept
SG15 (92)	W/SG12/top50_SG15-B	W/SG12/top50	1	ref 5	0.7967	0.7967	+0.00	kept
Temporal, structural, CA-weighting and multi-scale studies								
SG17 (95)	screen	W/P1	3	ref 5	0.7998	0.7998	+0.00	kept
SG18 (97)	screen	W/REF	3	W/T1 5	0.7873	0.7939	+0.66	promoted
SG18-T1+R2 (98)	screen	W/P1+R2	1	ref 5	0.8135	0.8135	+0.00	kept
SG21 (101)	screen	SG18/T1	4	ref 5	0.7939	0.7939	+0.00	kept
SG22 (102)	raw	ALL-P2	3	ref 3; MID-P2 2	0.7997	0.8003	+0.07	kept
SG22 (102)	crf	ALL-P2+R2	3	MID-P2+R2 4; ref 1	0.8132	0.8158	+0.26	promoted
SG23 (102)	raw	MID-R0	3	MID-SG2 5	0.7939	0.7999	+0.60	promoted
SG23 (102)	crf	ALL-P2+R2	8	ref 4; MID-SG2+R2 1	0.8133	0.8126	-0.07	kept
SG24-A (103)	CA	A0/CA	4	ref 5	0.7135	0.7135	+0.00	kept
SG24-A (103)	FINAL	A0/FINAL	4	A3/FINAL 3; ref 2	0.8169	0.8162	-0.07	kept (votes)
SG24-B (103)	CA	B0-A3/CA	1	ref 5	0.7050	0.7050	+0.00	kept
SG24-B (103)	FINAL	B0-A3/FINAL	1	ref 5	0.8230	0.8230	+0.00	kept
SG25 (104)	raw	FINE-P2	5	ref 5	0.8097	0.8097	+0.00	kept
SG25 (104)	final	FINE-P2+R2	5	ref 5	0.8228	0.8228	+0.00	kept
SG26 (105)	raw	S0	6	ref 5	0.8097	0.8097	+0.00	kept
SG26 (105)	final	S0+R2	6	ref 5	0.8228	0.8228	+0.00	kept
SG27 (109)	raw	S0	41	ref 5	0.8097	0.8097	+0.00	kept
SG27 (109)	final	S0+R2	41	ref 5	0.8228	0.8228	+0.00	kept
SG28 (112)	raw/all	P0	11	ref 5	0.8097	0.8097	+0.00	kept
SG28 (112)	raw/PA	P0	3	ref 5	0.8097	0.8097	+0.00	kept
SG28 (112)	raw/plain_and_boundary	P0	2	ref 5	0.8097	0.8097	+0.00	kept
SG28 (112)	raw/confidence	P0	3	ref 5	0.8097	0.8097	+0.00	kept
SG28 (112)	raw/confidence_and_boundary	P0	3	ref 5	0.8097	0.8097	+0.00	kept
SG28 (112)	final/all	P0+R2	11	ref 5	0.8228	0.8228	+0.00	kept
SG28 (112)	final/PA	P0+R2	3	ref 5	0.8228	0.8228	+0.00	kept
SG28 (112)	final/plain_and_boundary	P0+R2	2	ref 5	0.8228	0.8228	+0.00	kept
SG28 (112)	final/confidence	P0+R2	3	ref 5	0.8228	0.8228	+0.00	kept
SG28 (112)	final/confidence_and_boundary	P0+R2	3	ref 5	0.8228	0.8228	+0.00	kept
Paper-inspired propagation, hierarchy and combined studies								
SG29 (115)	raw	N0	6	ref 5	0.8098	0.8098	+0.00	kept
SG29 (115)	final	N0+R2	6	ref 5	0.8229	0.8229	+0.00	kept
SG30 (118)	raw	S0	7	ref 5	0.8098	0.8098	+0.00	kept
SG30 (118)	final	S0+R2	7	ref 5	0.8231	0.8231	+0.00	kept
SG31 (119, 120)	raw	S0	10	ref 5	0.8097	0.8097	+0.00	kept
SG31 (119, 120)	final	S0+R2	10	ref 5	0.8228	0.8228	+0.00	kept

Pkg. (Sec.)	Screen	Ref.	Cand.	Fold picks	OOF ref	OOF sel.	Gain	Outcome
SG32 (121)	raw	S0	11	ref 5	0.8097	0.8097	+0.00	kept
SG32 (121)	final	S0+R2	11	ref 5	0.8229	0.8229	+0.00	kept
SG33 (122)	raw	S0	11	M2 4; ref 1	0.8098	0.8159	+0.62	promoted
SG33 (122)	final	S0+R2	11	ref 5	0.8229	0.8229	+0.00	kept
SG-combined-v1 (123)	raw	C0	9	ref 5	0.8097	0.8097	+0.00	kept
SG-combined-v1 (123)	final	C0+R2	9	ref 5	0.8228	0.8228	+0.00	kept
SG-combined C10 (124, 127)	raw	S0	3	C10a 4; ref 1	0.8098	0.8159	+0.61	promoted
SG-combined C10 (124, 127)	final	S0+R2	3	ref 5	0.8231	0.8231	+0.00	kept
SG-combined-v2 (126, 127)	raw	V0	6	ref 5	0.8098	0.8098	+0.00	kept
SG-combined-v2 (126, 127)	final	V0+R2	6	ref 5	0.8231	0.8231	+0.00	kept
SG-A3G (128, 131)	raw	V0	3	ref 4; G3 1	0.8097	0.8098	+0.02	kept
SG-A3G (128, 131)	final	V0+R2	3	ref 5	0.8228	0.8228	+0.00	kept
SG-combined-v3 (129, 132)	raw	R0	4	ref 4; W2 1	0.8098	0.8093	-0.05	kept
SG-combined-v3 (129, 132)	final	R0+R2	4	ref 5	0.8230	0.8230	+0.00	kept
SG-threshold (133, 134)	raw	R0.T0	12	ref 5	0.8098	0.8098	+0.00	kept
SG-threshold (133, 134)	final	R0.T0+R2	12	ref 4; R1.T6+R2 1	0.8230	0.8232	+0.02	kept

12.2. The main pipelines, class by class

Class	VOC-Sim					COCO-Cap				
	1 pass	2 passes	FTF	MID	A3	1 pass	2 passes	FTF	MID	A3
aeroplane	83.5	86.7	89.6	89.7	88.4	56.4	58.7	63.4	64.9	71.7
bicycle	73.7	74.8	76.2	76.0	76.3	58.1	56.9	57.6	59.1	57.1
bird	84.0	83.9	88.4	89.8	91.3	71.3	72.2	75.6	75.8	77.9
boat	81.2	83.6	84.0	85.4	83.9	51.7	51.7	52.7	53.4	56.3
bottle	85.6	86.4	87.7	89.4	85.3	45.7	46.4	47.8	48.9	49.6
bus	90.9	90.5	90.6	91.0	89.7	92.0	90.5	90.6	91.0	90.3
car	82.5	82.6	82.7	83.2	84.6	56.7	55.7	55.5	56.1	63.3
cat	81.3	82.6	83.3	82.7	83.9	82.9	84.6	86.0	85.6	85.4
chair	62.9	70.9	77.6	80.1	77.9	41.9	43.6	45.0	47.1	44.6
cow	84.3	84.4	85.5	86.2	92.2	80.1	81.4	84.2	84.7	85.5
dining table	45.3	44.9	45.0	46.4	45.6	37.4	38.6	36.5	35.6	47.0
dog	90.3	90.2	91.5	91.8	93.2	74.5	74.3	75.6	75.6	75.6
horse	86.6	87.5	89.6	89.8	89.8	63.5	64.5	67.3	68.0	67.4
motorbike	84.0	85.1	84.9	85.7	85.7	61.7	61.8	62.9	63.9	62.4
person	70.6	73.4	73.8	72.2	77.5	22.0	23.5	23.2	23.8	26.6
potted plant	73.1	74.9	75.2	75.3	73.9	64.3	67.1	69.8	69.8	69.7
sheep	80.0	79.9	81.0	80.7	84.0	84.6	86.3	87.5	88.4	87.3
sofa	93.7	94.8	95.8	96.0	96.4	59.0	60.5	60.8	62.1	66.2
train	64.3	66.3	66.4	67.7	68.9	60.8	61.4	62.5	63.7	68.7
tv	79.3	77.9	77.6	77.9	80.2	69.7	69.2	70.0	70.1	71.5
mean	78.9	80.1	81.3	81.8	82.4	61.7	62.5	63.7	64.4	66.2

Table 11. **Per-class IoU of the main pipelines** (IoU $\times 100$), each at its full-VOC threshold (1 pass 0.425, 2 passes 0.450, FTF 0.450, MID 0.450, A3 0.425). 1 pass, 2 passes, FTF (two passes + DenseCRF) and MID (FTF with SA calls 10–20) are cells of the SG22 run; A3 is the SG24-A working default (owner override; not promoted by its selector). COCO-Cap uses the VOC threshold; nothing is chosen on it.

12.3. Early controlled phases: CA seed, SA module, propagation, tokens

The first phases chose the CA seed (A1–A3), the SA module (A4–A7), the propagation and refinement (A8, A9, P3-C) and the few-shot token pipeline (A11, T-C, A12). Their screens hold up to 1,279 cells each, so they are summarized by structured subsets: every single module or head, every leave-one-out set, the best subsets, and the full grids as figures. The tables and figures below are generated by the evidence package (`scripts/narmyn/attention/build_evidence.py`).

Setting	Value used so far	Chosen in	Images used to choose	Evidence	Controlled status
CA layers	12 of 16 (drop 1, 3, 4, 8)	B4 exhaustive search	99 dev images	ledger B4; Fig. a1_exhaustive	A1 CA-L (CV): drop modules 10, 11; A3 CA-C: subsumed by CA-R (frozen: 11 modules at 8, 16, 64)
CA heads	fixed top-4 per layer	B6 stage-1 suite	99 dev images	ledger B6	A1 CA-H (CV): drop head 1; A3 CA-C: adds nothing on top of CA-R (tie), so all 8 heads frozen
CA attribution	OVAM: unconditional query, unpadded softmax, first piece	inherited from OVAM	—	Table a1_ca_a	A1 CA-A (CV): reference (conditional, padded, all pieces) kept; OVAM's cell is −0.0156 CA-only
CA calls	all 31 (0–30)	B1/B8 windows; B14	99 dev / 490	ledger B1, B8, B14; Fig. a2_ca_t_calls	A2 CA-T (CV): keep all calls 0–30
CA resolutions	all four (8, 16, 32, 64)	B1/B8 resolution screens	99 dev / 490	Table a2_ca_r; Fig. a2_ca_r_subsets	A2 CA-R (CV): keep 8+16+64; A3 CA-C: frozen (4/5 votes, OOF +0.026)
CA normalization, threshold	percentile 1–99, $\geq$ 0.45	B5	99 dev images	ledger B5	protocol fixes min–max, 0.45 for the CA phase
SA module	U3.2 (up_blocks.3.attentions.2 attn1, 64×64)	B8 SA screens	490 (in-sample)	ledger B8	A4 SA-L (CV): only module 15 (U3.2); A6 SA-C: frozen (5/5, OOF +0.033)
SA passes	4 (revised: 1)	B10; revised from B11/ B13	490; 1-pass informed by COCO-Cap	Fig. b_joint_passes_time	A8 P1 (CV): two passes promoted (OOF +0.012); P3-C: frozen with DenseCRF (FTF)
SA heads	{0,1,2,4,5,6}	B10	490 (in-sample)	Fig. b_joint_heads_threshold	A4 SA-H (CV): heads {1, 4, 6}; A6 SA-C: does not survive, all 8 heads frozen
SA calls	15–30	B10	490 (in-sample)	Fig. b_joint_passes_time	A5 SA-T (CV): call 30 only; A6 SA-C: does not survive, all calls 0–30 frozen
Mask threshold	0.45	B5, B9, B10	99 dev / 490	Fig. b_joint_heads_threshold	CA/SA phases: fixed at 0.45; phase 3: chosen in the folds (0.30-0.70), REF picks 0.425-0.45
Cleanup	remove components ¡ 1%	B10	490 (in-sample)	Table b_component_ablations	deferred phase
DenseCRF	not in the final method	B17 (nothing selected)	—	Table b_token_crf	A9 R2 (CV): promoted (OOF +0.017, 20/20 classes); P3-C: frozen with two passes (FTF, OOF 0.813)
Optimized token	not in the final method	B17 (nothing selected; tokens trained by OVAM on separate images)	—	Table b_token_crf	A11 B1 (CV): released token promoted, OOF +0.063 (few-shot track); T-C: token + one pass + DenseCRF frozen (OOF 0.864); A12: retrained tokens do not replace it

Table 12. Where each setting of the method came from

Screen	Package	Reference	Vote winner (votes)	OOF mIoU reference	OOF mIoU choice	OOF gain	Paired Δ image IoU [95% CI]	Δ mIoU @0.35 / 0.45 / 0.55	Outcome
CA-T	A (Codex)	time/all_00_30	time/all_00_30 (5/5)	0.6661	0.6661	0.0000	+0.0000 [+0.0000, +0.0000]	+0.0000 / +0.0000 / +0.0000	reference won the fold vote
CA-R	A (Codex)	resolution/r8_r16_r32_r64	resolution/r8_r16_r64 (5/5)	0.6661	0.6957	0.0296	+0.0211 [+0.0139, +0.0286]	+0.0009 / +0.0296 / +0.0442	fold vote and OOF guard passed
CA-L	B (Claude)	L/all_16	L/leave2_10_11 (5/5)	0.6661	0.6923	0.0262	+0.0209 [+0.0155, +0.0265]	-0.0021 / +0.0262 / +0.0501	fold vote and OOF guard passed
CA-H	B (Claude)	H/0_1_2_3_4_5_6_7	H/0_2_3_4_5_6_7 (4/5)	0.6661	0.6741	0.0080	+0.0089 [+0.0047, +0.0131]	-0.0002 / +0.0104 / +0.0193	fold vote and OOF guard passed
CA-A	B (Claude)	A/cond_padded_all	A/cond_padded_all (5/5)	0.6661	0.6661	0.0000	+0.0000 [+0.0000, +0.0000]	+0.0000 / +0.0000 / +0.0000	reference won the fold vote
CA-C	C (Claude)	C/reference	C/R (4/5)	0.6653	0.6914	0.0260	+0.0217 [+0.0145, +0.0293]	+0.0010 / +0.0303 / +0.0447	fold vote and OOF guard passed

Table 13. Controlled CA screens (VOC-Sim, 5 folds, selection at threshold 0.45). The single-axis screens promote candidates; only the combined confirmation (CA-C) freezes the CA.

Configuration	mIoU@0.45	Δ vs reference	mIoU@0.35	mIoU@0.55	Mean AP	Fold votes
reference (16 modules, 8 heads)	0.6653	0.0000	0.7057	0.5430	0.9218	0/5
CA-R: without 32×32 (11 modules)	0.6956	0.0303	0.7067	0.5877	0.9131	4/5
CA-L: without modules 10, 11	0.6921	0.0267	0.7035	0.5938	0.9201	1/5
CA-H: without head 1	0.6761	0.0108	0.7056	0.5623	0.9205	0/5
CA-R + CA-H	0.6960	0.0307	0.7058	0.5879	0.9127	0/5
CA-L + CA-H	0.6924	0.0270	0.7020	0.5948	0.9190	0/5

Table 14. CA-C: the combined confirmation (one run, in-sample values; fold votes from the 5-fold selection)

Combined candidate	mIoU@0.45	Δ vs combined	Reading
combined CA-R + CA-H	0.6960	0.0000	—
without CA-R (= CA-L + CA-H)	0.6924	-0.0036	CA-R contributes
without CA-H (= CA-R alone)	0.6956	-0.0004	within the 0.002 tie: CA-H is not kept
without CA-L	0.6960	0.0000	CA-L is contained in CA-R

Table 15. CA-C leave-one-out of the combined candidate (in-sample)

Held-out fold	Chosen on the other four	Train mIoU chosen	Train mIoU reference	Held-out mIoU chosen	Held-out mIoU reference
0	CA-R: without 32×32 (11 modules)	0.6967	0.6644	0.6905	0.6708
1	CA-R: without 32×32 (11 modules)	0.6912	0.6610	0.7116	0.6827
2	CA-R: without 32×32 (11 modules)	0.6957	0.6641	0.6973	0.6714
3	CA-R: without 32×32 (11 modules)	0.6975	0.6661	0.6868	0.6676
4	CA-L: without modules 10, 11	0.6977	0.6715	0.6702	0.6429

Table 16. CA-C: the choice made on each set of four folds and its held-out score

Configuration	mIoU@0.45	Δ vs reference	mIoU@0.35	mIoU@0.55	Mean AP	SD across folds
all 00 30	0.6661	0.0000	0.7059	0.5434	0.9218	0.0133
window 00 04	0.5622	-0.1039	0.5373	0.5393	0.8041	0.0168
window 05 09	0.6469	-0.0192	0.6742	0.5465	0.8976	0.0131
window 10 14	0.6224	-0.0437	0.6943	0.4670	0.9086	0.0122
window 15 20	0.5796	-0.0865	0.6887	0.4031	0.9048	0.0096
window 21 25	0.5224	-0.1437	0.6623	0.3409	0.8979	0.0122
window 26 30	0.4524	-0.2137	0.6159	0.2789	0.8955	0.0081
trailing 05 30	0.6320	-0.0341	0.7077	0.4803	0.9206	0.0106
trailing 10 30	0.5893	-0.0768	0.6941	0.4209	0.9132	0.0097
trailing 15 30	0.5491	-0.1170	0.6767	0.3730	0.9064	0.0114
trailing 21 30	0.4998	-0.1663	0.6509	0.3215	0.8997	0.0108
trailing 26 30	0.4524	-0.2137	0.6159	0.2789	0.8955	0.0081
cumulative 00 04	0.5622	-0.1039	0.5373	0.5393	0.8041	0.0168
cumulative 00 09	0.6387	-0.0274	0.6137	0.5860	0.8755	0.0153
cumulative 00 14	0.6668	0.0007	0.6617	0.5899	0.9034	0.0134
cumulative 00 20	0.6752	0.0091	0.6886	0.5763	0.9157	0.0136
cumulative 00 25	0.6727	0.0066	0.7001	0.5601	0.9198	0.0138

Table 17. CA-T: call windows (in-sample; all 79 configurations, including every single call and leave-one-call-out, in controlled_ca_package_a.csv)

Configuration	mIoU@0.45	Δ vs reference	mIoU@0.35	mIoU@0.55	Mean AP	SD across folds
8+16+64	0.6957	0.0296	0.7068	0.5876	0.9131	0.0092
8+16	0.6933	0.0272	0.7148	0.5942	0.9171	0.0094
8+16+32+64	0.6661	0.0000	0.7059	0.5434	0.9218	0.0133
8+16+32	0.6562	-0.0099	0.7082	0.5323	0.9240	0.0112
16	0.6489	-0.0172	0.7099	0.5086	0.9059	0.0142
16+64	0.6449	-0.0212	0.6906	0.4925	0.8902	0.0130
16+32+64	0.6138	-0.0523	0.6849	0.4590	0.9031	0.0125
16+32	0.6004	-0.0657	0.6877	0.4461	0.9092	0.0127
8+32+64	0.5632	-0.1029	0.6403	0.4193	0.8929	0.0093
8	0.5626	-0.1035	0.6021	0.4787	0.8462	0.0084
8+64	0.5435	-0.1226	0.6001	0.4002	0.8307	0.0037
8+32	0.5328	-0.1333	0.6212	0.4106	0.8969	0.0089
32+64	0.3693	-0.2968	0.5075	0.2045	0.8043	0.0109
32	0.2995	-0.3666	0.4422	0.1740	0.8221	0.0115
64	0.1734	-0.4927	0.3285	0.0631	0.5604	0.0084

Table 18. CA-R: all 15 resolution subsets (in-sample; reference 8+16+32+64)

Table 19. CA-L: structured groups, single modules and leave-one-out (in-sample mIoU; all 159 configurations in controlled_ca_package_b.csv)

Configuration	mIoU@0.45	Δ vs all 16	mIoU@0.35	mIoU@0.55	Mean AP	SD across folds
all_16	0.6661	0.0000	0.7058	0.5435	0.9218	0.0134
non_middle_15	0.6139	-0.0521	0.6848	0.4590	0.9031	0.0125
down_only	0.6125	-0.0535	0.6589	0.4798	0.8603	0.0212
middle_only	0.5626	-0.1034	0.6021	0.4788	0.8462	0.0084
up_only	0.4186	-0.2475	0.5741	0.2580	0.8709	0.0055
down_middle	0.6819	0.0158	0.6842	0.5953	0.9025	0.0142
middle_up	0.5688	-0.0972	0.6720	0.4147	0.9066	0.0061
only 00 D0.0 (64^2)	0.0487	-0.6173	0.1040	0.0208	0.5032	0.0031
only 01 D0.1 (64^2)	0.1171	-0.5489	0.2396	0.0354	0.3713	0.0041
only 02 D1.0 (32^2)	0.2999	-0.3662	0.4240	0.1899	0.7657	0.0155
only 03 D1.1 (32^2)	0.1168	-0.5493	0.2255	0.0535	0.5648	0.0074
only 04 D2.0 (16^2)	0.5066	-0.1594	0.5931	0.3739	0.8481	0.0105
only 05 D2.1 (16^2)	0.4227	-0.2433	0.5552	0.2784	0.8224	0.0175
only 06 M (8^2)	0.5626	-0.1034	0.6021	0.4788	0.8462	0.0084
only 07 U1.0 (16^2)	0.4265	-0.2396	0.5272	0.3059	0.8349	0.0095
only 08 U1.1 (16^2)	0.2729	-0.3931	0.3845	0.1758	0.7389	0.0152
only 09 U1.2 (16^2)	0.1986	-0.4675	0.2944	0.1248	0.7807	0.0097
only 10 U2.0 (32^2)	0.0188	-0.6472	0.0310	0.0117	0.6985	0.0016
only 11 U2.1 (32^2)	0.0885	-0.5776	0.1556	0.0481	0.8507	0.0084
only 12 U2.2 (32^2)	0.1097	-0.5564	0.1974	0.0581	0.8283	0.0056
only 13 U3.0 (64^2)	0.0773	-0.5887	0.1870	0.0260	0.5357	0.0058
only 14 U3.1 (64^2)	0.0236	-0.6424	0.0613	0.0088	0.5582	0.0024
only 15 U3.2 (64^2)	0.0238	-0.6422	0.0659	0.0087	0.5604	0.0022
without 00 D0.0	0.6649	-0.0012	0.7052	0.5425	0.9214	0.0128
without 01 D0.1	0.6608	-0.0053	0.7094	0.5322	0.9236	0.0132
without 02 D1.0	0.6485	-0.0176	0.7066	0.5137	0.9180	0.0102
without 03 D1.1	0.6655	-0.0006	0.7080	0.5420	0.9228	0.0134
without 04 D2.0	0.6442	-0.0219	0.6983	0.5082	0.9169	0.0120
without 05 D2.1	0.6582	-0.0078	0.7016	0.5278	0.9174	0.0119
without 06 M	0.6139	-0.0521	0.6848	0.4590	0.9031	0.0125
without 07 U1.0	0.6443	-0.0217	0.6921	0.5207	0.9141	0.0144
without 08 U1.1	0.6557	-0.0104	0.7004	0.5275	0.9220	0.0132
without 09 U1.2	0.6677	0.0016	0.7042	0.5460	0.9200	0.0137
without 10 U2.0	0.6788	0.0127	0.7060	0.5671	0.9218	0.0128
without 11 U2.1	0.6806	0.0145	0.7052	0.5691	0.9202	0.0150
without 12 U2.2	0.6750	0.0089	0.7059	0.5580	0.9200	0.0134
without 13 U3.0	0.6541	-0.0120	0.7020	0.5300	0.9210	0.0115
without 14 U3.1	0.6716	0.0055	0.7093	0.5509	0.9237	0.0137
without 15 U3.2	0.6710	0.0049	0.7075	0.5500	0.9223	0.0141

Configuration	mIoU@0.45	Δ vs all 16	mIoU@0.35	mIoU@0.55	Mean AP	SD across folds
without 10 U2.0 & 11 U2.1	0.6923	0.0262	0.7036	0.5937	0.9201	0.0132
without 10 U2.0 & 12 U2.2	0.6870	0.0209	0.7060	0.5832	0.9199	0.0124
without 11 U2.1 & 12 U2.2	0.6860	0.0200	0.7016	0.5825	0.9179	0.0147
without 11 U2.1 & 14 U3.1	0.6852	0.0191	0.7076	0.5771	0.9220	0.0156
without 11 U2.1 & 15 U3.2	0.6851	0.0190	0.7062	0.5756	0.9207	0.0154
without 10 U2.0 & 14 U3.1	0.6831	0.0170	0.7088	0.5743	0.9236	0.0131
without 10 U2.0 & 15 U3.2	0.6828	0.0168	0.7070	0.5731	0.9222	0.0134
without 09 U1.2 & 11 U2.1	0.6826	0.0165	0.7026	0.5725	0.9183	0.0151
without 09 U1.2 & 10 U2.0	0.6806	0.0145	0.7037	0.5697	0.9200	0.0131
without 03 D1.1 & 11 U2.1	0.6803	0.0142	0.7078	0.5678	0.9213	0.0148

Table 20. CA-L: the ten best leave-two-out configurations (in-sample)

Head	Alone	All but this head	Δ vs all 8	Folds retaining it
0	0.2557	0.6484	-0.0177	5/5
1	0.0975	0.6765	0.0104	1/5
2	0.3638	0.6471	-0.0190	5/5
3	0.3556	0.6549	-0.0112	5/5
4	0.4344	0.6448	-0.0213	5/5
5	0.4179	0.6516	-0.0144	5/5
6	0.3208	0.6456	-0.0205	5/5
7	0.3165	0.6581	-0.0080	5/5

Table 21. CA-H: each head alone and leave-one-out (in-sample mIoU at 0.45), and fold retention

Round	Removed	Heads left	mIoU@0.45
0	—	0 1 2 3 4 5 6 7	0.6661
1	1	0 2 3 4 5 6 7	0.6765
2	5	0 2 3 4 6 7	0.6718
3	7	0 2 3 4 6	0.6609
4	0	2 3 4 6	0.6388
5	2	3 4 6	0.6005
6	6	3 4	0.5444
7	3	4	0.4344

Table 22. CA-H: greedy backward elimination (in-sample)

Head subset	Heads	mIoU@0.45	Δ vs all 8	mIoU@0.35	mIoU@0.55	Mean AP
0 2 3 4 5 6 7	7	0.6765	0.0104	0.7055	0.5629	0.9206
0 2 3 4 6 7	6	0.6718	0.0058	0.7110	0.5406	0.9121
0 2 3 4 5 6	6	0.6676	0.0015	0.7044	0.5514	0.9138
0 1 2 3 4 5 6 7	8	0.6661	-0.0000	0.7057	0.5435	0.9218
0 2 3 4 5 7	6	0.6629	-0.0031	0.7038	0.5377	0.9173
0 3 4 5 6 7	6	0.6626	-0.0035	0.6928	0.5373	0.9084
2 3 4 5 6 7	6	0.6625	-0.0035	0.7035	0.5405	0.9236
0 2 4 5 6 7	6	0.6615	-0.0046	0.7011	0.5424	0.9140
0 2 3 4 6	5	0.6609	-0.0052	0.7095	0.5248	0.9005
0 1 2 3 4 5 6	7	0.6581	-0.0080	0.7044	0.5371	0.9167
0 2 3 5 6 7	6	0.6571	-0.0090	0.6935	0.5325	0.9092
2 3 4 5 6	5	0.6564	-0.0097	0.7032	0.5324	0.9173
0 1 2 4 5 6 7	7	0.6549	-0.0112	0.7021	0.5274	0.9155
0 2 3 4 5	5	0.6541	-0.0120	0.7022	0.5275	0.9137
0 1 2 3 4 6 7	7	0.6516	-0.0144	0.7059	0.5060	0.9142

Table 23. CA-H: the 15 best of 255 head subsets (in-sample; all 255 in controlled_ca_package_b.csv)

Query branch	Softmax	Class pieces	mIoU@0.45	Δ vs reference	mIoU@0.35	mIoU@0.55	Mean AP
conditional	unpadded	all	0.6678	0.0018	0.7054	0.5480	0.9245
conditional	padded	all	0.6661	0.0000	0.7057	0.5435	0.9218
unconditional	padded	all	0.6612	-0.0048	0.6992	0.5489	0.9222
unconditional	unpadded	all	0.6610	-0.0050	0.6986	0.5506	0.9241
conditional	unpadded	last	0.6597	-0.0063	0.6998	0.5414	0.9246
conditional	padded	last	0.6579	-0.0082	0.7000	0.5365	0.9219
unconditional	padded	last	0.6541	-0.0119	0.6938	0.5436	0.9223
unconditional	unpadded	last	0.6539	-0.0122	0.6935	0.5454	0.9243
unconditional	unpadded	first	0.6505	-0.0156	0.6970	0.5252	0.9127
conditional	unpadded	first	0.6479	-0.0182	0.6962	0.5172	0.9107
unconditional	padded	first	0.6474	-0.0187	0.6967	0.5199	0.9106
conditional	padded	first	0.6451	-0.0210	0.6953	0.5112	0.9080

Table 24. CA-A: the 12 attribution conventions (in-sample; reference = conditional, padded, all)

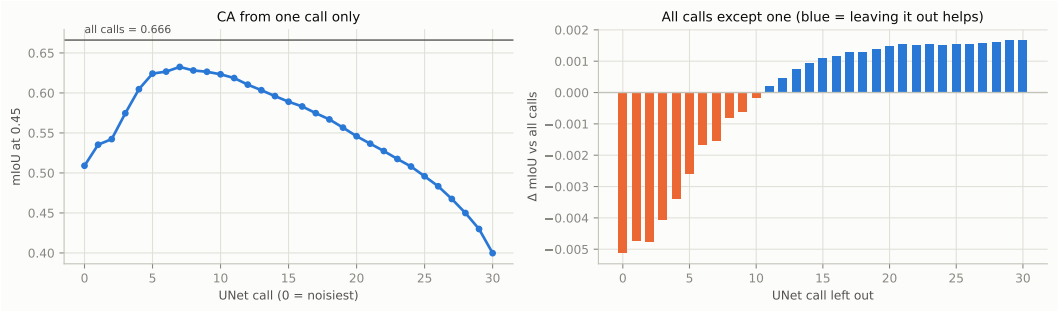


Figure 15. CA-T: CA from a single UNet call, and all calls except one.

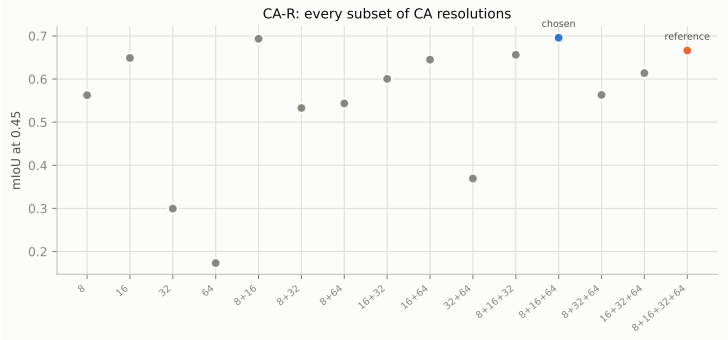


Figure 16. CA-R: every subset of CA resolutions.

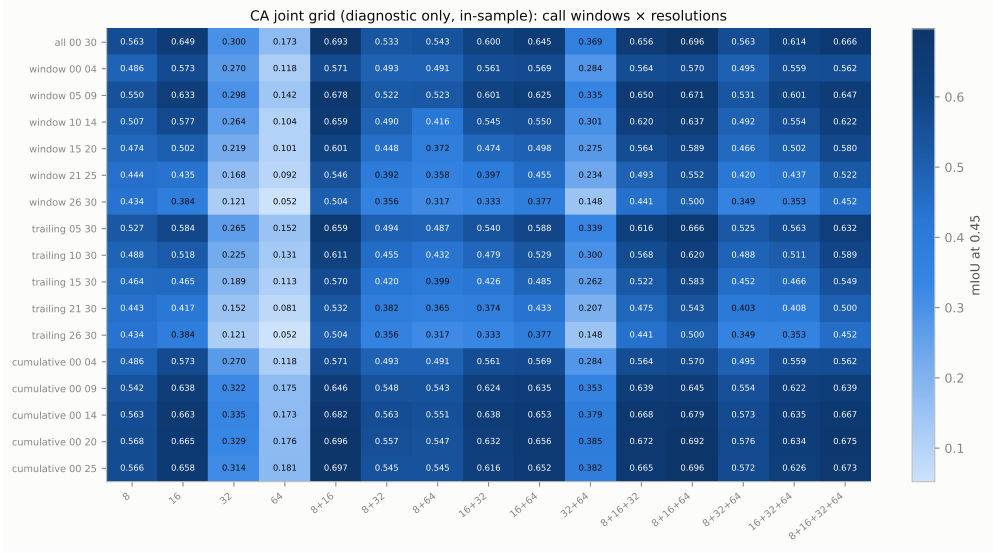


Figure 17. Joint grid of CA call windows and resolutions (diagnostic only).

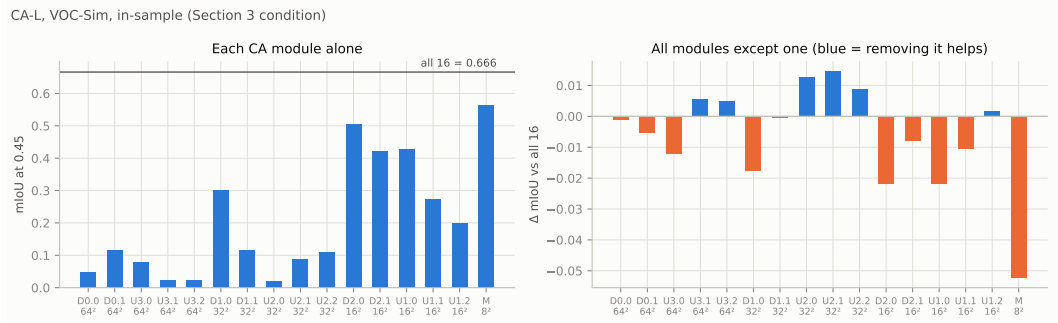


Figure 18. CA-L: each CA module alone (left) and all modules except one (right).

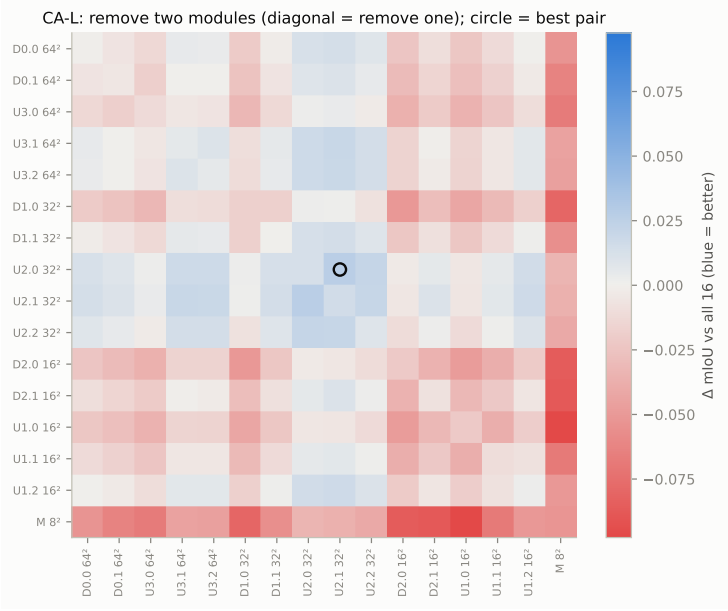


Figure 19. CA-L: all 120 leave-two-out configurations; the diagonal is leave-one-out.

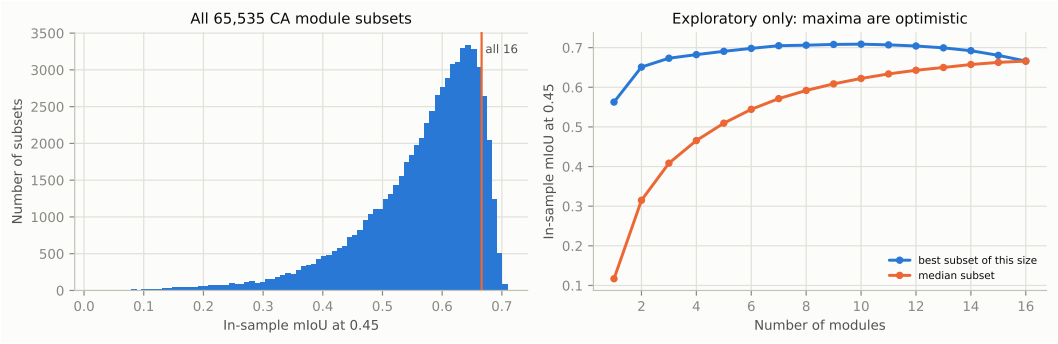


Figure 20. All 65,535 CA module subsets (exploratory; the maximum is not eligible for selection).

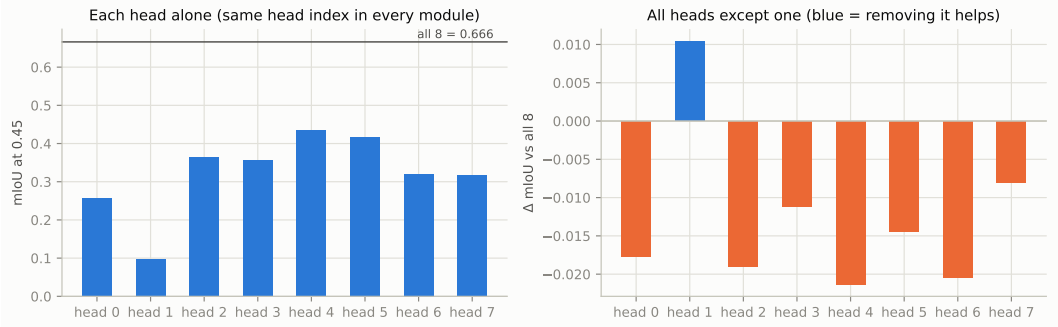


Figure 21. CA-H: each head alone and all heads except one (the same head index in every module).

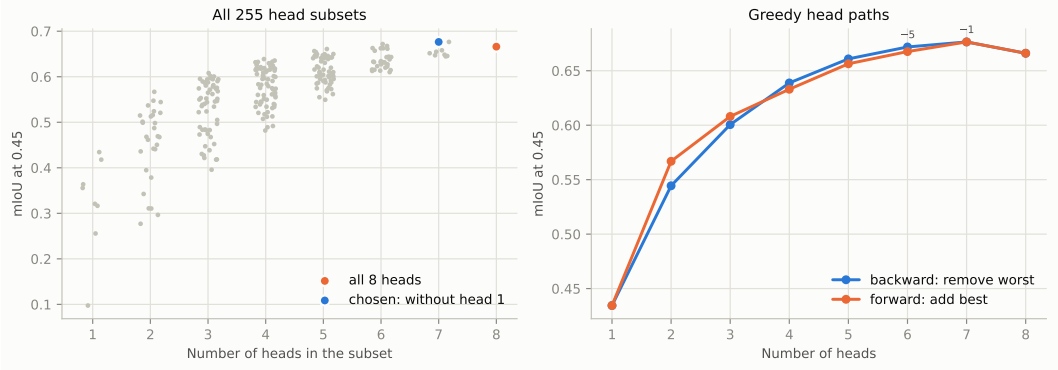


Figure 22. CA-H: all 255 head subsets by size, and the greedy removal and addition paths.

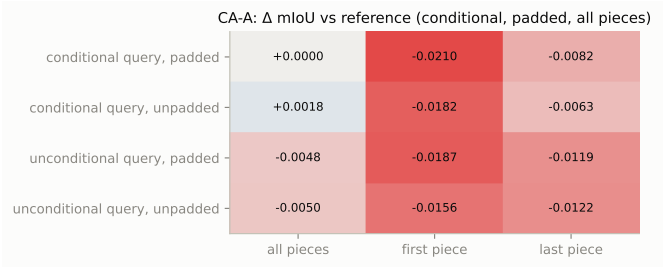


Figure 23. CA-A: change in mIoU of each attribution convention against the reference.

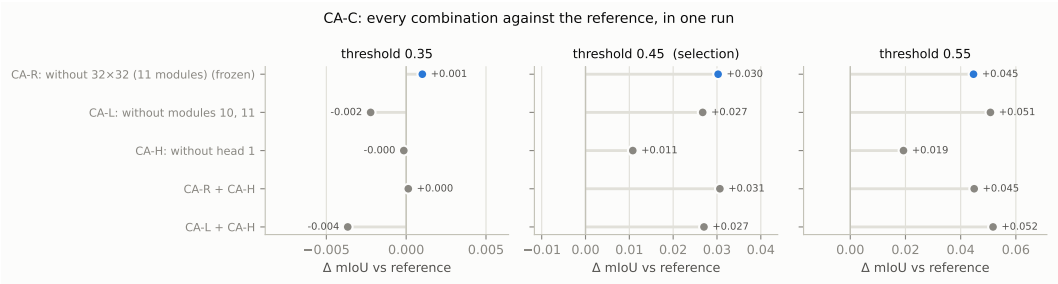


Figure 24. CA-C: change in mIoU of every combination against the reference at thresholds 0.35, 0.45 and 0.55 (blue = frozen).

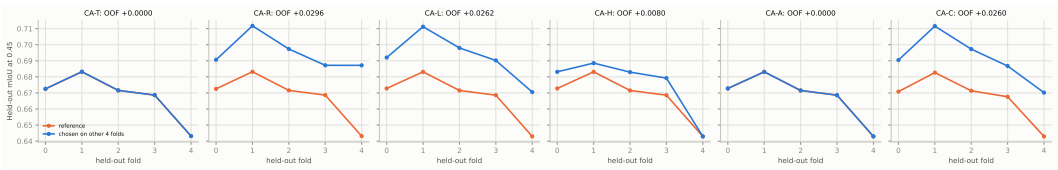


Figure 25. Controlled CA screens: held-out mIoU of the setting chosen on the other four folds, against the reference, per fold.

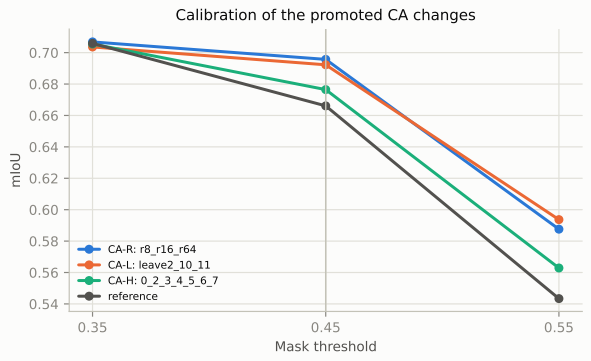


Figure 26. Calibration of the promoted CA changes at thresholds 0.35, 0.45 and 0.55.

CA seed (A1–A3).

Configuration	mIoU@0.45	Δ vs reference	mIoU@0.35	mIoU@0.55	Mean AP	Fold votes
reference (16 SA modules, 31 calls, 8 heads)	0.7556	0.0000	0.7213	0.7466	0.9536	0/5
T: call 30 only	0.7710	0.0153	0.7465	0.7457	0.9551	0/5
R: resolutions 8+64	0.7676	0.0119	0.7414	0.7399	0.9542	0/5
L: module 15 only (U3.2)	0.7887	0.0330	0.7691	0.7532	0.9560	5/5
H: heads 1, 4, 6	0.7613	0.0057	0.7295	0.7470	0.9552	0/5
T + R	0.7749	0.0193	0.7659	0.7244	0.9517	0/5
T + L	0.7820	0.0264	0.7704	0.7260	0.9474	0/5
T + H	0.7732	0.0175	0.7534	0.7412	0.9563	0/5
R + H	0.7656	0.0099	0.7389	0.7343	0.9534	0/5
L + H	0.7871	0.0314	0.7637	0.7554	0.9587	0/5
T + R + H	0.7692	0.0136	0.7669	0.7098	0.9512	0/5
T + L + H	0.7825	0.0268	0.7658	0.7330	0.9514	0/5

Table 25. SA-C: the combined confirmation (one run, in-sample values; fold votes from the 5-fold selection)

Component	With	Without	Δ mIoU@0.45
T within T+L+H	T + L + H	L + H	-0.0046
L within T+L+H	T + L + H	T + H	0.0093
H within T+L+H	T + L + H	T + L	0.0005
R within T+R+H	T + R + H	T + H	-0.0040

Table 26. SA-C: removing one component from the full combination (in-sample)

Held-out fold	Chosen on the other four	Train mIoU chosen	Train mIoU reference	Held-out mIoU chosen	Held-out mIoU reference
0	L: module 15 only (U3.2)	0.7892	0.7580	0.7892	0.7465
1	L: module 15 only (U3.2)	0.7825	0.7490	0.8116	0.7780
2	L: module 15 only (U3.2)	0.7899	0.7571	0.7869	0.7518
3	L: module 15 only (U3.2)	0.7908	0.7578	0.7825	0.7460
4	L: module 15 only (U3.2)	0.7910	0.7559	0.7757	0.7484

Table 27. SA-C: the choice made on each set of four folds and its held-out score

Class	Reference	Module 15 only	Δ
chair	0.5289	0.6503	0.1214
aeroplane	0.7309	0.8450	0.1141
bicycle	0.6788	0.7471	0.0683
boat	0.7638	0.8318	0.0680
bird	0.7859	0.8531	0.0673
bottle	0.8126	0.8665	0.0539
horse	0.8300	0.8696	0.0396
motorbike	0.8157	0.8527	0.0370
dog	0.8692	0.8960	0.0268
car	0.8152	0.8420	0.0268
train	0.6470	0.6697	0.0227
sofa	0.9230	0.9447	0.0217
dining-table	0.4511	0.4677	0.0166
cow	0.8055	0.8184	0.0129
cat	0.7688	0.7816	0.0127
bus	0.9064	0.9184	0.0120
sheep	0.7564	0.7660	0.0096
potted-plant	0.7064	0.7091	0.0027
tv	0.7953	0.7768	-0.0184
person	0.7218	0.6664	-0.0554

Table 28. SA-C: per-class IoU of the selected SA configuration against the inclusive SA reference (threshold 0.45)

Configuration	mIoU@0.45	Δ vs reference	mIoU@0.35	mIoU@0.55	Mean AP	SD across folds
all 00 30	0.7556	0.0000	0.7213	0.7466	0.9536	0.0121
window 00 04	0.6105	-0.1452	0.5513	0.6432	0.8907	0.0175
window 05 09	0.7290	-0.0266	0.6893	0.7286	0.9424	0.0130
window 10 14	0.7569	0.0012	0.7241	0.7470	0.9535	0.0118
window 15 20	0.7636	0.0079	0.7354	0.7477	0.9548	0.0110
window 21 25	0.7661	0.0105	0.7412	0.7453	0.9544	0.0100
window 26 30	0.7687	0.0131	0.7447	0.7447	0.9545	0.0098
trailing 05 30	0.7617	0.0060	0.7318	0.7473	0.9549	0.0114
trailing 10 30	0.7650	0.0093	0.7376	0.7474	0.9552	0.0108
trailing 15 30	0.7663	0.0107	0.7407	0.7465	0.9549	0.0103
trailing 21 30	0.7675	0.0119	0.7432	0.7452	0.9545	0.0098
trailing 26 30	0.7687	0.0131	0.7447	0.7447	0.9545	0.0098
cumulative 00 04	0.6105	-0.1452	0.5513	0.6432	0.8907	0.0175
cumulative 00 09	0.6956	-0.0600	0.6390	0.7114	0.9318	0.0151
cumulative 00 14	0.7276	-0.0280	0.6813	0.7337	0.9454	0.0136
cumulative 00 20	0.7441	-0.0116	0.7042	0.7429	0.9508	0.0132
cumulative 00 25	0.7510	-0.0046	0.7143	0.7454	0.9526	0.0126

Table 29. SA-T: call windows (in-sample; all 79 configurations, including every single call and leave-one-call-out, in controlled_sa_package_a.csv)

Configuration	mIoU@0.45	Δ vs reference	mIoU@0.35	mIoU@0.55	Mean AP	SD across folds
64	0.7678	0.0121	0.7474	0.7309	0.9539	0.0124
8+64	0.7676	0.0119	0.7414	0.7399	0.9542	0.0128
32+64	0.7627	0.0070	0.7388	0.7369	0.9550	0.0125
8+32+64	0.7619	0.0063	0.7356	0.7399	0.9549	0.0125
16+64	0.7578	0.0021	0.7199	0.7507	0.9531	0.0120
16+32+64	0.7568	0.0012	0.7237	0.7457	0.9538	0.0120
8+16+32+64	0.7556	0.0000	0.7213	0.7466	0.9536	0.0121
8+16+64	0.7554	-0.0002	0.7158	0.7516	0.9529	0.0123
32	0.7532	-0.0024	0.7275	0.7353	0.9534	0.0119
8+32	0.7510	-0.0047	0.7214	0.7387	0.9528	0.0122
16+32	0.7434	-0.0123	0.7066	0.7426	0.9503	0.0113
8+16+32	0.7413	-0.0144	0.7033	0.7430	0.9499	0.0116
16	0.7187	-0.0370	0.6749	0.7352	0.9402	0.0108
8+16	0.7156	-0.0400	0.6706	0.7343	0.9399	0.0112
8	0.6764	-0.0793	0.6237	0.6992	0.9030	0.0137

Table 30. SA-R: all 15 resolution subsets (in-sample; reference 8+16+32+64)

Screen	Reference	Vote winner (votes)	OOF mIoU reference	OOF mIoU choice	OOF gain	Paired Δ image IoU [95% CI]	Δ mIoU @0.35 / 0.45 / 0.55	Outcome
SA-L	SL/all_16	SL/single_15 (5/5)	0.7553	0.7884	0.0331	+0.0417 [+0.0365, +0.0469]	+0.0477 / +0.0331 / +0.0068	fold vote and OOF guard passed
SA-H	SH/0_1_2_3_4_5_6_7	SH/1_4_6 (5/5)	0.7553	0.7610	0.0057	+0.0065 [+0.0056, +0.0074]	+0.0083 / +0.0057 / +0.0004	fold vote and OOF guard passed

Table 31. Controlled SA package B (frozen CA seed; VOC-Sim, 5 folds, selection at 0.45). SA-C combines these with SA-T/SA-R.

Table 32. SA-L: structured groups, single SA modules and leave-one-out (in-sample; all 159 in controlled_sa_package_b.csv)

Configuration	mIoU@0.45	Δ vs all 16	mIoU@0.35	mIoU@0.55	Mean AP	SD across folds
all_16	0.7553	0.0000	0.7212	0.7463	0.9535	0.0124
non_middle_15	0.7565	0.0012	0.7237	0.7454	0.9537	0.0122
down_only	0.7472	-0.0081	0.7038	0.7503	0.9507	0.0124
middle_only	0.6763	-0.0790	0.6236	0.6990	0.9029	0.0138
up_only	0.7556	0.0003	0.7301	0.7357	0.9525	0.0120
down_middle	0.7448	-0.0105	0.6990	0.7534	0.9506	0.0126
middle_up	0.7539	-0.0014	0.7261	0.7376	0.9523	0.0121
only 00 D0.0 (64^2)	0.7311	-0.0243	0.6994	0.6993	0.9292	0.0149
only 01 D0.1 (64^2)	0.7458	-0.0095	0.7121	0.7224	0.9384	0.0142
only 02 D1.0 (32^2)	0.7371	-0.0182	0.6891	0.7442	0.9485	0.0127
only 03 D1.1 (32^2)	0.7412	-0.0141	0.7007	0.7425	0.9500	0.0113
only 04 D2.0 (16^2)	0.7116	-0.0437	0.6703	0.7232	0.9400	0.0132
only 05 D2.1 (16^2)	0.7112	-0.0442	0.6671	0.7274	0.9403	0.0110
only 06 M (8^2)	0.6763	-0.0790	0.6236	0.6990	0.9029	0.0138
only 07 U1.0 (16^2)	0.7203	-0.0351	0.6686	0.7411	0.9395	0.0114
only 08 U1.1 (16^2)	0.7125	-0.0428	0.6714	0.7253	0.9280	0.0107
only 09 U1.2 (16^2)	0.7155	-0.0398	0.6718	0.7316	0.9368	0.0101
only 10 U2.0 (32^2)	0.7387	-0.0166	0.7303	0.7017	0.9494	0.0112
only 11 U2.1 (32^2)	0.7416	-0.0138	0.7325	0.7045	0.9473	0.0110
only 12 U2.2 (32^2)	0.7698	0.0145	0.7375	0.7532	0.9557	0.0120
only 13 U3.0 (64^2)	0.7397	-0.0156	0.7394	0.6857	0.9476	0.0127
only 14 U3.1 (64^2)	0.7546	-0.0007	0.7440	0.7120	0.9526	0.0130
only 15 U3.2 (64^2)	0.7884	0.0331	0.7689	0.7531	0.9559	0.0123
without 00 D0.0	0.7536	-0.0018	0.7201	0.7448	0.9531	0.0121
without 01 D0.1	0.7533	-0.0020	0.7199	0.7442	0.9532	0.0124
without 02 D1.0	0.7556	0.0003	0.7224	0.7455	0.9534	0.0123
without 03 D1.1	0.7556	0.0003	0.7216	0.7460	0.9534	0.0124
without 04 D2.0	0.7570	0.0016	0.7236	0.7462	0.9537	0.0123
without 05 D2.1	0.7566	0.0012	0.7233	0.7459	0.9537	0.0123
without 06 M	0.7565	0.0012	0.7237	0.7454	0.9537	0.0122
without 07 U1.0	0.7559	0.0005	0.7233	0.7449	0.9537	0.0125
without 08 U1.1	0.7571	0.0018	0.7247	0.7453	0.9542	0.0126
without 09 U1.2	0.7565	0.0011	0.7233	0.7455	0.9537	0.0124
without 10 U2.0	0.7560	0.0006	0.7193	0.7494	0.9535	0.0125
without 11 U2.1	0.7556	0.0003	0.7193	0.7489	0.9537	0.0125
without 12 U2.2	0.7540	-0.0013	0.7195	0.7454	0.9531	0.0123
without 13 U3.0	0.7550	-0.0003	0.7177	0.7502	0.9537	0.0124
without 14 U3.1	0.7543	-0.0011	0.7181	0.7482	0.9532	0.0122
without 15 U3.2	0.7517	-0.0036	0.7171	0.7440	0.9526	0.0123

Configuration	mIoU@0.45	Δ vs SA reference	mIoU@0.35	mIoU@0.55	Mean AP
frozen CA seed alone (no SA)	0.6956	-0.0597	0.7067	0.5877	0.9131
SA reference: all 16 SA modules, 8 heads, y = A x	0.7553	0.0000	0.7212	0.7463	0.9535
operator check: y = Ax	0.4741	-0.2813	0.5914	0.3273	0.9179
SA-L choice: single_15	0.7884	0.0331	0.7689	0.7531	0.9559
SA-H choice: heads 1, 4, 6	0.7610	0.0057	0.7294	0.7467	0.9552

Table 33. SA package B: what one SA pass adds to the frozen CA seed, the transposed operator, and the two choices (in-sample)

Head	Alone	All but this head	Δ vs all 8	Folds retaining it
0	0.7541	0.7552	-0.0002	0/5
1	0.7577	0.7548	-0.0005	5/5
2	0.7512	0.7558	0.0005	0/5
3	0.7486	0.7561	0.0007	0/5
4	0.7625	0.7540	-0.0014	5/5
5	0.7460	0.7567	0.0014	0/5
6	0.7605	0.7544	-0.0010	5/5
7	0.7513	0.7555	0.0002	0/5

Table 34. SA-H: each SA head alone and leave-one-out (in-sample mIoU at 0.45), and fold retention

Head subset	Heads	mIoU@0.45	Δ vs all 8	mIoU@0.35	mIoU@0.55	Mean AP
4	1	0.7625	0.0072	0.7301	0.7473	0.9564
4 6	2	0.7623	0.0070	0.7288	0.7498	0.9550
1 4 6	3	0.7610	0.0057	0.7294	0.7467	0.9552
1 4	2	0.7606	0.0053	0.7302	0.7446	0.9561
6	1	0.7605	0.0052	0.7255	0.7499	0.9523
0 4 6	3	0.7604	0.0051	0.7238	0.7522	0.9544
0 1 4 6	4	0.7600	0.0047	0.7258	0.7494	0.9547
4 6 7	3	0.7597	0.0044	0.7278	0.7463	0.9544
0 1 4	3	0.7594	0.0040	0.7249	0.7490	0.9553
1 6	2	0.7594	0.0040	0.7281	0.7453	0.9539

Table 35. SA-H: the 10 best of 255 SA head subsets (in-sample; all 255 in controlled_sa_package_b.csv)

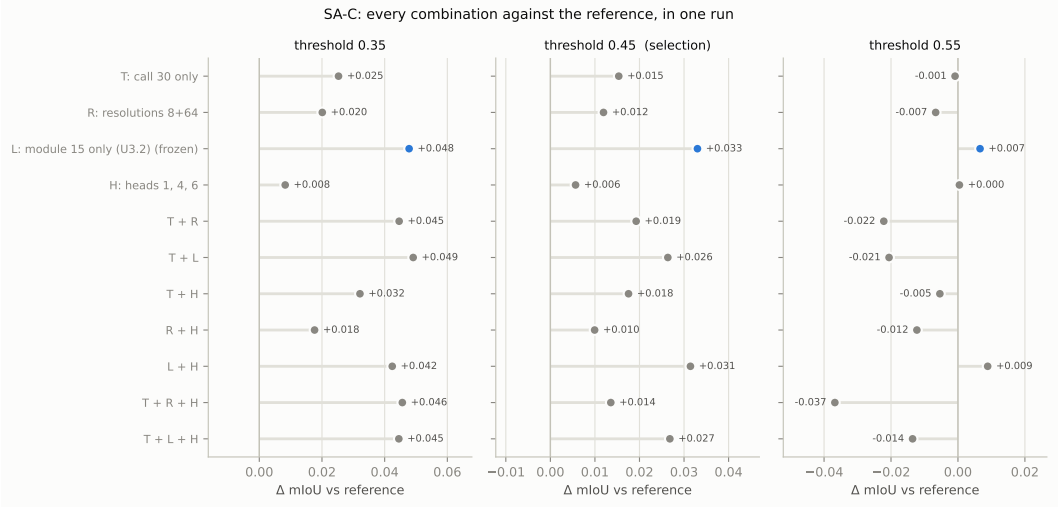


Figure 27. SA-C: change in mIoU of every combination against the SA reference at thresholds 0.35, 0.45 and 0.55 (blue = frozen).

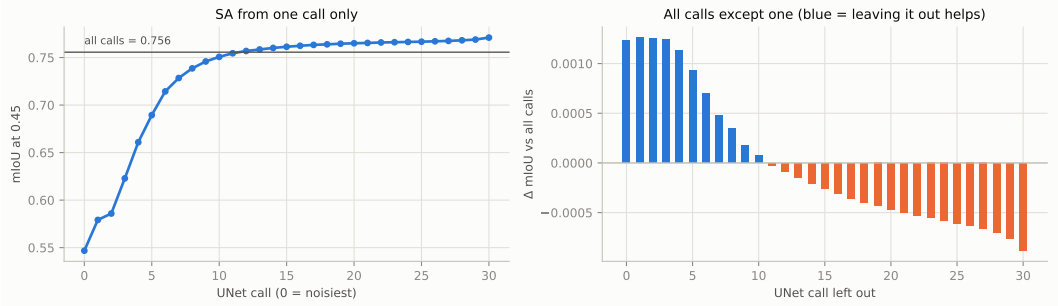


Figure 28. SA-T: SA from a single UNet call, and all calls except one.

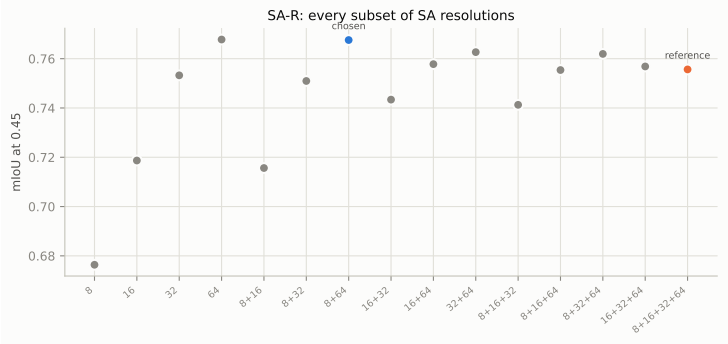


Figure 29. SA-R: every subset of SA resolutions.

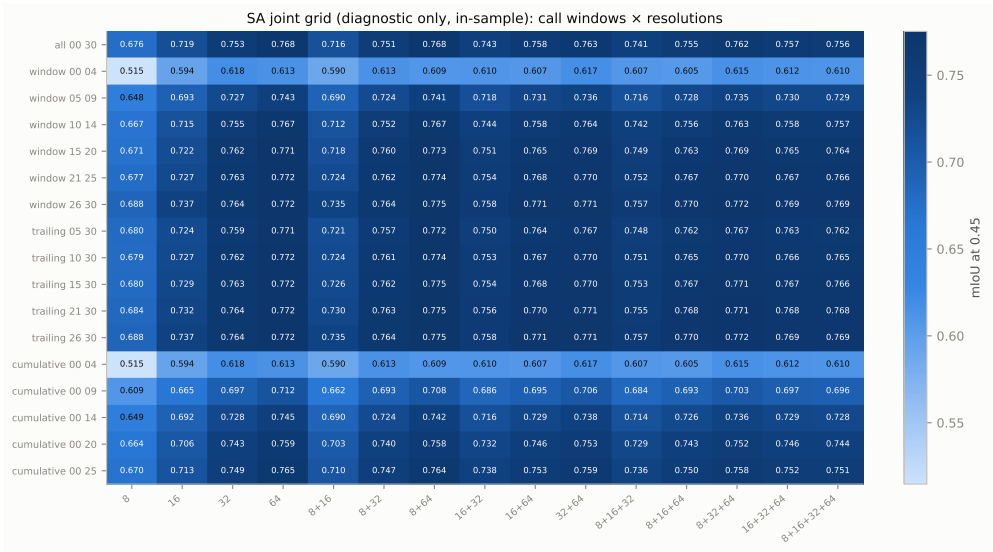


Figure 30. SA joint grid of call windows and resolutions (diagnostic only).

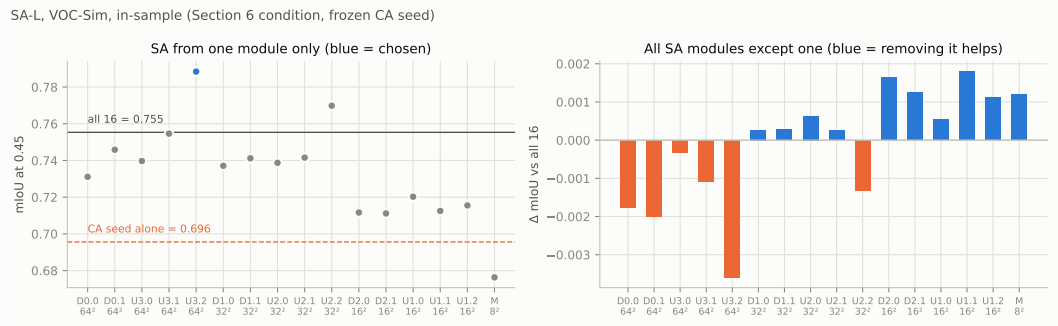


Figure 31. SA-L: SA from one module only, and all SA modules except one (frozen CA seed).

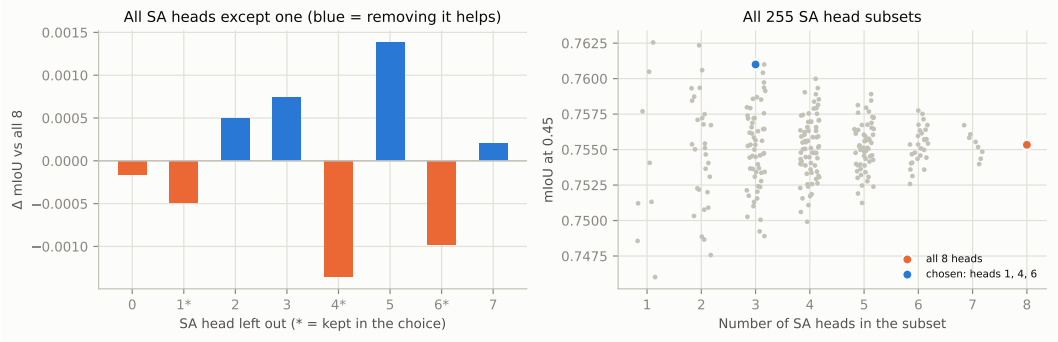


Figure 32. SA-H: leaving out one SA head, and all 255 SA head subsets by size.

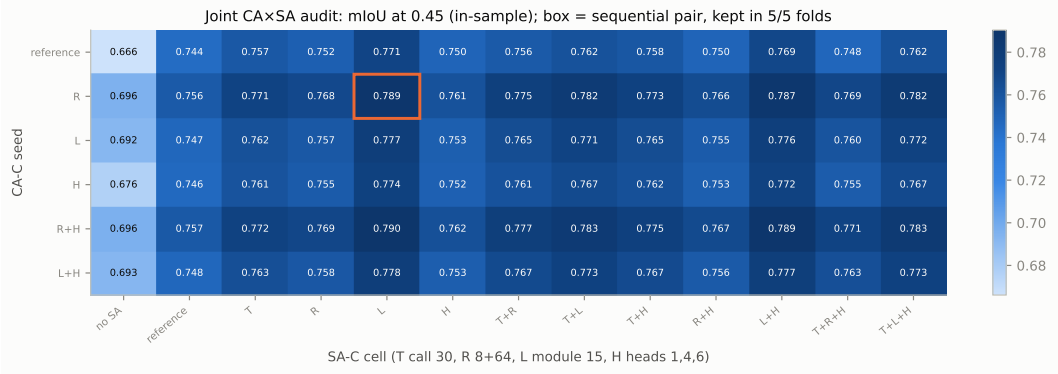


Figure 33. Joint CA×SA audit: mIoU of all 78 cells (in-sample); the box is the frozen pair.

SA module (A4–A7).

Screen	Fold choice (votes)	OOF mIoU REF	OOF mIoU choice	OOF gain	Paired Δ image IoU [95% CI]; better/worse	Δ mIoU @0.35 / 0.45 / 0.55	Candidates	Outcome
P1 observation-wise powers	P1/obs_n2 (4/5)	0.7875	0.7991	0.0116	+0.0140 [+0.0097, +0.0180]; 359/131	+0.0087 / +0.0121 / +0.0243	3	fold vote and OOF guard passed
W3 average-before powers	REF/one_pass (4/5)	0.7875	0.7875	-0.0000	—	—	4	reference won the fold vote
P2 observation-wise restart	REF/one_pass (4/5)	0.7875	0.7877	0.0002	—	—	12	reference won the fold vote
P2 average-before restart	REF/one_pass (5/5)	0.7875	0.7875	0.0000	—	—	12	reference won the fold vote
P6 sparse affinity	REF/one_pass (5/5)	0.7875	0.7875	0.0000	—	—	7	reference won the fold vote

Table 36. Phase 3 propagation package (Codex) (VOC-Sim, 5 folds; method parameter and threshold chosen on the training folds)

Screen	Fold choice (votes)	OOF mIoU REF	OOF mIoU choice	OOF gain	Paired Δ image IoU [95% CI]; better/worse	Δ mIoU @0.35 / 0.45 / 0.55	Candidates	Outcome
P4 residual mix	REF/one_pass (5/5)	0.7875	0.7875	0.0000	—	—	4	reference won the fold vote
F3/F4 gating	REF/one_pass (5/5)	0.7875	0.7875	0.0000	—	—	2	reference won the fold vote
N2-N4 calibration	REF/one_pass (5/5)	0.7875	0.7875	0.0000	—	—	3	reference won the fold vote
R2 DenseCRF	R2/crf (5/5)	0.7875	0.8049	0.0175	+0.0222 [+0.0185, +0.0260]; 361/129	+0.0215 / +0.0141 / +0.0063	1	fold vote and OOF guard passed

Table 37. Phase 3 fusion and calibration package (VOC-Sim, 5 folds; method parameter and threshold chosen on the training folds)

Screen	Fold choice (votes)	OOF mIoU REF	OOF mIoU choice	OOF gain	Paired Δ image IoU [95% CI]; better/worse	Δ mIoU @0.35 / 0.45 / 0.55	Candidates	Outcome
B1 released token, OVAM form (unconditional query)	TU/REF (5/5)	0.7873	0.8428	0.0555	+0.0470 [+0.0380, +0.0560]; 369/121	+0.0324 / +0.0518 / +0.0819	1	fold vote and OOF guard passed
B1 released token, conditional query	TC/REF (5/5)	0.7873	0.8507	0.0634	+0.0529 [+0.0440, +0.0621]; 375/115	+0.0414 / +0.0596 / +0.0852	1	fold vote and OOF guard passed

Table 38. Token track B1 (released token vs class word on REF) (VOC-Sim, 5 folds; method parameter and threshold chosen on the training folds)

Pipeline	Class word	Token, OVAM form	Token, conditional
seed	0.6956	0.7893	0.7962
REF	0.7884	0.8402	0.8480
P1	0.8006	0.8444	0.8519
R2	0.8024	0.8567	0.8650
P1+R2	0.8132	0.8589	0.8664

Table 39. Token track: class word vs OVAM's released token on every pipeline (mIoU at 0.45, in-sample; OOF values in protocol Section 50)

Token	Cell	VOC-Sim mIoU (T-C, 0.475)	Images
released (OVAM, 2-3 supports)	REL	0.8653	490
support 1	A1	0.8333	490
support 2	A2	0.8630	490
support 3 (19 classes)	A3	0.8451	468
supports 1+2	A12	0.8689	490
supports 1+3 (19 classes)	A13	0.8589	468
supports 2+3 (19 classes)	A23	0.8713	468
all supports	AALL	0.8684	490
fold-safe one-support (B2b)	B2B	0.8564	490

Table 40. A12: retrained tokens in the frozen T-C pipeline (linear-objective variant of OVAM's recipe)

Comparison	Δ mIoU	Paired Δ image IoU	95% CI	Classes improved
all supports vs released, VOC-Sim	0.0032	0.0018	[-0.0033, +0.0064]	14
fold-safe one-support vs released, VOC-Sim	-0.0089	-0.0035	[-0.0119, +0.0042]	5
all supports vs released, COCO-Cap	0.0075	0.0096	[+0.0039, +0.0154]	14

Table 41. A12: retrained against released tokens (paired bootstrap over images)

Attention objective (unconditional, no middle block)	Epoch-0 loss vs OVAM, median abs. diff. (57 images)	Token cosine to OVAM after 800 epochs (6 images)
token softmax (OVAM's library default)	1.8e-06	0.9961
linear (B2 recipe)	1.8e-02	0.8731

Table 42. A12: which attention objective reproduces OVAM's released optimization (protocol Section 69)

Passes	mIoU@0.45	Recall	Precision	FP per object pixel
seed	0.6957	0.8220	0.8434	0.1996
1 pass	0.7887	0.9123	0.8650	0.1827
2 passes	0.8007	0.9239	0.8655	0.1822
3 passes	0.8014	0.9271	0.8624	0.1874
4 passes	0.8004	0.9291	0.8589	0.1936

Table 43. AN1: error decomposition by propagation pass (VOC-Sim, class word, threshold 0.45, in-sample)

Setting	Seed recall	Mean second-pass gain (image IoU)	Spearman: seed recall vs gain
voc_word	0.8403	0.0127	-0.2369
voc_token	0.9424	0.0054	-0.0340
coco_word	0.7582	0.0020	-0.0929
coco_token	0.9131	0.0001	0.0994

Table 44. AN1: seed coverage and the second-pass gain

Transition	New TP (¿ 16 px from previous mask)	New FP (separate; ¿ 16 px from object)	FP removed	New FP per new TP	Net per object pixel
VOC-Sim, word, 1→2	1.09 M (26%)	0.70 M (7%; 61%)	0.62 M	0.6437	0.0159
VOC-Sim, word, 2→3	0.46 M (12%)	0.51 M (8%; 66%)	0.26 M	1.1227	0.0005
VOC-Sim, token, 1→2	0.34 M (15%)	0.51 M (3%; 54%)	0.53 M	1.4793	0.0033
COCO-Cap, word, 1→2	1.20 M (23%)	2.73 M (14%; 83%)	0.61 M	2.2679	-0.0251
COCO-Cap, word, 2→3	0.44 M (15%)	1.53 M (18%; 82%)	0.33 M	3.5165	-0.0212
COCO-Cap, token, 1→2	0.36 M (18%)	1.65 M (17%; 83%)	0.70 M	4.6210	-0.0195

Table 45. AN1 part 2: where each pass's new pixels land (threshold 0.45, annotation resolution, summed over images)

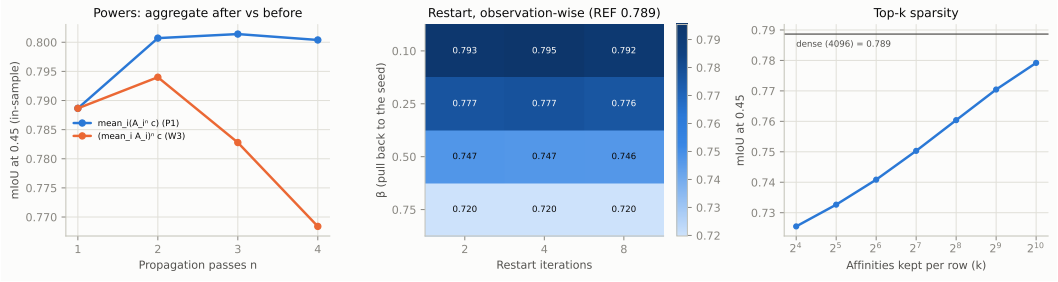


Figure 34. Propagation: passes under both aggregation orders, restart, and top-k sparsity (in-sample).

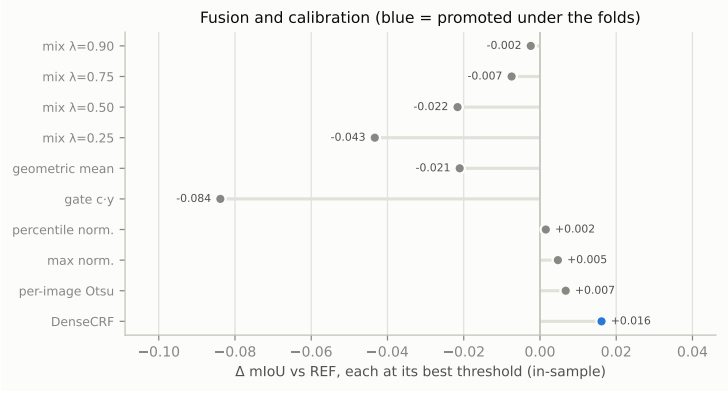


Figure 35. Fusion and calibration: change against REF, each at its best threshold (in-sample).

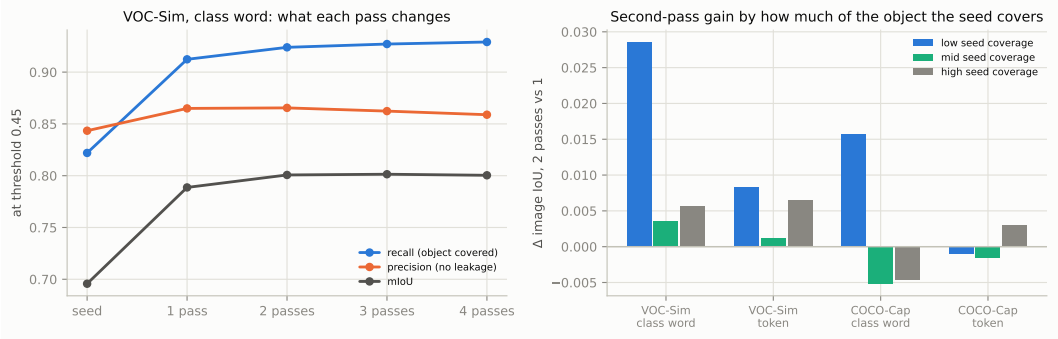


Figure 36. AN1: what each pass changes, and the second-pass gain by seed coverage.

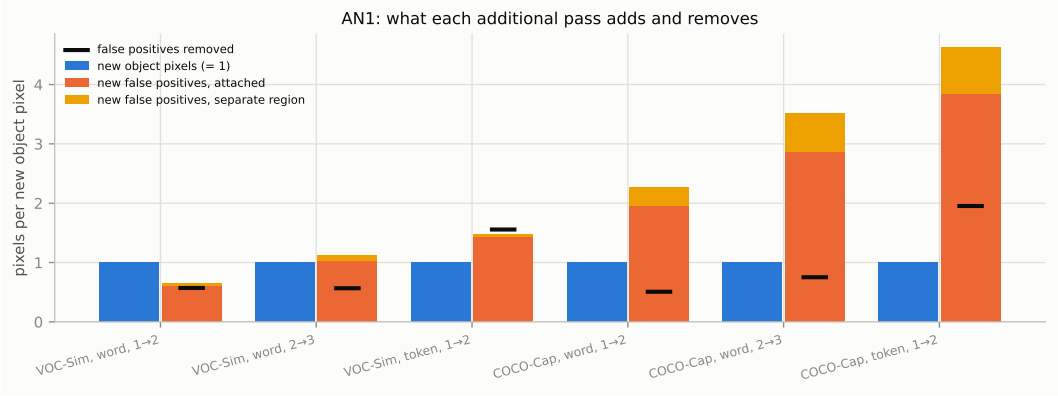


Figure 37. AN1 part 2: new object pixels, new false positives (attached or separate) and removed false positives, per new object pixel.

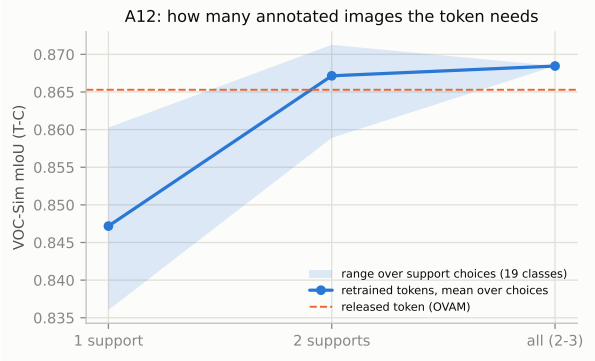


Figure 38. A12: VOC-Sim mIoU against the number of annotated support images per class (T-C pipeline).

Propagation, refinement and tokens (A8–A12).

12.4. Every cell of every later package

Each package below is described in its own registration and result sections of the protocol (Sec. numbers in the headings). For every cell the tables give its best full-VOC threshold t^*, VOC-Sim mIoU there (in-sample), the change against the reference of the screen that lists the cell, a threshold-only OOF mIoU (t^* chosen on four folds, scored on the fifth; the cell itself is fixed, so this is not the OOF of a selection), and COCO-Cap mIoU at the VOC t^*. Small differences from the paper's tables come from the one-pass reference being recomputed in each run (0.7873–0.7875 OOF across environments).

12.4.1. Cross-attention seed

Table 46. **A3: every cell** (6 cells). Screen references (`C/reference`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF
`C/reference`	0.350	0.7057	+0.00	0.7057
`C/R`⋆	0.350	0.7067	+0.10	0.7067
`C/L`	0.350	0.7035	-0.22	0.7035
`C/H`	0.350	0.7056	-0.01	0.7056
`C/R+H`	0.350	0.7058	+0.01	0.7058
`C/L+H`	0.350	0.7020	-0.37	0.7020

A3: A3 (Sec. 27).

12.4.2. Self-attention module

Table 47. **A6: every cell** (13 cells). Screen references (`C/reference`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF
`CHECK/ca_seed`	0.350	0.7068	-4.88	0.7068
`C/reference`	0.450	0.7556	+0.00	0.7556
`C/T`	0.450	0.7710	+1.53	0.7710
`C/R`	0.450	0.7676	+1.19	0.7676
`C/L`⋆	0.450	0.7887	+3.30	0.7887
`C/H`	0.450	0.7613	+0.57	0.7613
`C/T+R`	0.450	0.7749	+1.93	0.7749
`C/T+L`	0.450	0.7820	+2.64	0.7820
`C/T+H`	0.450	0.7732	+1.75	0.7732
`C/R+H`	0.450	0.7656	+0.99	0.7656
`C/L+H`	0.450	0.7871	+3.14	0.7871
`C/T+R+H`	0.450	0.7692	+1.36	0.7692
`C/T+L+H`	0.450	0.7825	+2.68	0.7825

A6: A6 (Sec. 33).

Table 48. **A7: every cell** (78 cells). Screen references (CA=C/R|SA=C/L) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF
CA=C/reference|SA=none	0.350	0.7059	-8.28	0.7059
CA=C/reference|SA=C/reference	0.450	0.7444	-4.43	0.7444
CA=C/reference|SA=C/T	0.450	0.7573	-3.14	0.7573
CA=C/reference|SA=C/R	0.450	0.7521	-3.65	0.7521
CA=C/reference|SA=C/L	0.450	0.7705	-1.81	0.7705
CA=C/reference|SA=C/H	0.450	0.7495	-3.91	0.7495
CA=C/reference|SA=C/T+R	0.350	0.7580	-3.07	0.7497
CA=C/reference|SA=C/T+L	0.450	0.7616	-2.71	0.7546
CA=C/reference|SA=C/T+H	0.450	0.7582	-3.05	0.7582
CA=C/reference|SA=C/R+H	0.450	0.7496	-3.91	0.7496
CA=C/reference|SA=C/L+H	0.450	0.7688	-1.98	0.7688
CA=C/reference|SA=C/T+R+H	0.350	0.7581	-3.05	0.7581
CA=C/reference|SA=C/T+L+H	0.450	0.7621	-2.66	0.7567
CA=C/R|SA=none	0.350	0.7068	-8.18	0.7068
CA=C/R|SA=C/reference	0.450	0.7556	-3.30	0.7556
CA=C/R|SA=C/T	0.450	0.7710	-1.77	0.7710
CA=C/R|SA=C/R	0.450	0.7676	-2.11	0.7676
CA=C/R|SA=C/L	0.450	0.7887	+0.00	0.7887
CA=C/R|SA=C/H	0.450	0.7613	-2.74	0.7613
CA=C/R|SA=C/T+R	0.450	0.7749	-1.38	0.7749
CA=C/R|SA=C/T+L	0.450	0.7820	-0.66	0.7820
CA=C/R|SA=C/T+H	0.450	0.7732	-1.55	0.7732
CA=C/R|SA=C/R+H	0.450	0.7656	-2.31	0.7656
CA=C/R|SA=C/L+H	0.450	0.7871	-0.16	0.7871
CA=C/R|SA=C/T+R+H	0.450	0.7692	-1.94	0.7692
CA=C/R|SA=C/T+L+H	0.450	0.7825	-0.62	0.7825
CA=C/L|SA=none	0.350	0.7037	-8.49	0.7037
CA=C/L|SA=C/reference	0.450	0.7473	-4.14	0.7473
CA=C/L|SA=C/T	0.450	0.7622	-2.65	0.7622
CA=C/L|SA=C/R	0.450	0.7571	-3.15	0.7571
CA=C/L|SA=C/L	0.450	0.7773	-1.13	0.7773
CA=C/L|SA=C/H	0.450	0.7529	-3.57	0.7529
CA=C/L|SA=C/T+R	0.450	0.7650	-2.36	0.7650
CA=C/L|SA=C/T+L	0.450	0.7714	-1.73	0.7714
CA=C/L|SA=C/T+H	0.450	0.7651	-2.36	0.7651
CA=C/L|SA=C/R+H	0.450	0.7552	-3.34	0.7552
CA=C/L|SA=C/L+H	0.450	0.7760	-1.27	0.7760
CA=C/L|SA=C/T+R+H	0.450	0.7603	-2.84	0.7564
CA=C/L|SA=C/T+L+H	0.450	0.7720	-1.67	0.7720
CA=C/H|SA=none	0.350	0.7057	-8.30	0.7057
CA=C/H|SA=C/reference	0.450	0.7465	-4.22	0.7465
CA=C/H|SA=C/T	0.450	0.7606	-2.81	0.7606
CA=C/H|SA=C/R	0.450	0.7551	-3.35	0.7551
CA=C/H|SA=C/L	0.450	0.7738	-1.48	0.7738
CA=C/H|SA=C/H	0.450	0.7519	-3.67	0.7519
CA=C/H|SA=C/T+R	0.450	0.7612	-2.75	0.7564
CA=C/H|SA=C/T+L	0.450	0.7670	-2.17	0.7670
CA=C/H|SA=C/T+H	0.450	0.7624	-2.63	0.7624
CA=C/H|SA=C/R+H	0.450	0.7528	-3.59	0.7528
CA=C/H|SA=C/L+H	0.450	0.7722	-1.64	0.7722
CA=C/H|SA=C/T+R+H	0.350	0.7588	-2.99	0.7588
CA=C/H|SA=C/T+L+H	0.450	0.7674	-2.13	0.7674
CA=C/R+H|SA=none	0.350	0.7059	-8.28	0.7059
CA=C/R+H|SA=C/reference	0.450	0.7568	-3.19	0.7568
CA=C/R+H|SA=C/T	0.450	0.7723	-1.63	0.7723
CA=C/R+H|SA=C/R	0.450	0.7691	-1.96	0.7691
CA=C/R+H|SA=C/L	0.450	0.7904	+0.17	0.7904
CA=C/R+H|SA=C/H	0.450	0.7623	-2.63	0.7623
CA=C/R+H|SA=C/T+R	0.450	0.7765	-1.22	0.7765
CA=C/R+H|SA=C/T+L	0.450	0.7826	-0.60	0.7826
CA=C/R+H|SA=C/T+H	0.450	0.7748	-1.39	0.7748
CA=C/R+H|SA=C/R+H	0.450	0.7671	-2.16	0.7671
CA=C/R+H|SA=C/L+H	0.450	0.7888	+0.01	0.7888
CA=C/R+H|SA=C/T+R+H	0.450	0.7711	-1.76	0.7711
CA=C/R+H|SA=C/T+L+H	0.450	0.7831	-0.55	0.7831
CA=C/L+H|SA=none	0.350	0.7023	-8.64	0.7023
CA=C/L+H|SA=C/reference	0.450	0.7475	-4.11	0.7475
CA=C/L+H|SA=C/T	0.450	0.7632	-2.54	0.7632
CA=C/L+H|SA=C/R	0.450	0.7576	-3.11	0.7576
CA=C/L+H|SA=C/L	0.450	0.7780	-1.07	0.7780
CA=C/L+H|SA=C/H	0.450	0.7533	-3.54	0.7533
CA=C/L+H|SA=C/T+R	0.450	0.7670	-2.17	0.7670
CA=C/L+H|SA=C/T+L	0.450	0.7730	-1.57	0.7730

continued

Cell	t^*	VOC	Δ	OOF
CA=C/L+H|SA=C/T+H	0.450	0.7666	-2.21	0.7666
CA=C/L+H|SA=C/R+H	0.450	0.7558	-3.29	0.7558
CA=C/L+H|SA=C/L+H	0.450	0.7765	-1.22	0.7765
CA=C/L+H|SA=C/T+R+H	0.450	0.7631	-2.55	0.7631
CA=C/L+H|SA=C/T+L+H	0.450	0.7733	-1.54	0.7733

A7: A7 (Sec. 36).

12.4.3. Propagation, refinement and tokens

Table 49. **A8: every cell** (40 cells). Screen references (REF/one_pass) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF
CHECK/ca_seed	0.400	0.7136	-7.52	0.7130
REF/one_pass	0.425	0.7888	+0.00	0.7875
P1/obs_n2⋆	0.450	0.8007	+1.19	0.7998
P1/obs_n3	0.475	0.8019	+1.31	0.8004
P1/obs_n4	0.475	0.8005	+1.17	0.7992
W3/mean_n1	0.425	0.7888	+0.00	0.7875
W3/mean_n2	0.475	0.7950	+0.62	0.7950
W3/mean_n3	0.475	0.7857	-0.31	0.7841
W3/mean_n4	0.475	0.7706	-1.82	0.7706
P2/obs_b10_n2	0.450	0.7934	+0.46	0.7922
P2/obs_b10_n4	0.450	0.7954	+0.66	0.7954
P2/obs_b10_n8	0.450	0.7922	+0.34	0.7922
P2/obs_b25_n2	0.425	0.7788	-1.00	0.7788
P2/obs_b25_n4	0.425	0.7791	-0.97	0.7791
P2/obs_b25_n8	0.425	0.7775	-1.12	0.7775
P2/obs_b50_n2	0.400	0.7536	-3.52	0.7536
P2/obs_b50_n4	0.400	0.7532	-3.56	0.7524
P2/obs_b50_n8	0.400	0.7530	-3.58	0.7522
P2/obs_b75_n2	0.400	0.7321	-5.67	0.7321
P2/obs_b75_n4	0.400	0.7320	-5.67	0.7320
P2/obs_b75_n8	0.400	0.7320	-5.68	0.7320
P2/mean_b10_n2	0.450	0.7888	-0.00	0.7888
P2/mean_b10_n4	0.450	0.7825	-0.63	0.7814
P2/mean_b10_n8	0.450	0.7763	-1.25	0.7763
P2/mean_b25_n2	0.425	0.7752	-1.36	0.7752
P2/mean_b25_n4	0.425	0.7706	-1.81	0.7706
P2/mean_b25_n8	0.425	0.7676	-2.12	0.7676
P2/mean_b50_n2	0.400	0.7515	-3.73	0.7508
P2/mean_b50_n4	0.400	0.7498	-3.90	0.7490
P2/mean_b50_n8	0.400	0.7495	-3.93	0.7486
P2/mean_b75_n2	0.400	0.7316	-5.72	0.7316
P2/mean_b75_n4	0.400	0.7313	-5.75	0.7313
P2/mean_b75_n8	0.400	0.7313	-5.75	0.7313
P6/topk_0016	0.425	0.7270	-6.18	0.7270
P6/topk_0032	0.425	0.7335	-5.53	0.7335
P6/topk_0064	0.425	0.7409	-4.79	0.7402
P6/topk_0128	0.450	0.7503	-3.85	0.7503
P6/topk_0256	0.450	0.7604	-2.84	0.7604
P6/topk_0512	0.450	0.7704	-1.83	0.7704
P6/topk_1024	0.450	0.7792	-0.96	0.7792

A8: A8 (Sec. 43).

Table 50. **A9: every cell** (12 cells). Screen references (REF/one_pass) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF
CHECK/ca_seed	0.400	0.7134	-7.54	0.7127
REF/one_pass	0.425	0.7888	+0.00	0.7875
P4/lam25	0.400	0.7454	-4.33	0.7454
P4/lam50	0.425	0.7671	-2.16	0.7659
P4/lam75	0.425	0.7813	-0.74	0.7813
P4/lam90	0.425	0.7863	-0.24	0.7851

continued

Cell	t^*	VOC	Δ	OOF
F3/gate	0.300	0.7049	-8.38	0.7049
F4/geomean	0.425	0.7677	-2.11	0.7664
N2/percentile	0.475	0.7903	+0.15	0.7890
N3/max	0.600	0.7935	+0.47	0.7935
N4/otsu	0.450	0.7955	+0.68	0.7955
R2/crf⋆	0.425	0.8049	+1.61	0.8049

A9: A9 (Sec. 44).

Table 51. **P3-C: every cell** (30 cells). Screen references (REF/one_pass|N0/global|R0/none) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF
REF/one_pass|N0/global|R0/none	0.425	0.7888	+0.00	0.7875
REF/one_pass|N0/global|R2/crf	0.425	0.8050	+1.62	0.8050
REF/one_pass|N2/percentile|R0/none	0.450	0.7903	+0.15	0.7890
REF/one_pass|N2/percentile|R2/crf	0.450	0.8066	+1.78	0.8056
REF/one_pass|N4/otsu|R0/none	0.450	0.7955	+0.67	0.7955
REF/one_pass|N4/otsu|R2/crf	0.450	0.8131	+2.43	0.8131
P1/obs_n2|N0/global|R0/none	0.450	0.8007	+1.19	0.7998
P1/obs_n2|N0/global|R2/crf⋆	0.450	0.8135	+2.47	0.8135
P1/obs_n2|N2/percentile|R0/none	0.475	0.8019	+1.31	0.8006
P1/obs_n2|N2/percentile|R2/crf	0.450	0.8156	+2.68	0.8143
P1/obs_n2|N4/otsu|R0/none	0.450	0.8051	+1.63	0.8051
P1/obs_n2|N4/otsu|R2/crf⋆	0.450	0.8207	+3.19	0.8207
P1/obs_n3|N0/global|R0/none	0.475	0.8019	+1.31	0.8004
P1/obs_n3|N0/global|R2/crf	0.450	0.8136	+2.48	0.8125
P1/obs_n3|N2/percentile|R0/none	0.475	0.8038	+1.51	0.8022
P1/obs_n3|N2/percentile|R2/crf	0.450	0.8167	+2.80	0.8153
P1/obs_n3|N4/otsu|R0/none	0.450	0.8017	+1.29	0.8017
P1/obs_n3|N4/otsu|R2/crf	0.450	0.8168	+2.80	0.8168
P2/obs_b10_n4|N0/global|R0/none	0.450	0.7954	+0.66	0.7954
P2/obs_b10_n4|N0/global|R2/crf	0.425	0.8102	+2.14	0.8090
P2/obs_b10_n4|N2/percentile|R0/none	0.475	0.7978	+0.90	0.7978
P2/obs_b10_n4|N2/percentile|R2/crf	0.450	0.8136	+2.49	0.8136
P2/obs_b10_n4|N4/otsu|R0/none	0.450	0.7993	+1.05	0.7993
P2/obs_b10_n4|N4/otsu|R2/crf	0.450	0.8168	+2.80	0.8168
W3/mean_n2|N0/global|R0/none	0.475	0.7950	+0.62	0.7950
W3/mean_n2|N0/global|R2/crf	0.450	0.8064	+1.76	0.8064
W3/mean_n2|N2/percentile|R0/none	0.500	0.7957	+0.69	0.7944
W3/mean_n2|N2/percentile|R2/crf	0.475	0.8072	+1.84	0.8066
W3/mean_n2|N4/otsu|R0/none	0.450	0.7973	+0.85	0.7973
W3/mean_n2|N4/otsu|R2/crf	0.450	0.8115	+2.27	0.8115

P3-C: P3-C (Sec. 51).

Table 52. **A11: every cell** (27 cells). Screen references (W/REF) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF
W/seed	0.400	0.7134	-7.52	0.7128
W/REF	0.425	0.7886	+0.00	0.7873
W/P1	0.450	0.8006	+1.20	0.7997
W/R2	0.425	0.8047	+1.61	0.8047
W/P1+R2	0.450	0.8132	+2.46	0.8132
TU/seed	0.475	0.7915	+0.29	0.7915
TU/REF⋆	0.475	0.8435	+5.48	0.8428
TU/P1	0.475	0.8476	+5.90	0.8462
TU/R2	0.475	0.8580	+6.94	0.8580
TU/P1+R2	0.475	0.8603	+7.17	0.8603
TC/seed	0.475	0.7967	+0.81	0.7967
TC/REF⋆	0.475	0.8507	+6.21	0.8507
TC/P1	0.475	0.8543	+6.57	0.8543
TC/R2	0.475	0.8652	+7.66	0.8643
TC/P1+R2	0.475	0.8669	+7.83	0.8654
T1U/REF	0.475	0.8233	+3.47	0.8233

continued

Cell	t^*	VOC	Δ	OOF
T1U/P1	0.500	0.8247	+3.61	0.8238
T2U/REF	0.500	0.8500	+6.14	0.8500
T2U/P1	0.500	0.8546	+6.60	0.8546
T3U/REF	0.500	0.8385	+4.99	0.8385
T3U/P1	0.500	0.8403	+5.16	0.8403
T1C/REF	0.475	0.8309	+4.23	0.8309
T1C/P1	0.500	0.8338	+4.52	0.8334
T2C/REF	0.500	0.8536	+6.50	0.8529
T2C/P1	0.500	0.8578	+6.92	0.8572
T3C/REF	0.500	0.8419	+5.33	0.8419
T3C/P1	0.500	0.8430	+5.43	0.8430

A11: A11 (Sec. 50).

12.4.4. Graph, seed and observation operators

SG5: SG5 contrastive cross-attention seed (Sec. 66, 70). Tests whether contrasting the class-word cross-attention (CA) probability against other prompt positions closes the coverage gap between the class-word seed (seed recall 0.82) and OVAM's released token (0.93) without losing precision. Only the per-head map entering the frozen CA average changes; every seed is then run through the same five fixed pipelines. Training-free for the W-variants; the token variants use OVAM's released token, frozen.

Cell names. Each name is <seed>/<pipeline>. Seeds: W = class word, p_T (mean softmax probability of the class word pieces over the 77 padded prompt keys; frozen convention); WS = log p_T - log p_SoT (contrast with the start token); WK = p_T/(p_T + p_SoT) (class word in OVAM's two-key form); WN = log p_T - log mean of p over all non-target positions; WE = log p_T - log mean of p over <EoT> and padding positions; WY = mean of p_T over the class word and its fixed synonyms (same template; 8 classes have no synonym, so WY = W for them); TC = OVAM released token with keys [<SoT>, token], p_tok (few-shot reference, T-C); TS = log p_tok - log p_SoT. Pipelines: seed = CA map alone, no propagation; REF = one observation-wise SA pass through module 15, mean_i(A_i c); P1 = two passes, mean_i(A_iˆ2 c); R2 = REF followed by OVAM's fixed DenseCRF (note: here 'R2' alone means REF+DenseCRF); P1+R2 = two passes + DenseCRF (FTF for W). 8 seeds x 5 pipelines = 40 cells.

Reference. Four separate screens: W/REF (seed effect under one pass; candidates WS/WK/WN/WE/WY at REF), W/P1+R2 = FTF (candidates the same five at P1+R2), TC/REF (candidate TS/REF) and TC/R2 = T-C few-shot pipeline (candidate TS/R2). The seed and P1 cells of non-screened pipelines are reported only.

Decision. The reference won 5/5 folds in all four screens and nothing was frozen (OOF W/REF 0.7873, W/P1+R2 0.8134, TC/REF 0.8496, TC/R2 0.8647). The log-ratio seeds raise seed recall to 0.96-0.98 but cut precision to 0.47-0.67 (Section 70).

Table 53. **SG5: every cell** (40 cells). Screen references (TC/R2, TC/REF, W/P1+R2, W/REF) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/seed	0.400	0.7133	-7.53	0.7127	0.5629	-5.44
W/REF	0.425	0.7886	+0.00	0.7873	0.6173	+0.00
W/P1	0.450	0.8005	+1.19	0.7997	0.6244	+0.71
W/R2	0.425	0.8049	+1.63	0.8049	0.6349	+1.76
W/P1+R2	0.450	0.8134	+0.00	0.8134	0.6370	+0.00
WS/seed	0.550	0.7002	-8.84	0.7002	0.5408	-7.65
WS/REF	0.525	0.7839	-0.47	0.7820	0.5778	-3.95
WS/P1	0.525	0.7926	+0.40	0.7912	0.5772	-4.00
WS/R2	0.525	0.8019	+1.33	0.7983	0.5933	-2.40
WS/P1+R2	0.500	0.8059	-0.75	0.8028	0.5742	-6.29
WK/seed	0.375	0.7090	-7.96	0.7090	0.5514	-6.59
WK/REF	0.450	0.7733	-1.53	0.7719	0.6144	-0.29
WK/P1	0.450	0.7850	-0.37	0.7850	0.6091	-0.82
WK/R2	0.425	0.7886	-0.00	0.7869	0.6208	+0.35
WK/P1+R2	0.425	0.7978	-1.57	0.7956	0.6093	-2.77
WN/seed	0.600	0.6999	-8.87	0.6999	0.5461	-7.12
WN/REF	0.550	0.7948	+0.62	0.7933	0.5907	-2.66
WN/P1	0.550	0.8024	+1.37	0.8001	0.5894	-2.79
WN/R2	0.550	0.8127	+2.41	0.8127	0.6050	-1.23
WN/P1+R2	0.525	0.8165	+0.31	0.8165	0.5880	-4.91
WE/seed	0.650	0.5930	-19.56	0.5930	0.4582	-15.91
WE/REF	0.625	0.7375	-5.11	0.7364	0.5569	-6.04
WE/P1	0.625	0.7470	-4.16	0.7459	0.5552	-6.21

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
WE/R2	0.600	0.7499	-3.88	0.7491	0.5599	-5.74
WE/P1+R2	0.600	0.7542	-5.92	0.7527	0.5547	-8.23
WY/seed	0.400	0.7111	-7.75	0.7111	0.5589	-5.84
WY/REF	0.450	0.7822	-0.64	0.7813	0.6201	+0.28
WY/P1	0.450	0.7939	+0.53	0.7929	0.6178	+0.05
WY/R2	0.425	0.7978	+0.92	0.7978	0.6283	+1.10
WY/P1+R2	0.425	0.8059	-0.75	0.8053	0.6202	-1.68
TC/seed	0.475	0.7960	+0.73	0.7960	0.6380	+2.07
TC/REF	0.475	0.8501	+0.00	0.8496	0.6680	+0.00
TC/P1	0.475	0.8534	+6.48	0.8534	0.6606	+4.33
TC/R2	0.475	0.8647	+0.00	0.8647	0.6848	+0.00
TC/P1+R2	0.475	0.8661	+7.75	0.8661	0.6743	+5.70
TS/seed	0.575	0.8014	+1.27	0.8014	0.6307	+1.34
TS/REF	0.550	0.8517	+0.16	0.8517	0.6547	-1.33
TS/P1	0.550	0.8528	+6.41	0.8521	0.6513	+3.40
TS/R2	0.550	0.8645	-0.02	0.8636	0.6687	-1.61
TS/P1+R2	0.525	0.8646	+7.60	0.8646	0.6517	+3.44

SG1/SG3: SG1/SG3 anchored graph refinement (Sec. 68, 71). Asks whether confident foreground/background anchors can keep the object completion of SA propagation while limiting leakage. SG1 solves a foreground-before-background absorbing walk (hitting probability) on the mean SA graph; SG3 is a finite-step confidence-locked baseline. The anchor quantiles, an optional image-border background band and (for SG3) the number of steps are varied, on both the class-word (W) and released-token (T) tracks.

Cell names. <track>/<family>/f<FF>_b<BB>_r<R>[_n<N>]. Track W = class-word CA seed, T = OVAM released token. The graph is the mean row-stochastic SA affinity (all 31 calls x 8 heads of module 15). fFF = foreground anchors are pixels at or above quantile 0.FF of the min-max-normalized 64x64 seed (f80/f90/f95 = top 20/10/5%). bBB = background anchors at or below quantile 0.BB (b05/b10 = bottom 5/10%). rR = border width: r0 none, r1 also adds a one-pixel image border to the background anchors (foreground wins at overlap). SG1 = clamp anchors to 1/0 and iterate the mean graph, re-clamping each step, until max change $<$ 1e-4 or 200 iterations. SG3 = start from the CA seed, apply nN mean-graph steps (n1 or n2), restoring the original CA values at both anchor sets after each step. REF = observation-wise one pass mean_i(A_i c); P1 = observation-wise two passes mean_i(A_i^2 c). 12 SG1 + 24 SG3 + 2 references = 38 cells per track, 76 total.

Reference. W/REF for screens W-SG1 and W-SG3, T/REF for T-SG1 and T-SG3 (each family screened independently against its track's one-pass map). P1 is recomputed and reported only.

Decision. No family met the 4/5 rule; W-SG1, W-SG3 and T-SG1 kept REF 5/5 (OOF 0.7873 word, 0.8507 token); T-SG3 kept T/REF with 3/5 votes (candidate chosen in 2 folds, selected-path OOF 0.8507 → 0.8534, +0.0027). Best token cell T/SG3/f95_b05_r0_n1 led to the locked confirmation (Section 72).

Table 54. **SG1/SG3: every cell** (76 cells). Screen references (T/REF, W/REF) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7886	+0.00	0.7873	0.6172	+0.00
W/P1	0.450	0.8006	+1.20	0.7997	0.6245	+0.73
W/SG1/f80_b05_r0	0.700	0.6494	-13.92	0.6494	0.3784	-23.88
W/SG1/f80_b05_r1	0.625	0.7095	-7.91	0.7095	0.4758	-14.14
W/SG1/f80_b10_r0	0.650	0.7172	-7.14	0.7133	0.4526	-16.45
W/SG1/f80_b10_r1	0.575	0.7178	-7.08	0.7178	0.4814	-13.58
W/SG1/f90_b05_r0	0.650	0.7137	-7.50	0.7137	0.4906	-12.66
W/SG1/f90_b05_r1	0.475	0.7372	-5.14	0.7372	0.5322	-8.49
W/SG1/f90_b10_r0	0.525	0.7375	-5.11	0.7375	0.5054	-11.17
W/SG1/f90_b10_r1	0.425	0.7456	-4.30	0.7456	0.5350	-8.22
W/SG1/f95_b05_r0	0.500	0.6909	-9.77	0.6861	0.4838	-13.34
W/SG1/f95_b05_r1	0.325	0.7268	-6.18	0.7178	0.5356	-8.16
W/SG1/f95_b10_r0	0.375	0.7224	-6.62	0.7224	0.5051	-11.21
W/SG1/f95_b10_r1	0.300	0.7366	-5.20	0.7366	0.5597	-5.74
W/SG3/f80_b05_r0_n1	0.400	0.7624	-2.62	0.7624	0.5970	-2.01
W/SG3/f80_b05_r0_n2	0.425	0.7583	-3.03	0.7583	0.5953	-2.19
W/SG3/f80_b05_r1_n1	0.400	0.7602	-2.85	0.7602	0.5954	-2.18
W/SG3/f80_b05_r1_n2	0.400	0.7567	-3.19	0.7557	0.5975	-1.96
W/SG3/f80_b10_r0_n1	0.400	0.7622	-2.64	0.7622	0.5964	-2.07
W/SG3/f80_b10_r0_n2	0.425	0.7596	-2.90	0.7587	0.5942	-2.29
W/SG3/f80_b10_r1_n1	0.400	0.7601	-2.85	0.7601	0.5949	-2.23
W/SG3/f80_b10_r1_n2	0.400	0.7588	-2.99	0.7588	0.5976	-1.95
W/SG3/f90_b05_r0_n1	0.400	0.7747	-1.39	0.7747	0.6131	-0.41

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/SG3/f90_b05_r0_n2	0.400	0.7678	-2.08	0.7678	0.6121	-0.50
W/SG3/f90_b05_r1_n1	0.400	0.7724	-1.62	0.7724	0.6111	-0.60
W/SG3/f90_b05_r1_n2	0.400	0.7683	-2.03	0.7683	0.6114	-0.57
W/SG3/f90_b10_r0_n1	0.400	0.7745	-1.41	0.7745	0.6125	-0.47
W/SG3/f90_b10_r0_n2	0.400	0.7705	-1.81	0.7705	0.6117	-0.55
W/SG3/f90_b10_r1_n1	0.400	0.7723	-1.63	0.7723	0.6106	-0.65
W/SG3/f90_b10_r1_n2	0.400	0.7699	-1.87	0.7699	0.6108	-0.64
W/SG3/f95_b05_r0_n1	0.400	0.7767	-1.20	0.7767	0.6193	+0.21
W/SG3/f95_b05_r0_n2	0.400	0.7657	-2.29	0.7657	0.6044	-1.28
W/SG3/f95_b05_r1_n1	0.400	0.7744	-1.42	0.7744	0.6172	+0.01
W/SG3/f95_b05_r1_n2	0.400	0.7656	-2.30	0.7656	0.6038	-1.34
W/SG3/f95_b10_r0_n1	0.400	0.7765	-1.21	0.7765	0.6187	+0.15
W/SG3/f95_b10_r0_n2	0.400	0.7678	-2.08	0.7678	0.6030	-1.42
W/SG3/f95_b10_r1_n1	0.400	0.7743	-1.43	0.7743	0.6167	-0.04
W/SG3/f95_b10_r1_n2	0.375	0.7680	-2.06	0.7665	0.6173	+0.02
T/REF	0.475	0.8507	+0.00	0.8507	0.6681	+0.00
T/P1	0.475	0.8543	+6.57	0.8543	0.6606	+4.35
T/SG1/f80_b05_r0	0.700	0.6732	-17.75	0.6732	0.3921	-27.60
T/SG1/f80_b05_r1	0.600	0.7216	-12.91	0.7216	0.4751	-19.30
T/SG1/f80_b10_r0	0.650	0.7266	-12.41	0.7244	0.4710	-19.70
T/SG1/f80_b10_r1	0.550	0.7295	-12.12	0.7265	0.4825	-18.56
T/SG1/f90_b05_r0	0.625	0.7363	-11.44	0.7363	0.4950	-17.30
T/SG1/f90_b05_r1	0.475	0.7602	-9.05	0.7542	0.5568	-11.12
T/SG1/f90_b10_r0	0.500	0.7559	-9.47	0.7544	0.5103	-15.78
T/SG1/f90_b10_r1	0.400	0.7672	-8.35	0.7640	0.5330	-13.51
T/SG1/f95_b05_r0	0.475	0.7197	-13.09	0.7138	0.4863	-18.18
T/SG1/f95_b05_r1	0.325	0.7559	-9.48	0.7559	0.5693	-9.88
T/SG1/f95_b10_r0	0.375	0.7448	-10.59	0.7448	0.5338	-13.42
T/SG1/f95_b10_r1	0.300	0.7607	-9.00	0.7607	0.5870	-8.11
T/SG3/f80_b05_r0_n1	0.450	0.8396	-1.11	0.8396	0.6635	-0.46
T/SG3/f80_b05_r0_n2	0.475	0.8357	-1.50	0.8344	0.6662	-0.19
T/SG3/f80_b05_r1_n1	0.450	0.8372	-1.35	0.8372	0.6616	-0.65
T/SG3/f80_b05_r1_n2	0.450	0.8348	-1.58	0.8348	0.6581	-0.99
T/SG3/f80_b10_r0_n1	0.450	0.8391	-1.16	0.8391	0.6631	-0.49
T/SG3/f80_b10_r0_n2	0.450	0.8362	-1.45	0.8362	0.6589	-0.92
T/SG3/f80_b10_r1_n1	0.450	0.8368	-1.39	0.8368	0.6613	-0.67
T/SG3/f80_b10_r1_n2	0.450	0.8350	-1.57	0.8350	0.6582	-0.99
T/SG3/f90_b05_r0_n1	0.450	0.8535	+0.29	0.8535	0.6839	+1.58
T/SG3/f90_b05_r0_n2	0.450	0.8473	-0.34	0.8473	0.6804	+1.24
T/SG3/f90_b05_r1_n1	0.450	0.8510	+0.03	0.8510	0.6818	+1.38
T/SG3/f90_b05_r1_n2	0.450	0.8458	-0.49	0.8447	0.6792	+1.11
T/SG3/f90_b10_r0_n1	0.450	0.8531	+0.24	0.8531	0.6835	+1.55
T/SG3/f90_b10_r0_n2	0.425	0.8474	-0.32	0.8463	0.6736	+0.56
T/SG3/f90_b10_r1_n1	0.450	0.8507	+0.00	0.8507	0.6816	+1.35
T/SG3/f90_b10_r1_n2	0.425	0.8474	-0.32	0.8474	0.6736	+0.56
T/SG3/f95_b05_r0_n1	0.450	0.8562	+0.56	0.8562	0.6920	+2.39
T/SG3/f95_b05_r0_n2	0.425	0.8461	-0.46	0.8461	0.6812	+1.31
T/SG3/f95_b05_r1_n1	0.450	0.8537	+0.30	0.8537	0.6899	+2.18
T/SG3/f95_b05_r1_n2	0.425	0.8467	-0.40	0.8467	0.6813	+1.32
T/SG3/f95_b10_r0_n1	0.450	0.8558	+0.51	0.8558	0.6916	+2.35
T/SG3/f95_b10_r0_n2	0.425	0.8487	-0.19	0.8487	0.6816	+1.35
T/SG3/f95_b10_r1_n1	0.450	0.8534	+0.27	0.8534	0.6896	+2.15
T/SG3/f95_b10_r1_n2	0.425	0.8482	-0.25	0.8482	0.6814	+1.34

SG3-T: Locked token-SG3 confirmation (Sec. 72, 78, 80). A locked confirmation of the strongest SG1/SG3 signal: the fixed token confidence-lock T/SG3/f95_b05_r0_n1, with and without OVAM's DenseCRF, against the one- and two-pass token references. No graph parameter is selected; only each fixed cell's mask threshold is chosen inside the folds.

Cell names. All on the T track (OVAM released optimized token, keys [<SoT>, token]). REF = observation-wise one SA pass; P1 = observation-wise two passes; SG3 = fixed confidence-locked candidate f95_b05_r0_n1: top-5% foreground and bottom-5% background anchors, no border band, one mean-graph step through SA module 15 with anchor values restored to the CA seed; +R2 = followed by OVAM's fixed DenseCRF. 6 cells.

Reference. Registered fixed comparisons: P1 vs REF, P1+R2 vs REF+R2, SG3 vs REF, SG3+R2 vs REF+R2, and SG3+R2 vs P1+R2; the key test is T/SG3+R2 vs T/REF+R2.

Decision. T/SG3 vs T/REF OOF 0.8507 → 0.8562 (+0.0056, paired +0.0059 [+0.0013, +0.0105]); T/SG3+R2 vs T/REF+R2 0.8643 → 0.8674 (+0.0030, paired CI [-0.0020, +0.0076]); vs T/P1+R2 +0.0020 (CI crosses zero). COCO T/SG3+R2 0.7006 vs T/REF+R2 0.6850, transfer only; not promoted (Sections 78, 80).

Table 55. **SG3-T: every cell** (6 cells). Screen references (none recorded) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
T/REF	0.475	0.8507	–	0.8507	0.6681	–
T/REF+R2	0.475	0.8652	–	0.8643	0.6850	–
T/P1	0.475	0.8543	–	0.8543	0.6606	–
T/P1+R2	0.475	0.8669	–	0.8654	0.6743	–
T/SG3	0.450	0.8562	–	0.8562	0.6920	–
T/SG3+R2	0.425	0.8684	–	0.8674	0.7006	–

SG8: SG8 observation-consensus propagation (Sec. 73, 78, 80). Asks whether agreement among the 248 individual SA observations (31 denoising calls x 8 heads of module 15) identifies pixels where propagation is reliable. Each z_i = A_i c is computed separately (affinities are never averaged before propagation) and combined by four independent families: robust aggregation, variance-guided retention of the seed, symmetric sign agreement, and expansion-only agreement. Screened separately on the class-word (W) and released-token (T) tracks.

Cell names. <track>/<family>/<parameter>; track W = class word, T = OVAM released token. REF = ordinary mean mu = mean_i z_i (one pass). A = per-pixel robust aggregate instead of the mean: trim10 / trim20 = trimmed mean removing floor(0.1*248)=24 / floor(0.2*248)=49 observations from each tail; median = median (mean of the two central values). B/tauXpYY = variance-guided CA retention, u*c + (1-u)*mu with u = sigma/(sigma+tau), sigma = per-pixel population std of the z_i; tau in 0.02, 0.05, 0.10, 0.20 ('p' = decimal point). C/rhoXpYY = symmetric sign agreement: mu where r_up $\geq$ rho or r_down $\geq$ rho, else keep the seed c, where r_up / r_down = fraction of observations with z_i $>$ c / z_i $<$ c; rho in 0.60, 0.70, 0.80, 0.90. D/rhoXpYY = expansion-only agreement: max(c, mu) where r_up $\geq$ rho, else c; same rho values. 16 cells per track, 32 total.

Reference. W/REF and T/REF (the ordinary observation-wise one-pass mean); each family A-D is its own screen against its track's REF.

Decision. Two screens froze. W-C froze W/C/rho0p80 (4/5, OOF 0.7875 → 0.7970, +0.0095; COCO 0.6213 vs 0.6178) and T-A froze T/A/median (4/5, OOF 0.8508 → 0.8544, +0.0036; COCO 0.6742 vs 0.6683). W-A had a positive path (+0.0067) but only 3/5 votes; B and D lost; T-C kept REF.

Table 56. **SG8: every cell** (32 cells). Screen references (T/REF, W/REF) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7888	+0.00	0.7875	0.6178	+0.00
W/A/trim10	0.425	0.7924	+0.37	0.7913	0.6187	+0.09
W/A/trim20	0.450	0.7952	+0.64	0.7941	0.6294	+1.16
W/A/median	0.425	0.7971	+0.83	0.7956	0.6230	+0.53
W/B/tau0p02	0.375	0.7410	-4.78	0.7410	0.5861	-3.17
W/B/tau0p05	0.375	0.7574	-3.14	0.7574	0.6003	-1.75
W/B/tau0p10	0.375	0.7688	-1.99	0.7688	0.6103	-0.75
W/B/tau0p20	0.375	0.7773	-1.15	0.7773	0.6157	-0.21
W/C/rho0p60	0.450	0.7949	+0.61	0.7949	0.6239	+0.61
W/C/rho0p70	0.450	0.7970	+0.82	0.7953	0.6219	+0.41
W/C/rho0p80⋆	0.425	0.7969	+0.81	0.7969	0.6213	+0.35
W/C/rho0p90	0.400	0.7896	+0.08	0.7896	0.6132	-0.46
W/D/rho0p60	0.300	0.7284	-6.04	0.7284	0.5732	-4.46
W/D/rho0p70	0.300	0.7295	-5.93	0.7295	0.5719	-4.59
W/D/rho0p80	0.300	0.7298	-5.90	0.7298	0.5680	-4.98
W/D/rho0p90	0.300	0.7269	-6.19	0.7260	0.5631	-5.47
T/REF	0.475	0.8508	+0.00	0.8508	0.6683	+0.00
T/A/trim10	0.475	0.8532	+0.24	0.8532	0.6701	+0.18
T/A/trim20	0.475	0.8549	+0.42	0.8549	0.6718	+0.35
T/A/median⋆	0.475	0.8560	+0.52	0.8560	0.6742	+0.59
T/B/tau0p02	0.450	0.8197	-3.11	0.8197	0.6583	-1.00
T/B/tau0p05	0.450	0.8327	-1.80	0.8327	0.6709	+0.26
T/B/tau0p10	0.450	0.8415	-0.93	0.8415	0.6780	+0.97
T/B/tau0p20	0.450	0.8475	-0.32	0.8475	0.6805	+1.22
T/C/rho0p60	0.475	0.8495	-0.13	0.8495	0.6691	+0.08
T/C/rho0p70	0.475	0.8526	+0.19	0.8526	0.6741	+0.58
T/C/rho0p80	0.450	0.8522	+0.15	0.8522	0.6721	+0.38
T/C/rho0p90	0.450	0.8476	-0.32	0.8476	0.6760	+0.77
T/D/rho0p60	0.375	0.8030	-4.78	0.8030	0.6345	-3.38
T/D/rho0p70	0.375	0.8049	-4.58	0.8049	0.6336	-3.47
T/D/rho0p80	0.400	0.8048	-4.60	0.8048	0.6367	-3.16

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
T/D/rho0p90	0.400	0.8046	-4.61	0.8046	0.6345	-3.38

SG9: SG9 reciprocal SA propagation (Sec. 74, 78, 80). Tests whether propagation edges supported in both spatial directions are more reliable than asymmetric long-range shortcuts. Each of the 248 per-observation affinities A_i is made reciprocal before a single observation-wise pass; three reciprocity operators are compared with the original A on the class-word and released-token tracks.

Cell names. <track>/<operator>; W = class word, T = OVAM released token. REF = original row-stochastic A_i (one pass, mean_i A_i c). R1_geometric = sqrt(A_i * A_iˆT) elementwise, then row-normalized. R2_minimum = min(A_i, A_iˆT) elementwise, then row-normalized. R3_soft_gate = A_i * sqrt(A_iˆT), then row-normalized. Every operator is formed per call/head before propagation; one pass only. 4 cells per track, 8 total.

Reference. W/REF and T/REF (one track-wise screen each); tie preference R3 > R1 > R2.

Decision. REF won 5/5 on both tracks and nothing froze (OOF W 0.7875, T 0.8508). R1/R2 changed full-VOC mIoU only slightly (W 0.7913/0.7904 vs 0.7888) and R3 lowered it (0.7743).

Table 57. **SG9: every cell** (8 cells). Screen references (T/REF, W/REF) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7888	+0.00	0.7875	0.6178	+0.00
W/R1_geometric	0.425	0.7913	+0.26	0.7901	0.6187	+0.09
W/R2_minimum	0.425	0.7904	+0.16	0.7889	0.6197	+0.19
W/R3_soft_gate	0.400	0.7743	-1.45	0.7731	0.6079	-0.98
T/REF	0.475	0.8508	+0.00	0.8508	0.6683	+0.00
T/R1_geometric	0.475	0.8528	+0.20	0.8528	0.6703	+0.20
T/R2_minimum	0.475	0.8524	+0.16	0.8524	0.6713	+0.30
T/R3_soft_gate	0.475	0.8439	-0.69	0.8439	0.6717	+0.34

SG10: SG10 seed-connected bottleneck propagation (Sec. 75, 77, 79). Tests whether pixels that propagation correctly adds are more strongly connected to the confident seed (by the widest-path, max-min bottleneck strength in the mean SA graph) than false-positive leakage. The ordinary one-pass map is kept as evidence and modulated by the rank-normalized bottleneck score; the operator, graph and seed-source quantile are varied on the class-word and released-token tracks.

Cell names. <track>/<cell>; W = class word, T = OVAM released token. seed = CA map alone; REF = mu = mean_i(A_i c), one observation-wise pass; P1 = two passes (reported only). Candidates <Op>_<G>_q<QQ>: qQQ = seed sources S_F are pixels with c ≥ the QQth percentile (q90/q95 = top 10/5%), used as sources only, never clamped. G1 = mean affinity A_bar as a directed graph; G2 = sqrt(A_bar * A_barˆT) (symmetric). Bn = exact widest-path strength from S_F on the dense 4096-node graph, converted to percentile rank (frozen from a smoke test before any score). Op: B1 = mu*Bn; B2a25 / B2a50 = mu*(alpha + (1-alpha)*Bn) with alpha 0.25 / 0.50; B3 = c + Bn*(mu - c) where mu > c, else mu (expansion gated by connectivity). L_<Op>_q<QQ> = the same operators on a local graph (G2 weights restricted to 4-neighbour edges); diagnostic only, never selectable. 16 candidates + seed, REF, P1 + 8 L cells = 27 per track, 54 total.

Reference. W/REF for SG10-W and T/REF for SG10-T, over the 16 candidates each (seed, P1 and L_* are reported only).

Decision. The reference won 5/5 in both screens and nothing froze (OOF W/REF 0.7873, T/REF 0.8496). The best cell, B3, stays below REF (W 0.7824-0.7827 vs 0.7886), and the bottleneck AUC for new TP vs new FP is at or below chance (Section 79).

Table 58. **SG10: every cell** (54 cells). Screen references (T/REF, W/REF) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/seed	0.400	0.7133	-7.53	0.7127	0.5629	-5.44
W/REF	0.425	0.7886	+0.00	0.7873	0.6173	+0.00
W/P1	0.450	0.8005	+1.19	0.7997	0.6244	+0.71
W/B1_G1_q90	0.375	0.6657	-12.30	0.6646	0.5098	-10.75
W/B2a25_G1_q90	0.350	0.6970	-9.16	0.6960	0.5276	-8.97
W/B2a50_G1_q90	0.350	0.7351	-5.35	0.7341	0.5552	-6.21
W/B3_G1_q90	0.550	0.7824	-0.62	0.7824	0.6148	-0.25
W/B1_G1_q95	0.350	0.6585	-13.01	0.6585	0.4936	-12.37

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/B2a25_G1_q95	0.350	0.6925	-9.62	0.6925	0.5250	-9.23
W/B2a50_G1_q95	0.350	0.7334	-5.52	0.7334	0.5555	-6.18
W/B3_G1_q95	0.550	0.7827	-0.59	0.7827	0.6148	-0.25
W/B1_G2_q90	0.375	0.6642	-12.44	0.6627	0.5103	-10.70
W/B2a25_G2_q90	0.350	0.6955	-9.31	0.6955	0.5274	-8.99
W/B2a50_G2_q90	0.350	0.7336	-5.50	0.7326	0.5555	-6.18
W/B3_G2_q90	0.550	0.7824	-0.62	0.7824	0.6151	-0.22
W/B1_G2_q95	0.350	0.6565	-13.21	0.6565	0.4939	-12.34
W/B2a25_G2_q95	0.350	0.6907	-9.79	0.6907	0.5249	-9.24
W/B2a50_G2_q95	0.350	0.7317	-5.69	0.7317	0.5561	-6.12
W/B3_G2_q95	0.550	0.7827	-0.59	0.7827	0.6151	-0.22
W/L_B1_q90	0.375	0.6737	-11.50	0.6737	0.5232	-9.41
W/L_B2a25_q90	0.375	0.7015	-8.71	0.7008	0.5494	-6.79
W/L_B2a50_q90	0.375	0.7374	-5.12	0.7365	0.5767	-4.06
W/L_B3_q90	0.550	0.7819	-0.67	0.7805	0.6182	+0.09
W/L_B1_q95	0.350	0.6656	-12.30	0.6649	0.5072	-11.01
W/L_B2a25_q95	0.350	0.6968	-9.18	0.6956	0.5387	-7.86
W/L_B2a50_q95	0.375	0.7352	-5.34	0.7339	0.5773	-4.00
W/L_B3_q95	0.550	0.7821	-0.65	0.7813	0.6191	+0.18
T/seed	0.475	0.7960	+0.73	0.7960	0.6380	+2.07
T/REF	0.475	0.8501	+0.00	0.8496	0.6680	+0.00
T/P1	0.475	0.8534	+6.48	0.8534	0.6606	+4.33
T/B1_G1_q90	0.350	0.7082	-14.19	0.7082	0.5331	-13.49
T/B2a25_G1_q90	0.350	0.7452	-10.49	0.7452	0.5606	-10.73
T/B2a50_G1_q90	0.375	0.7935	-5.66	0.7935	0.6025	-6.55
T/B3_G1_q90	0.575	0.8448	-0.53	0.8448	0.6675	-0.04
T/B1_G1_q95	0.350	0.7005	-14.96	0.7005	0.5255	-14.25
T/B2a25_G1_q95	0.350	0.7404	-10.97	0.7404	0.5573	-11.07
T/B2a50_G1_q95	0.375	0.7915	-5.86	0.7915	0.6027	-6.52
T/B3_G1_q95	0.575	0.8449	-0.52	0.8449	0.6674	-0.06
T/B1_G2_q90	0.350	0.7072	-14.29	0.7072	0.5325	-13.54
T/B2a25_G2_q90	0.350	0.7437	-10.64	0.7431	0.5596	-10.84
T/B2a50_G2_q90	0.375	0.7929	-5.72	0.7929	0.6015	-6.64
T/B3_G2_q90	0.575	0.8450	-0.51	0.8450	0.6674	-0.06
T/B1_G2_q95	0.350	0.6994	-15.07	0.6994	0.5257	-14.22
T/B2a25_G2_q95	0.350	0.7389	-11.12	0.7389	0.5563	-11.17
T/B2a50_G2_q95	0.375	0.7908	-5.93	0.7908	0.6016	-6.64
T/B3_G2_q95	0.575	0.8451	-0.50	0.8451	0.6673	-0.06
T/L_B1_q90	0.350	0.7113	-7.73	0.7104	0.5423	-7.50
T/L_B2a25_q90	0.350	0.7479	-4.07	0.7472	0.5677	-4.96
T/L_B2a50_q90	0.375	0.7927	+0.40	0.7927	0.6063	-1.10
T/L_B3_q90	0.575	0.8448	+5.61	0.8448	0.6697	+5.24
T/L_B1_q95	0.350	0.7032	-8.54	0.7032	0.5365	-8.08
T/L_B2a25_q95	0.350	0.7434	-4.53	0.7434	0.5651	-5.22
T/L_B2a50_q95	0.375	0.7911	+0.25	0.7911	0.6071	-1.02
T/L_B3_q95	0.575	0.8449	+5.62	0.8449	0.6692	+5.19

SG2: SG2 consensus-edge SA graph (Sec. 81, 85). Tests whether reweighting each SA edge by how consistently the 248 observations agree on it reduces leakage while keeping object completion. Edges are reweighted, never pruned, by a reliability score r raised to a power gamma, either in one consensus mean graph or per observation, with one or two propagation passes; run on class-word and released-token tracks (the token track is diagnostic only under Section 82).

Cell names. <track>/<form>_<rel>_g<GG>_n<N>; W = class word, T = OVAM released token (diagnostic). rel = per-edge reliability: cv = mean_i A_i / sqrt(mean_i A_i^2) = 1/sqrt(1+CV^2) (1 when all observations agree); occ = (number of observations with A_i(p,q) $>$ 1/4096, plus 0.5) / 249 (share of observations where the edge is above uniform). gGG = exponent gamma: g05 = 0.5, g10 = 1, g20 = 2; weights A * r^gamma. form: G = one consensus graph W = rownorm(A_bar * r^gamma), map W c (n1) or W(W c) (n2); O = per observation A~_i = rownorm(A_i * r^gamma), map mean_i A~_i c (n1) or mean_i A~_i^2 c (n2). nN = passes (1 or 2). Other cells: seed = CA alone; REF = mean_i(A_i c); P1 = mean_i(A_i^2 c); MG2 = A_bar^2 c (mean graph squared; diagnostic). 24 candidates + 4 = 28 per track, 56 total.

Reference. Four screens: one-pass candidates (n1) against REF and two-pass candidates (n2) against P1, per track. Tie order gamma 0.5 $>$ 1 $>$ 2, then cv $>$ occ, then G $>$ O.

Decision. SG2-W-1pass froze W/G_cv_g20_n1 (5/5, OOF 0.7873 $\rightarrow$ 0.7984, +0.0111; COCO 0.6234 vs 0.6173, paired +0.0158 [+0.0129, +0.0187]); SG2-W-2pass kept W/P1 (0.7997). The token diagnostic also froze T/G_cv_g20_n1 (0.8496 $\rightarrow$ 0.8568); T 2-pass kept T/P1. The frozen one-pass cell matches, but does not beat, P1 (0.8005 full VOC).

Table 59. **SG2: every cell** (56 cells). Screen references (T/P1, T/REF, W/P1, W/REF) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/seed	0.400	0.7133	-7.53	0.7127	0.5629	-5.44
W/REF	0.425	0.7886	+0.00	0.7873	0.6173	+0.00
W/P1	0.450	0.8005	+0.00	0.7997	0.6244	+0.00
W/MG2	0.475	0.7949	+0.63	0.7949	0.6179	+0.06
W/G_cv_g05_n1	0.450	0.7918	+0.32	0.7910	0.6260	+0.87
W/G_cv_g10_n1	0.450	0.7946	+0.60	0.7946	0.6252	+0.79
W/G_cv_g20_n1⋆	0.450	0.7984	+0.98	0.7984	0.6234	+0.61
W/G_occ_g05_n1	0.425	0.7859	-0.28	0.7841	0.6182	+0.09
W/G_occ_g10_n1	0.425	0.7821	-0.65	0.7812	0.6189	+0.16
W/G_occ_g20_n1	0.400	0.7760	-1.26	0.7760	0.6102	-0.71
W/O_cv_g05_n1	0.450	0.7912	+0.26	0.7900	0.6262	+0.89
W/O_cv_g10_n1	0.450	0.7936	+0.50	0.7936	0.6260	+0.87
W/O_cv_g20_n1	0.450	0.7972	+0.85	0.7972	0.6245	+0.73
W/O_occ_g05_n1	0.450	0.7852	-0.34	0.7842	0.6226	+0.53
W/O_occ_g10_n1	0.425	0.7834	-0.52	0.7820	0.6149	-0.24
W/O_occ_g20_n1	0.425	0.7799	-0.87	0.7789	0.6163	-0.10
W/G_cv_g05_n2	0.475	0.7952	-0.54	0.7952	0.6135	-1.10
W/G_cv_g10_n2	0.475	0.7948	-0.58	0.7938	0.6095	-1.49
W/G_cv_g20_n2	0.450	0.7929	-0.76	0.7915	0.5885	-3.60
W/G_occ_g05_n2	0.475	0.7983	-0.22	0.7983	0.6283	+0.39
W/G_occ_g10_n2	0.450	0.7987	-0.18	0.7970	0.6260	+0.16
W/G_occ_g20_n2	0.425	0.7952	-0.54	0.7936	0.6226	-0.18
W/O_cv_g05_n2	0.475	0.8009	+0.04	0.7998	0.6320	+0.76
W/O_cv_g10_n2	0.475	0.8013	+0.08	0.8002	0.6301	+0.57
W/O_cv_g20_n2	0.475	0.8015	+0.09	0.8001	0.6261	+0.17
W/O_occ_g05_n2	0.475	0.7980	-0.25	0.7980	0.6303	+0.59
W/O_occ_g10_n2	0.450	0.7972	-0.33	0.7958	0.6222	-0.22
W/O_occ_g20_n2	0.450	0.7962	-0.43	0.7952	0.6263	+0.19
T/seed	0.475	0.7960	+0.73	0.7960	0.6380	+2.07
T/REF	0.475	0.8501	+0.00	0.8496	0.6680	+0.00
T/P1	0.475	0.8534	+0.00	0.8534	0.6606	+0.00
T/MG2	0.475	0.8450	+5.64	0.8439	0.6421	+2.48
T/G_cv_g05_n1	0.475	0.8525	+0.24	0.8520	0.6681	+0.01
T/G_cv_g10_n1	0.475	0.8544	+0.43	0.8544	0.6679	-0.00
T/G_cv_g20_n1⋆	0.475	0.8568	+0.67	0.8568	0.6675	-0.05
T/G_occ_g05_n1	0.475	0.8486	-0.15	0.8486	0.6713	+0.34
T/G_occ_g10_n1	0.475	0.8467	-0.34	0.8467	0.6737	+0.58
T/G_occ_g20_n1	0.475	0.8426	-0.75	0.8426	0.6747	+0.67
T/O_cv_g05_n1	0.475	0.8518	+0.17	0.8513	0.6678	-0.01
T/O_cv_g10_n1	0.475	0.8532	+0.31	0.8527	0.6676	-0.04
T/O_cv_g20_n1	0.475	0.8552	+0.51	0.8552	0.6666	-0.13
T/O_occ_g05_n1	0.500	0.8471	-0.30	0.8466	0.6746	+0.66
T/O_occ_g10_n1	0.475	0.8461	-0.40	0.8454	0.6680	+0.00
T/O_occ_g20_n1	0.475	0.8441	-0.60	0.8441	0.6712	+0.32
T/G_cv_g05_n2	0.475	0.8459	-0.76	0.8459	0.6398	-2.09
T/G_cv_g10_n2	0.475	0.8455	-0.79	0.8455	0.6374	-2.33
T/G_cv_g20_n2	0.475	0.8433	-1.01	0.8433	0.6334	-2.72
T/G_occ_g05_n2	0.500	0.8509	-0.25	0.8509	0.6660	+0.53
T/G_occ_g10_n2	0.500	0.8524	-0.10	0.8524	0.6754	+1.47
T/G_occ_g20_n2	0.500	0.8505	-0.29	0.8501	0.6805	+1.99
T/O_cv_g05_n2	0.475	0.8538	+0.04	0.8538	0.6588	-0.18
T/O_cv_g10_n2	0.475	0.8540	+0.05	0.8540	0.6570	-0.36
T/O_cv_g20_n2	0.475	0.8534	+0.00	0.8527	0.6536	-0.70
T/O_occ_g05_n2	0.500	0.8518	-0.16	0.8518	0.6687	+0.80
T/O_occ_g10_n2	0.500	0.8513	-0.21	0.8513	0.6713	+1.07
T/O_occ_g20_n2	0.500	0.8511	-0.23	0.8511	0.6761	+1.55

SG4: SG4 SA regions labelled by CA (Sec. 81, 85). Segments the image into regions using self-attention only (spectral or row clustering of the mean SA graph), then gives each region a single score from cross-attention evidence, producing a piecewise-constant map. Partition type, number of regions K and the region-labelling statistic are varied on class-word and released-token tracks (the token track is diagnostic only). A GT-majority oracle labelling measures how good the partitions themselves are.

Cell names. <track>/<PART><K>_<label>; W = class word, T = OVAM released token (diagnostic). PART: SPEC = k-means on the K leading row-normalized eigenvectors of Dˆ-1/2 S Dˆ-1/2 with S = (A_bar + A_barˆT)/2 (spectral clustering); ROWS = k-means on the L2-normalized rows of A_bar. K in 8, 16, 32 regions (deterministic k-means++, seed 0, 3 restarts, 30 iterations). label: cmean = region mean of the CA seed c; cmed = region median of c; rmean = region mean of the one-pass REF map. seed = CA alone; REF = mean_i(A_i c). 18 candidates + 2 = 20 per track, 40 total.

Reference. W/REF and T/REF (one screen per track). Tie order cmean > cmed > rmean, then SPEC > ROWS, then K 8 > 16 > 32.

Decision. REF won 5/5 on both tracks (OOF W 0.7873, T 0.8496); the best W cell SPEC32_rmean reaches 0.7898 vs 0.7886 in-sample. The oracle-labelled partitions reach 0.80-0.88 mean-image IoU on VOC-Sim (SPEC32 0.8761) and 0.61-0.80 on COCO-Cap, so the bottleneck is labelling, not the partition.

Table 60. **SG4: every cell** (40 cells). Screen references (`T/REF`, `W/REF`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/seed	0.400	0.7133	-7.53	0.7127	0.5629	-5.44
W/REF	0.425	0.7886	+0.00	0.7873	0.6173	+0.00
W/SPEC8_cmean	0.425	0.7775	-1.11	0.7775	0.5808	-3.65
W/SPEC8_cmed	0.425	0.7779	-1.08	0.7779	0.5824	-3.49
W/SPEC8_rmean	0.475	0.7843	-0.43	0.7843	0.5897	-2.76
W/SPEC16_cmean	0.425	0.7656	-2.30	0.7637	0.5994	-1.79
W/SPEC16_cmed	0.400	0.7687	-1.99	0.7687	0.5948	-2.25
W/SPEC16_rmean	0.475	0.7884	-0.02	0.7884	0.6084	-0.89
W/SPEC32_cmean	0.425	0.7507	-3.79	0.7495	0.5983	-1.90
W/SPEC32_cmed	0.400	0.7512	-3.74	0.7497	0.5957	-2.16
W/SPEC32_rmean	0.450	0.7898	+0.12	0.7875	0.6107	-0.66
W/ROWS8_cmean	0.500	0.7489	-3.97	0.7489	0.5420	-7.53
W/ROWS8_cmed	0.475	0.7468	-4.18	0.7455	0.5320	-8.53
W/ROWS8_rmean	0.550	0.7625	-2.61	0.7601	0.5521	-6.52
W/ROWS16_cmean	0.425	0.7527	-3.59	0.7464	0.5767	-4.06
W/ROWS16_cmed	0.400	0.7554	-3.33	0.7543	0.5709	-4.64
W/ROWS16_rmean	0.475	0.7842	-0.44	0.7827	0.5927	-2.46
W/ROWS32_cmean	0.425	0.7493	-3.93	0.7493	0.5971	-2.02
W/ROWS32_cmed	0.400	0.7475	-4.11	0.7460	0.5940	-2.33
W/ROWS32_rmean	0.475	0.7892	+0.05	0.7878	0.6189	+0.16
T/seed	0.475	0.7960	+0.73	0.7960	0.6380	+2.07
T/REF	0.475	0.8501	+0.00	0.8496	0.6680	+0.00
T/SPEC8_cmean	0.475	0.8225	-2.76	0.8225	0.6285	-3.95
T/SPEC8_cmed	0.500	0.8244	-2.57	0.8220	0.6329	-3.50
T/SPEC8_rmean	0.475	0.8238	-2.63	0.8226	0.6071	-6.09
T/SPEC16_cmean	0.500	0.8246	-2.55	0.8246	0.6554	-1.26
T/SPEC16_cmed	0.500	0.8258	-2.43	0.8251	0.6556	-1.24
T/SPEC16_rmean	0.500	0.8342	-1.59	0.8342	0.6432	-2.48
T/SPEC32_cmean	0.475	0.8234	-2.67	0.8234	0.6577	-1.02
T/SPEC32_cmed	0.475	0.8229	-2.72	0.8222	0.6598	-0.82
T/SPEC32_rmean	0.500	0.8417	-0.84	0.8409	0.6559	-1.21
T/ROWS8_cmean	0.500	0.7946	-5.55	0.7907	0.5827	-8.53
T/ROWS8_cmed	0.500	0.7969	-5.32	0.7945	0.5824	-8.55
T/ROWS8_rmean	0.525	0.7981	-5.20	0.7981	0.5732	-9.47
T/ROWS16_cmean	0.475	0.8207	-2.94	0.8199	0.6335	-3.44
T/ROWS16_cmed	0.500	0.8233	-2.68	0.8233	0.6434	-2.45
T/ROWS16_rmean	0.500	0.8320	-1.81	0.8314	0.6299	-3.80
T/ROWS32_cmean	0.475	0.8198	-3.03	0.8191	0.6561	-1.19
T/ROWS32_cmed	0.475	0.8214	-2.87	0.8214	0.6592	-0.88
T/ROWS32_rmean	0.475	0.8410	-0.91	0.8410	0.6446	-2.33

BL: Training-free Track-A backlog (14 families) (Sec. 82, 83, 86). One package that screens the 14 remaining training-free backlog ideas (diffusion kernels, multi-resolution propagation, background suppression, head weighting, DiffSeg-style and query-feature segmentation, SA mean field) against the class-word one-pass reference. Only frozen SD 1.5 class-word CA and module-15 SA (all 31 calls x 8 heads) are used; each family independently selects its parameter and threshold in the five VOC-Sim folds.

Cell names. Format <family>/<parameter>; numeric suffixes are the value x 100 as 3 digits (e.g. t050 = 0.5, w1000 = 10). A_bar = mean module-15 SA affinity; c = min-max CA seed. REF/one_pass = mean_i(A_i c). P3/tXXX = heat-kernel diffusion exp(t(A_bar - I)) c truncated at 8 terms (k = 0..8), diffusion time t = 0.5/1/2/4. P5/nK = K = 1..4 applications of the symmetric normalized graph Dˆ-1/2 S Dˆ-1/2, S = (A_bar + A_barˆT)/2. P9/sXXX_bgB = coarse-to-fine propagation through the mean graph of the last SA module at 8 → 16 → 32 → 64 resolution, keeping the confident seed (c ≥ s, s = 0.5/0.7/0.9) as a floor after each step; bg1 multiplies by 1 - minmax(the same expansion of the start-token (BOS) CA map), bg0 does not. F2/aXXX = REF x (alpha + (1 - alpha) x minmax(column sums of A_bar)), alpha = 0/0.25/0.5/0.75. BG1/gXXX = binary mask REF > gamma x (A_bar applied to the BOS CA map), gamma = 0.5/0.75/1.0/1.25/1.5. W1/TXXX = weighted mean of the 248 observation maps z_i with weights proportional to (1/spatial entropy of z_i)ˆ(1/T), T = 0.5/1/2. W2/kK = average of only the K = 2/4/6 of 8 heads whose call-mean propagated map has the highest Pearson correlation with c. W4/converged =

iterate x <- minmax(A_bar x) from c until max change < 1e-4 (max 50 iterations). H1/proportional = one-step propagation of c through the last SA module at each of 8/16/32/64 resolution, averaged with weights proportional to resolution. G1/tauXXX = DiffSeg merging on A_bar (16x16 anchor grid, 3 symmetric-KL merge rounds, KL threshold tau = 0.9/1.1, no k-means refinement); each region scored by its mean REF. G4/sXXX = binary seeded region growing: 4-connected components of neighbouring pixels whose call-mean SA query features have cosine $\geq$ s (0.5/0.7/0.9), kept if they contain a CA peak (c $\geq$ 0.9). M1/ca_prototype = cosine of each pixel's SA query feature to one CA-weighted mean prototype. M2/kK = k-means (K = 2/3/4, 20 iterations) on query features of the top-20% CA pixels; score = max cosine to the K prototypes. R1/wXXX = 5 mean-field iterations q <- sigmoid(logit(minmax REF) + w A_bar (2q - 1)), pairwise weight w = 1/3/10. BG1 and G4 output binary masks, so the mask threshold does not affect them. 44 cells.

Reference. REF/one_pass (class-word one SA pass, mean_i(A_i c)), against which each of the 14 families is an independent screen.

Decision. REF/one_pass won 5/5 in all 14 screens; nothing promoted (OOF 0.7875, COCO 0.6178). Best full-VOC cells: R1/w100 0.7929 and P5/n2 0.7895 vs REF 0.7888. The label-using diagnostic C1 (per-class thresholds) reaches 0.8348 but is excluded from the label-free pipeline.

Table 61. **BL: every cell** (44 cells). Screen references (`REF/one_pass`) in bold; promoted cell marked $\star$. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
REF/one_pass	0.425	0.7888	+0.00	0.7875	0.6178	+0.00
P3/t050	0.400	0.7427	-4.61	0.7427	0.5881	-2.97
P3/t100	0.425	0.7623	-2.65	0.7617	0.6073	-1.05
P3/t200	0.450	0.7829	-0.59	0.7822	0.6245	+0.68
P3/t400	0.475	0.7872	-0.16	0.7858	0.6050	-1.27
P5/n1	0.400	0.7893	+0.05	0.7882	0.6144	-0.34
P5/n2	0.425	0.7895	+0.07	0.7895	0.6058	-1.19
P5/n3	0.400	0.7693	-1.95	0.7677	0.5531	-6.47
P5/n4	0.400	0.7416	-4.72	0.7416	0.5200	-9.77
P9/s050_bg0	0.600	0.7783	-1.05	0.7775	0.6490	+3.12
P9/s050_bg1	0.300	0.7033	-8.55	0.7033	0.6245	+0.68
P9/s070_bg0	0.525	0.7754	-1.34	0.7739	0.6232	+0.54
P9/s070_bg1	0.300	0.6973	-9.15	0.6973	0.6314	+1.37
P9/s090_bg0	0.525	0.7589	-2.99	0.7580	0.6068	-1.09
P9/s090_bg1	0.300	0.6861	-10.27	0.6861	0.6233	+0.56
F2/a000	0.300	0.4440	-34.48	0.4440	0.3988	-21.89
F2/a025	0.300	0.6879	-10.09	0.6879	0.5455	-7.22
F2/a050	0.325	0.7617	-2.71	0.7617	0.5794	-3.84
F2/a075	0.400	0.7818	-0.70	0.7818	0.6140	-0.37
BG1/g050	0.450	0.7339	-5.49	0.7339	0.5740	-4.38
BG1/g075	0.450	0.7100	-7.88	0.7100	0.5783	-3.95
BG1/g100	0.450	0.4370	-35.18	0.4370	0.3770	-24.08
BG1/g125	0.450	0.2044	-58.44	0.2044	0.1841	-43.36
BG1/g150	0.450	0.0864	-70.24	0.0864	0.0741	-54.36
W1/T050	0.425	0.7889	+0.02	0.7876	0.6181	+0.03
W1/T100	0.425	0.7889	+0.01	0.7876	0.6179	+0.02
W1/T200	0.425	0.7888	+0.00	0.7875	0.6178	+0.01
W2/k2	0.400	0.7703	-1.85	0.7703	0.6069	-1.08
W2/k4	0.400	0.7800	-0.88	0.7789	0.6075	-1.03
W2/k6	0.425	0.7848	-0.40	0.7848	0.6183	+0.05
W4/converged	0.425	0.6113	-17.75	0.6061	0.3670	-25.08
H1/proportional	0.450	0.7813	-0.75	0.7802	0.6270	+0.92
G1/tau090	0.475	0.6963	-9.25	0.6924	0.5356	-8.22
G1/tau110	0.450	0.5294	-25.94	0.5257	0.4323	-18.55
G4/s050	0.450	0.3711	-41.77	0.3711	0.2547	-36.31
G4/s070	0.450	0.3729	-41.59	0.3729	0.2554	-36.24
G4/s090	0.450	0.0469	-74.19	0.0469	0.0369	-58.08
M1/ca_prototype	0.400	0.3913	-39.75	0.3909	0.2729	-34.48
M2/k2	0.525	0.5822	-20.66	0.5801	0.4146	-20.32
M2/k3	0.525	0.5761	-21.27	0.5761	0.3977	-22.01
M2/k4	0.525	0.5735	-21.53	0.5725	0.3882	-22.96
R1/w100	0.400	0.7929	+0.41	0.7929	0.6300	+1.22
R1/w300	0.300	0.7748	-1.40	0.7748	0.6272	+0.95
R1/w1000	0.325	0.7250	-6.38	0.7250	0.5714	-4.64

SG11/SG12: SG11 robust consensus and SG12 reliable observations (Sec. 84, 89). Two class-word-only follow-ups to SG8. SG11 combines SG8's sign-agreement gate with a robust aggregate (median or 20% trimmed mean) in place of the plain mean. SG12 instead keeps only the most mutually consistent SA observations (by mean cosine similarity of their

propagated maps to all other observations) and averages those. Each is an independent screen against the one-pass reference, with SG8 cells recomputed as controls.

Cell names. All cells are class-word (W). REF = one pass mean_i(A_i c), no DenseCRF. Controls recomputed from SG8: C/rho0p80 = SG8-C sign consensus at rho 0.80; A/trim20 and A/median = SG8-A robust aggregates. SG11/<agg>/rhoXpYY: per pixel, use the robust aggregate where max(r_up, r_down) $\geq$ rho, otherwise keep the seed c; agg = median (mean of the two central values of 248) or trim20 (drop floor(0.2*248) = 49 observations from each tail); r_up / r_down = fraction of observations with z_i $>$ c / z_i $<$ c; rho in 0.70, 0.80, 0.90 ('p' = decimal point). SG12/topNN: rank the 248 observations by mean cosine similarity of their raw propagated vector to all other observations (self excluded) and average the raw vectors of the top NN% (ceil(fraction*248): top50 = 124, top70 = 174, top80 = 199, top90 = 224). 14 cells.

Reference. W/REF (one pass without DenseCRF) for both screens; the SG8 controls are reported for context, not as references.

Decision. Recovered after the finalizer audit (Sec. 89), from within-package V100 comparisons: SG11 froze W/SG11/trim20/rho0p80 (5/5, OOF 0.7873 $\rightarrow$ 0.7984, +0.0111; COCO 0.6219 vs 0.6172) and SG12 froze W/SG12/top50 (5/5, OOF 0.7873 $\rightarrow$ 0.7967, +0.0094; COCO 0.6272).

Table 62. **SG11/SG12: every cell** (14 cells). Screen references (`W/REF`) in bold; promoted cell marked $\star$. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7886	+0.00	0.7873	0.6172	+0.00
W/C/rho0p80	0.425	0.7967	+0.81	0.7967	0.6205	+0.33
W/A/trim20	0.450	0.7950	+0.63	0.7939	0.6288	+1.16
W/A/median	0.425	0.7969	+0.83	0.7954	0.6226	+0.54
W/SG11/median/rho0p70	0.425	0.7976	+0.90	0.7959	0.6144	-0.28
W/SG11/median/rho0p80	0.425	0.7984	+0.97	0.7984	0.6221	+0.49
W/SG11/median/rho0p90	0.400	0.7912	+0.26	0.7912	0.6155	-0.16
W/SG11/trim20/rho0p70	0.450	0.7980	+0.94	0.7964	0.6225	+0.53
W/SG11/trim20/rho0p80*	0.425	0.7984	+0.98	0.7984	0.6219	+0.48
W/SG11/trim20/rho0p90	0.400	0.7908	+0.22	0.7908	0.6152	-0.19
W/SG12/top50*	0.450	0.7986	+0.99	0.7967	0.6272	+1.00
W/SG12/top70	0.425	0.7958	+0.72	0.7941	0.6200	+0.28
W/SG12/top80	0.425	0.7941	+0.55	0.7924	0.6219	+0.47
W/SG12/top90	0.425	0.7910	+0.24	0.7894	0.6200	+0.28

SG2+R2: Locked SG2 + DenseCRF confirmation (Sec. 87, 89). A locked confirmation (no parameter search) of whether the promoted one-pass SG2 consensus graph, followed by OVAM's fixed DenseCRF, beats the final training-free pipeline FTF (two passes + DenseCRF). Six cells are computed from each single generation.

Cell names. All class-word (W). REF = mean_i(A_i c); P1 = mean_i(A_i^2 c); SG2 = the Section 85 winner W/G_cv_g20_n1 exactly: G = rownorm(A_bar * r^2), r = mean_i A_i / sqrt(mean_i A_i^2), map G c. +R2 = followed by OVAM's fixed DenseCRF on the thresholded mask; P1+R2 = FTF.

Reference. Primary: W/P1+R2 (FTF), candidate W/SG2+R2. Secondary: W/REF+R2, isolating SG2's contribution under identical postprocessing. The no-CRF cells check that SG2 reproduces.

Decision. Recovered after the finalizer audit (Sec. 89): primary kept FTF with 0/5 candidate votes; locked SG2+R2 OOF 0.8111 vs FTF 0.8132 (-0.0021); COCO 0.6261 vs 0.6371. Secondary vs REF+R2: +0.0064 OOF (0.8047 $\rightarrow$ 0.8111) but only 2/5 votes, so W/REF+R2 stays frozen.

Table 63. **SG2+R2: every cell** (6 cells). Screen references (`W/P1+R2`, `W/REF+R2`) in bold; promoted cell marked $\star$. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7886	-2.46	0.7873	0.6172	-2.00
W/P1	0.450	0.8006	-1.26	0.7997	0.6245	-1.27
W/SG2	0.450	0.7984	-1.48	0.7984	0.6235	-1.37
W/REF+R2	0.425	0.8047	+0.00	0.8047	0.6348	+0.00
W/P1+R2	0.450	0.8132	+0.00	0.8132	0.6371	+0.00
W/SG2+R2	0.425	0.8111	-0.21	0.8111	0.6261	-1.10

SG13: SG13 target-conditioned region labelling (Sec. 88, 90). Tests whether simple target-conditioned scores can label the fixed SPEC32 SA regions well enough to recover SG4's oracle headroom. Three fixed region-scoring rules (no learned

components) threshold raw region scores into hard region masks; class-word only.

Cell names. All class-word (W). REF = one pass; P1 = two passes; SG4/SPEC32_rmean = legacy SG4 control (32 spectral regions, region mean of REF, min-max then bilinear, thresholds 0.30-0.70); C/rho0p80 = SG8-C consensus control. SG13/R1-R3 on the same SPEC32 partition, with per-region C = mean c, P = mean of raw one-pass mu, Q = mean of q = max(fraction z_i $>$ c, fraction z_i $<$ c), M = max c: R1 = P; R2 = P*Q; R3 = 0.5*M + 0.5*P*Q. The raw region score is thresholded (grid 0-1 in steps of 0.005), then region labels are nearest-expanded to annotation size. 7 cells.

Reference. W/REF; R1, R2, R3 screened individually and as a joint family (tie order R1 $>$ R2 $>$ R3).

Decision. All four screens (R1, R2, R3, joint) kept W/REF 5/5, OOF 0.7875, gain 0. Full-VOC R1/R2/R3 = 0.7685/0.7537/0.7039 vs REF 0.7888; the same-generation majority oracle is 0.8758 (VOC) and 0.7928 (COCO) mean-image IoU.

Table 64. **SG13: every cell** (7 cells). Screen references (`W/REF`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7888	+0.00	0.7875	0.6178	+0.00
W/P1	0.450	0.8007	+1.19	0.7998	0.6249	+0.71
W/SG4/SPEC32_rmean	0.450	0.7888	+0.00	0.7869	0.6093	-0.85
W/C/rho0p80	0.425	0.7969	+0.81	0.7969	0.6213	+0.35
W/SG13/R1	0.400	0.7685	-2.03	0.7685	0.6012	-1.66
W/SG13/R2	0.345	0.7537	-3.50	0.7517	0.5924	-2.53
W/SG13/R3	0.480	0.7039	-8.49	0.7030	0.5431	-7.46

SG12+R2: Locked SG12 + DenseCRF confirmation (Sec. 91, 92). Locked confirmation of whether the frozen SG12 top-50% observation filter, followed by the fixed OVAM DenseCRF, beats the final training-free pipeline FTF (two passes + DenseCRF). No CRF or SG12 parameter is tuned.

Cell names. All class-word (W). REF = mean_i(A_i c); P1 = mean_i(A_iˆ2 c) (observation-wise, not (mean A)ˆ2 c); SG12/top50 = mean of the raw propagated maps of the 124 of 248 observations with the highest leave-one-out mean cosine similarity to the others; +R2 = fixed OVAM DenseCRF (parameters 10, 80, 13, 3, 3, 5) on the thresholded mask. P1+R2 = FTF. 5 cells.

Reference. W/P1+R2 (FTF); candidate W/SG12/top50+R2. Unrefined SG12 is diagnostic.

Decision. Protocol kept W/P1+R2 5/5; forced candidate OOF 0.811526 vs FTF 0.813209 (-0.0017); full VOC 0.813126 vs 0.813209; COCO 0.631503 vs 0.637141.

Table 65. **SG12+R2: every cell** (5 cells). Screen references (`W/P1+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7886	-2.46	0.7873	0.6172	-2.00
W/P1	0.450	0.8006	-1.26	0.7997	0.6245	-1.27
W/SG12/top50	0.450	0.7986	-1.47	0.7967	0.6272	-1.00
W/P1+R2	0.450	0.8132	+0.00	0.8132	0.6371	+0.00
W/SG12/top50+R2	0.425	0.8131	-0.01	0.8115	0.6315	-0.56

SG14: SG14 SG11 x SG12 complementarity (Sec. 91, 92). Tests whether SG12 (dropping globally unreliable observations) and SG11 (location-specific consensus gate with robust aggregation) are complementary. A single fixed combination is computed with no tuning: SG11's trim20/rho 0.80 rule applied only over SG12's top-50% observations. Success requires beating both parents, not just REF.

Cell names. All class-word (W). REF = one pass; SG11/trim20/rho0p80 = frozen SG11 cell (20% trimmed mean where max(r_up, r_down) $\geq$ 0.80, else seed c); SG12/top50 = frozen SG12 cell (mean of the 124 most consistent of 248 observations). SG14 = among SG12's 124 retained observations, compute r_up/r_down and a 20% trimmed mean (24 removed per tail); use the trimmed mean where max(r_up, r_down) $\geq$ 0.80, else c. No CRF. 4 cells.

Reference. Three screens: SG14 vs W/REF, vs W/SG11/trim20/rho0p80 and vs W/SG12/top50, plus a joint vote; success needed positive OOF and $\geq$ 4/5 votes against both parents.

Decision. SG14 froze against REF (5/5, OOF 0.798651 vs 0.787280) but each parent was retained 5/5 in its own screen (SG11 0.798400, SG12 0.796658), and the joint screen kept SG11; COCO SG14 0.621678 vs SG11 0.621916 and SG12

0.627174. The complementarity criterion is not met.

Table 66. **SG14: every cell** (4 cells). Screen references (`W/REF`, `W/SG11/trim20/rho0p80`, `W/SG12/top50`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7886	+0.00	0.7873	0.6172	+0.00
W/SG11/trim20/rho0p80⋆	0.425	0.7984	+0.00	0.7984	0.6219	+0.00
W/SG12/top50	0.450	0.7986	+0.00	0.7967	0.6272	+0.00
W/SG14⋆	0.450	0.7987	+1.00	0.7987	0.6217	+0.45

SG15: SG15 prior-driven SPEC32 region refinement (Sec. 91, 92, 93). Instead of recognising the target from region statistics (SG13), SG15 uses an already-good pixelwise mask as the semantic prior and the fixed SPEC32 SA partition only as a spatial regulariser. Each region's fraction of prior-foreground pixels decides whether the region is projected to foreground, snapped, or left as is. Three frozen priors are refined separately, each at its own inherited threshold; Section 93 is a 20-image visual diagnostic of the same rules against the GT-majority oracle, not part of the table.

Cell names. <prior>[/SG15-<rule>]; all class-word (W). Priors: P1 = two passes mean_i(A_i^2 c); SG11/trim20/rho0p80 and SG12/top50 = the frozen SG11 and SG12 cells; REF = one pass (control). For each SPEC32 region, f = fraction of its annotation-resolution pixels in the thresholded prior mask (threshold inherited from the unrefined prior, never re-selected). SG15-A025 / A050 / A075 = majority projection: the whole region is foreground if f $\geq$ q, q = 0.25 / 0.50 / 0.75, else background (q chosen inside the VOC training folds in the A screen). SG15-B = confidence snapping: f $\geq$ 0.75 $\rightarrow$ whole region foreground, f $\leq$ 0.25 $\rightarrow$ whole region background, otherwise keep the prior's pixels. 16 cells.

Reference. Each refinement is screened against its own unrefined prior (W/P1, W/SG11/trim20/rho0p80, W/SG12/top50): six screens (3 priors x A, B).

Decision. All six screens kept their prior 5/5. Best refinement SG11+B: OOF 0.798934 vs 0.798400, COCO 0.622023 vs 0.621916, no eligible vote; all A variants hurt OOF.

Table 67. **SG15: every cell** (16 cells). Screen references (`W/P1`, `W/SG11/trim20/rho0p80`, `W/SG12/top50`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7886	-1.20	0.7873	0.6172	-0.73
W/P1	0.450	0.8006	+0.00	0.7997	0.6245	+0.00
W/SG11/trim20/rho0p80	0.425	0.7984	+0.00	0.7984	0.6219	+0.00
W/SG12/top50	0.450	0.7986	+0.00	0.7967	0.6272	+0.00
W/P1/SG15-A025	0.475	0.7754	-2.52	0.7742	0.6001	-2.44
W/P1/SG15-A050	0.450	0.7869	-1.36	0.7869	0.6110	-1.35
W/P1/SG15-A075	0.400	0.7905	-1.01	0.7905	0.6039	-2.06
W/P1/SG15-B	0.450	0.7975	-0.31	0.7965	0.6229	-0.16
W/SG11/trim20/rho0p80/SG15-A025	0.475	0.7777	-2.07	0.7755	0.6082	-1.38
W/SG11/trim20/rho0p80/SG15-A050	0.400	0.7894	-0.90	0.7836	0.6053	-1.67
W/SG11/trim20/rho0p80/SG15-A075	0.350	0.7961	-0.23	0.7956	0.5966	-2.53
W/SG11/trim20/rho0p80/SG15-B	0.400	0.7989	+0.05	0.7977	0.6131	-0.88
W/SG12/top50/SG15-A025	0.475	0.7761	-2.24	0.7761	0.6028	-2.44
W/SG12/top50/SG15-A050	0.425	0.7875	-1.11	0.7862	0.6071	-2.01
W/SG12/top50/SG15-A075	0.400	0.7941	-0.44	0.7923	0.6111	-1.61
W/SG12/top50/SG15-B	0.450	0.7971	-0.14	0.7937	0.6260	-0.12

12.4.5. Temporal, structural, CA-weighting and multi-scale studies

SG17: SG17 reliability-filtered second-pass additions (Sec. 94, 95). Asks whether the second SA propagation pass can be kept only where the individual observations agree that it should add foreground. Seed is the frozen class-word CA map; the 248 observations are 31 calls x 8 heads of SA module 15. Where the second pass lowers the score it is kept as is; a raw increase is kept only at pixels where at least a cutoff fraction of observations increase from pass 1 to pass 2. The cutoff (top 30/20/10% of VOC positive-increment pixels) is the only thing varied; CRF was to run only if a candidate was promoted.

Cell names. Prefix W/ = frozen training-free class-word CA seed. W/P1 = ordinary observation-wise two-pass map mean_i(A_i^2 c) over all 248 observations, calls 0-30 (in this older naming 'P1' is the two-pass map; SG22 later calls the same thing ALL-P2). W/SG17-Q70, -Q80, -Q90 = gated map y1 + min(y2-y1,0) + 1[q_up $\geq$ cutoff]*max(y2-y1,0), where y1 is the one-pass mean, y2 the two-pass mean and q_up the fraction of the 248 observations whose second-pass value exceeds the first; Q70/Q80/Q90 set the cutoff so that the top 30/20/10% of VOC positive-increment pixels pass. All three collapsed

to the same cutoff (248/248 observations, 39.93% of increments retained), so the three rows are identical. No +R2 cells were run.

Reference. W/P1, the ungated two-pass map, i.e. the existing method that the gate tries to improve.

Decision. W/P1 won all 5 VOC folds and was frozen; no SG17 cell promoted. Forced OOF 0.7991 vs 0.7998 (gain -0.0007), full VOC 0.8001 vs 0.8007, frozen COCO 0.6196 vs 0.6249; DenseCRF confirmation skipped because no cell was promoted.

Table 68. **SG17: every cell** (4 cells). Screen references (`W/P1`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/P1	0.450	0.8007	+0.00	–	0.6249	+0.00
W/SG17-Q70	0.475	0.8001	-0.06	–	0.6196	-0.52
W/SG17-Q80	0.475	0.8001	-0.06	–	0.6196	-0.52
W/SG17-Q90	0.475	0.8001	-0.06	–	0.6196	-0.52

SG18: SG18 temporal-windowed one-pass propagation (Sec. 96, 97). Asks whether one-pass SA propagation improves when the per-observation maps are averaged over only part of the 31 denoising calls instead of all of them. Seed, SA module 15 and all 8 heads are fixed; only the call window changes (three preregistered windows). No CRF; the two-pass map is shown as an external benchmark but cannot win.

Cell names. Prefix W/ = class-word CA seed. All cells are one-pass means of A_i c over the listed calls x 8 heads of SA module 15. W/REF = calls 0-30 (248 observations). W/T1 = calls 10-20 (88 observations; the 'middle' window, later called MID-P1). W/T2 = calls 10-30 (168). W/T3 = calls 6-30 (200, drops the first six calls). W/P1 = two-pass all-call map mean_i(A_i^2 c) (later ALL-P2), external benchmark only, not eligible for selection.

Reference. W/REF, the one-pass all-call average the windows are meant to improve on.

Decision. W/T1 won 5/5 VOC folds and was promoted. OOF 0.7939 vs REF 0.7873 (+0.0066); full VOC 0.7959 vs 0.7886; frozen COCO 0.6285 vs 0.6172. T2 and T3 had positive forced OOF gains but 0/5 votes.

Table 69. **SG18: every cell** (5 cells). Screen references (`W/REF`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/REF	0.425	0.7886	+0.00	–	0.6172	+0.00
W/T1⋆	0.425	0.7959	+0.73	–	0.6285	+1.13
W/T2	0.425	0.7936	+0.50	–	0.6240	+0.68
W/T3	0.425	0.7935	+0.48	–	0.6227	+0.56
W/P1	0.450	0.8006	+1.20	–	0.6245	+0.73

SG18-T1+R2: SG18 follow-up: T1 window + DenseCRF (Sec. 98). Asks whether the promoted one-pass middle-window map (T1) plus the unchanged OVAM DenseCRF beats the established final pipeline, two-pass all-call propagation plus the same CRF (FTF). Four cells share each generated trajectory; no window or CRF parameter is selected.

Cell names. Prefix W/ = class-word CA seed. W/T1 = one-pass mean over SA calls 10-20 x 8 heads of module 15 (SG18 winner; later MID-P1). W/P1 = two-pass all-call map mean_i(A_i^2 c), calls 0-30 (later ALL-P2). Suffix +R2 = the fixed OVAM DenseCRF on the thresholded mask. W/P1+R2 is FTF.

Reference. W/P1+R2 (FTF), the then-final training-free pipeline.

Decision. W/P1+R2 received 5/5 VOC fold votes; T1+R2 not promoted. Forced OOF 0.8106 vs 0.8135 (-0.0029); full VOC 0.8122 vs 0.8135; frozen COCO 0.6465 vs 0.6375 (transfer only).

Table 70. **SG18-T1+R2: every cell** (4 cells). Screen references (`W/P1+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
W/T1	0.425	0.7960	-1.75	0.7939	0.6286	-0.90
W/T1+R2	0.425	0.8122	-0.13	0.8106	0.6465	+0.89
W/P1	0.450	0.8007	-1.28	0.7998	0.6249	-1.27
W/P1+R2	0.450	0.8135	+0.00	0.8135	0.6375	+0.00

SG21: SG21 structural SA-head weighting (Sec. 101). Keeps the SG18/T1 setup (one pass, calls 10-20) but replaces the equal average over the 8 heads within each call with weights from a GT-free structural descriptor of each head's attention matrix. Four preregistered weightings; the 11 call outputs are still averaged equally. No CRF.

Cell names. SG18/T1 = equal-head one-pass mean, SA module 15, calls 10-20. SG21-LM = head weights proportional to raw radius-1 local attention mass. SG21-LM-RANK = weights proportional to ascending local-mass rank 1-8 (ties by physical head ID). SG21-ENT = weights proportional to (largest head entropy in that call - head's mean row entropy + 1e-8), favouring sharp heads. SG21-DIST = weights proportional to inverse mean query-key distance (+1e-8). Weights are normalized to sum to 1 within each call.

Reference. SG18/T1, same-generation; the method whose head average is being reweighted.

Decision. T1 won 5/5 VOC folds; no structural weighting promoted. Forced OOF T1 0.7939, LM 0.7884, LM-RANK 0.7901, ENT 0.7872, DIST 0.7941 (+0.0002, below the promotion guard); DIST frozen COCO 0.6287 vs T1 0.6285.

Table 71. **SG21: every cell** (5 cells). Screen references (`SG18/T1`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
SG18/T1	0.425	0.7959	+0.00	–	0.6285	+0.00
SG21-LM	0.400	0.7903	-0.56	–	0.6193	-0.91
SG21-LM-RANK	0.400	0.7918	-0.42	–	0.6200	-0.85
SG21-ENT	0.400	0.7892	-0.67	–	0.6201	-0.84
SG21-DIST	0.425	0.7941	-0.19	–	0.6287	+0.02

SG22: SG22 call window x propagation depth factorial (Sec. 102). Full factorial crossing the SA call window (all 31 calls vs middle calls 10-20) with propagation depth (1, 2 or 3 passes of each observation's own SA matrix), each scored raw and after the fixed DenseCRF. Seed is the class-word CA map; module 15, all 8 heads. Raw and CRF levels are screened separately.

Cell names. Pattern WINDOW-Pk[+R2]. WINDOW: ALL = SA calls 0-30 (248 call/head observations); MID = calls 10-20 (88). Pk = mean over observations of A_iˆk c, k = 1, 2, 3 passes (P1 = one pass, P2 = two passes, P3 = three). +R2 = fixed OVAM DenseCRF. Correspondences with older names: ALL-P1 = W/REF, ALL-P2 = old W/P1 (two-pass), ALL-P2+R2 = FTF, MID-P1 = SG18 W/T1, MID-P1+R2 = W/T1+R2. MID-P2+R2 is the operator used downstream from SG24 on.

Reference. Raw: ALL-P2 (two-pass all-call map). CRF: ALL-P2+R2 (FTF, the final pipeline at the time).

Decision. Raw screen kept ALL-P2 (3/5 votes). CRF screen selected MID-P2+R2 (4/5 votes; selected OOF 0.8158 vs 0.8132); forced OOF 0.8168, full VOC 0.8185 vs 0.8132, frozen COCO 0.6438 vs 0.6372.

Table 72. **SG22: every cell** (12 cells). Screen references (`ALL-P2`, `ALL-P2+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
ALL-P1	0.425	0.7886	-1.20	0.7873	0.6172	-0.74
ALL-P1+R2	0.425	0.8047	+0.41	0.8047	0.6348	+1.02
ALL-P2	0.450	0.8006	+0.00	0.7997	0.6246	+0.00
ALL-P2+R2	0.450	0.8132	+0.00	0.8132	0.6372	+0.00
ALL-P3	0.475	0.8017	+0.12	0.8002	0.6282	+0.36
ALL-P3+R2	0.450	0.8134	+0.02	0.8122	0.6287	-0.85
MID-P1	0.425	0.7959	-0.46	0.7939	0.6284	+0.38
MID-P1+R2	0.425	0.8121	+1.15	0.8104	0.6464	+2.19
MID-P2	0.450	0.8055	+0.49	0.8035	0.6308	+0.62
MID-P2+R2⋆	0.450	0.8185	+0.53	0.8168	0.6438	+0.67
MID-P3	0.450	0.8044	+0.38	0.8024	0.6214	-0.32
MID-P3+R2	0.425	0.8168	+0.36	0.8146	0.6216	-1.56

SG23: SG23 call window x reliability rule factorial (Sec. 102). Asks whether three earlier reliability-based aggregation rules (SG2 graph, SG11 trimmed consensus, SG12 top-half observations) still help once propagation is restricted to the middle calls. 2 windows x 4 aggregation rules, all one pass, raw and after the fixed DenseCRF, plus a same-generation FTF reference outside the factorial.

Cell names. Pattern WINDOW-RULE[+R2]. WINDOW: ALL = calls 0-30 (248 observations), MID = calls 10-20 (88). RULE (all one pass, module 15, 8 heads): R0 = ordinary mean of A_i c (MID-R0 = T1, ALL-R0 = REF); SG2 = locked consensus-edge graph G_cv_g20_n1, one pass through W = rownorm(mean_i A_i * rˆ2) with r = mean_i A_i / sqrt(mean_i

A_i^2), i.e. each edge reweighted by its squared cross-observation agreement (gamma 2); SG11 = 20%-per-tail trimmed mean of the observation maps gated by directional consensus at rho 0.80; SG12 = mean of the top 50% observations by leave-one-out cosine (124 of 248 for ALL, 44 of 88 for MID). +R2 = fixed DenseCRF. ALL-P2 / ALL-P2+R2 = two-pass all-call map without / with CRF (ALL-P2+R2 = FTF), auxiliary references outside the factorial.

Reference. Raw: MID-R0 (= SG18/T1). Final: ALL-P2+R2 (FTF). MID-R0+R2 is reported separately.

Decision. Raw screen selected MID-SG2 (5/5; OOF 0.7999 vs T1 0.7939; COCO 0.6292 vs 0.6287). CRF screen kept ALL-P2+R2 (4/5); no final cell promoted. Window x rule interactions are negative on VOC OOF and COCO.

Table 73. **SG23: every cell** (18 cells). Screen references (`ALL-P2+R2`, `MID-R0`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
ALL-R0	0.425	0.7887	-0.72	0.7874	0.6175	-1.11
ALL-R0+R2	0.425	0.8049	-0.84	0.8049	0.6351	-0.22
ALL-SG2	0.450	0.7984	+0.25	0.7984	0.6237	-0.50
ALL-SG2+R2	0.425	0.8112	-0.22	0.8112	0.6263	-1.10
ALL-SG11	0.425	0.7984	+0.25	0.7984	0.6219	-0.67
ALL-SG11+R2	0.400	0.8153	+0.20	0.8153	0.6295	-0.79
ALL-SG12	0.450	0.7987	+0.28	0.7968	0.6274	-0.12
ALL-SG12+R2	0.425	0.8133	-0.00	0.8117	0.6317	-0.56
MID-R0	0.425	0.7959	+0.00	0.7939	0.6287	+0.00
MID-R0+R2	0.425	0.8122	-0.11	0.8105	0.6467	+0.93
MID-SG2⋆	0.425	0.8020	+0.61	0.7999	0.6292	+0.06
MID-SG2+R2	0.425	0.8168	+0.35	0.8155	0.6453	+0.80
MID-SG11	0.425	0.8012	+0.53	0.7994	0.6311	+0.24
MID-SG11+R2	0.400	0.8178	+0.45	0.8158	0.6405	+0.32
MID-SG12	0.425	0.7982	+0.23	0.7964	0.6271	-0.15
MID-SG12+R2	0.425	0.8146	+0.13	0.8146	0.6439	+0.66
ALL-P2	0.450	0.8006	+0.47	0.7997	0.6247	-0.40
ALL-P2+R2	0.450	0.8133	+0.00	0.8133	0.6373	+0.00

SG24-A: SG24-A automatic CA seed construction (Sec. 103). Changes only how the CA seed is built from the selected 11 CA modules (calls 0-30, all heads), testing four automatic weighting schemes adapted from GoCA-style ideas against equal aggregation. Each variant is scored as a CA-only map and through the final pipeline (MID-P2 + R2).

Cell names. Pattern Ak/LEVEL. Ak = CA construction: A0 = existing equal aggregation; A1 = per-pixel content-token mass rescaling (content tokens exclude special/padding tokens, punctuation and stopwords); A2 = projected-output CA head contribution weights (v_h = (A_h V_h) W_h^O, score s_h = v_h . sum_j v_j, GoCA dot-product normalization); A3 = pseudo-global / paired-SA similarity layer (module) weights: a pseudo-global map from the conditional Q of up_blocks.2.attentions.1 attn2 is compared with each layer's paired mean-head SA affinity, weights nonnegative and normalized within each call, calls averaged equally; A4 = A1+A2+A3 combined. LEVEL: /CA = CA-only seed map, thresholded directly (no SA, no CRF); /FINAL = seed → MID-P2 (SA module 15, calls 10-20, 8 heads, two observation-wise passes) → R2 DenseCRF.

Reference. A0/CA and A0/FINAL, same generation; A0/FINAL reproduces SG22 MID-P2+R2.

Decision. CA-only screen kept A0 5/5; final screen kept A0 (A3 got 3/5 votes, selected-path OOF 0.81616 vs 0.81689). Forced A3/FINAL won 4/5 held-out folds, OOF 0.82297 vs 0.81689, full VOC 0.8243 vs 0.8185, COCO 0.6620 vs 0.6437; not a formal promotion, but the owner later adopted A3 → MID-P2 → R2 as the exploratory working default (protocol Section 103). A1/A4 score 0 on COCO because class-only prompts make content rescaling spatially constant. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand.

Table 74. **SG24-A: every cell** (10 cells). Screen references (`A0/CA`, `A0/FINAL`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
A0/CA	0.400	0.7135	+0.00	0.7135	0.5627	+0.00
A0/FINAL	0.450	0.8185	+0.00	0.8169	0.6437	+0.00
A1/CA	0.700	0.6406	-7.30	0.6406	0.0000	-56.27
A1/FINAL	0.625	0.8022	-1.63	0.8005	0.0000	-64.37
A2/CA	0.300	0.6897	-2.39	0.6897	0.5559	-0.68
A2/FINAL	0.375	0.8039	-1.46	0.8033	0.6366	-0.70
A3/CA	0.400	0.7049	-0.86	0.7049	0.5836	+2.09

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
A3/FINAL	0.425	0.8243	+0.58	0.8230	0.6620	+1.83
A4/CA	0.700	0.6288	-8.48	0.6288	0.0000	-56.27
A4/FINAL	0.600	0.8027	-1.58	0.8004	0.0000	-64.37

SG24-B: SG24-B CA module set: 11 vs all 16 (Sec. 103). Asks whether adding back the five excluded CA modules (including all 32x32 modules) helps when layer weights are set automatically by A3. Because SG24-A promoted no automatic method, the preregistered fallback A3 was used, so this package is diagnostic and cannot promote anything.

Cell names. Pattern Bm-A3/LEVEL. B0 = the 11 selected CA modules; B1 = all 16 CA modules, including the 32x32 ones; -A3 = A3 automatic layer weighting (from SG24-A) recomputed over that module set; /CA = CA-only seed map; /FINAL = seed → MID-P2 → R2 (same as SG24-A).

Reference. B0-A3/CA and B0-A3/FINAL, the 11-module A3 seed (same as SG24-A's A3 cells, regenerated).

Decision. B1-A3/FINAL lost all five VOC folds: OOF 0.8152 vs 0.8230, full VOC 0.8152 vs 0.8244; frozen COCO rose 0.6622 → 0.6703 (transfer only, cannot override VOC). SG24-B is recorded as diagnostic (A3 fallback). Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand.

Table 75. **SG24-B: every cell** (4 cells). Screen references (B0−A3/CA, B0−A3/FINAL) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
B0−A3/CA	0.400	0.7050	+0.00	0.7050	0.5836	+0.00
B0−A3/FINAL	0.425	0.8244	+0.00	0.8230	0.6622	+0.00
B1−A3/CA	0.350	0.6932	-1.17	0.6932	0.5799	-0.37
B1−A3/FINAL	0.425	0.8152	-0.93	0.8152	0.6703	+0.82

SG25: SG25 multi-resolution SA propagation (+ Pre-SG26 first-arrival diagnostic) (Sec. 104). Asks whether coarser SA modules (16x16, 32x32), alone or paired with the fine 64x64 module, give better two-step propagation of the A3 seed than fine-only propagation. Each cell applies two propagation operations taken from the same diffusion call, with calls 10-20 and all heads; raw and after R2. The same run also records at which propagation depth (1-5) each fine-map pixel first becomes foreground (diagnostic only).

Cell names. All cells: A3 CA seed, SA calls 10-20, all 8 heads, two propagation operations, output at 64x64. FINE-P2 = both passes with the 64x64 module 15 (= MID-P2; A3/FINE-P2+R2 is the working default). COARSE16-P2 / COARSE32-P2 = seed area-downsampled to 16x16 / 32x32, both passes with the final up-path SA module at that size, result bilinearly upsampled to 64. 16-to-64 / 32-to-64 (shown as 16→64, 32→64) = first pass with the 16 / 32 module (its heads averaged), upsample, second pass with each 64x64 head. 64-to-16 = first pass 64x64, area-downsample, second pass 16x16, upsample. Modules: up_blocks.1/.2/.3 .attentions.2.transformer_blocks.0.attn1 for 16/32/64. +R2 = fixed DenseCRF.

Reference. FINE-P2 and FINE-P2+R2, the existing fine-only operator (same generation).

Decision. Raw and R2 screens kept the fine reference in all five folds; nothing promoted. FINE-P2+R2 0.82425 VOC / 0.66210 COCO. Frozen COCO favoured 16-to-64+R2 (0.6975) and COARSE32-P2+R2 (0.6948), but both lost on VOC (forced OOF 0.8170 and 0.8138 vs 0.8228). First-arrival TP rate fell with depth: VOC 0.865 → 0.223, COCO 0.708 → 0.117 (depth 1 → 5). Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand.

Table 76. **SG25: every cell** (12 cells). Screen references (FINE−P2, FINE−P2+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
FINE−P2	0.450	0.8114	+0.00	0.8097	0.6599	+0.00
FINE−P2+R2	0.425	0.8242	+0.00	0.8228	0.6621	+0.00
COARSE16−P2	0.575	0.7231	-8.83	0.7225	0.6282	-3.17
COARSE16−P2+R2	0.575	0.7538	-7.04	0.7538	0.6652	+0.31
COARSE32−P2	0.475	0.7924	-1.90	0.7924	0.6693	+0.94
COARSE32−P2+R2	0.475	0.8148	-0.95	0.8138	0.6948	+3.27

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
16-to-64	0.525	0.8032	-0.81	0.8020	0.6869	+2.70
16-to-64+R2	0.500	0.8183	-0.60	0.8170	0.6975	+3.54
32-to-64	0.475	0.8106	-0.08	0.8089	0.6695	+0.96
32-to-64+R2	0.450	0.8228	-0.14	0.8217	0.6717	+0.96
64-to-16	0.600	0.7513	-6.01	0.7505	0.6403	-1.96
64-to-16+R2	0.575	0.7825	-4.17	0.7825	0.6705	+0.84

SG26: SG26 coarse-gated fine SA (Sec. 105). Asks whether coarse SA can constrain fine SA without replacing it: the fine 64x64 attention matrices are multiplied elementwise by a gate lifted from the coarse (16 or 32) attention of the same call, then row-normalized. The gate is used on both passes or only on the second; raw and after R2, A3 seed, calls 10-20.

Cell names. S0 = FINE-P2 (two ordinary 64x64 passes). S1 = SG25 16-to-64. S2 = SG25 32-to-64 (repeat controls). Gated operator: per call, average the 8 coarse attention matrices at 16 (or 32), block-lift to the 64x64 grid (each fine pixel mapped to its coarse cell), multiply each fine head's matrix elementwise by the lifted gate, row-normalize. S3 / S4 = 16- / 32-gated fine SA on both passes. S5 / S6 = ordinary fine pass 1, then 16- / 32-gated fine pass 2. +R2 = fixed DenseCRF. These S numbers are unrelated to SG27's, except that S0-S2 are the same base maps.

Reference. S0 and S0+R2 (FINE-P2, the working default).

Decision. Raw and R2 screens chose S0 in 5/5 folds; no gate promoted. Final VOC/COCO: S0 0.8242/0.6621, S3 0.7992/0.6817, S4 0.7993/0.6746, S5 0.8201/0.6821, S6 0.8193/0.6773. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand.

Table 77. **SG26: every cell** (14 cells). Screen references (S0, S0+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S0	0.450	0.8114	+0.00	0.8097	0.6599	+0.00
S0+R2	0.425	0.8242	+0.00	0.8228	0.6621	+0.00
S1	0.525	0.8032	-0.81	0.8020	0.6869	+2.70
S1+R2	0.500	0.8183	-0.60	0.8170	0.6975	+3.54
S2	0.475	0.8106	-0.08	0.8089	0.6695	+0.96
S2+R2	0.450	0.8228	-0.14	0.8217	0.6717	+0.96
S3	0.450	0.7725	-3.89	0.7711	0.6502	-0.97
S3+R2	0.450	0.7992	-2.51	0.7983	0.6817	+1.96
S4	0.400	0.7755	-3.59	0.7748	0.6474	-1.25
S4+R2	0.400	0.7993	-2.49	0.7993	0.6746	+1.25
S5	0.475	0.8009	-1.04	0.7999	0.6701	+1.02
S5+R2	0.450	0.8201	-0.42	0.8185	0.6821	+2.00
S6	0.450	0.8021	-0.93	0.8021	0.6672	+0.73
S6+R2	0.425	0.8193	-0.50	0.8160	0.6773	+1.52

SG27: SG27 late fusion of three SA output maps (Sec. 106, 109). Asks whether a fixed, parameter-free pixelwise combination of three completed propagation maps (fine, 16→64, 32→64 from SG25) beats the fine map alone. Only the fusion formula varies (46 proposed, 42 numerically unique); every fused map is scored raw and after R2. No coarse attention modifies the fine SA matrices.

Cell names. Base maps, each min-max normalized before fusion: F = FINE-P2 (= S0), C16 = 16-to-64 (= S1), C32 = 32-to-64 (= S2); every fused map is min-max normalized again; +R2 = fixed DenseCRF. Pairwise operators in a fixed order: mean, geometric mean sqrt(ab), harmonic 2ab/(a+b+eps), RMS, max, min, product ab, soft OR a+b-ab, agreement-weighted first map a(1-|a-b|), agreement-weighted mean ((a+b)/2)(1-|a-b|). F with C16: S3 mean, S4 geometric, S5 harmonic, S6 RMS, S7 max, S8 min, S9 product, S10 soft OR, S13 F(1-|F-C16|), S14 agreement-weighted mean. F with C32: S16 mean, S17 geometric, S18 harmonic, S19 RMS, S20 max, S21 min, S22 product, S23 soft OR, S25 F(1-|F-C32|), S26 agreement-weighted mean. C16 with C32 (no fine map): S27 mean, S28 geometric, S29 harmonic, S30 max, S31 min, S32 soft OR. All three maps: S33 mean, S34 geometric (cube root of product), S35 harmonic, S36 RMS, S37 max, S38 min, S39 soft OR 1-(1-F)(1-C16)(1-C32), S40 median, S41 mean of the top two, S42 mean of the bottom two. Fine-anchored: S43 F(1-|F-(C16+C32)/2|), S44 F*sqrt((1-|F-C16|)(1-|F-C32|)), S45 max(F, (C16+C32)/2). S11, S12, S15, S24 were exact duplicates (S11 = S15 = S7, S12 = S8, S24 = S20) and were dropped, leaving 42 unique cells (84 with +R2), numbered S0-S45 with gaps.

Reference. S0 / S0+R2 = FINE-P2 alone (the working default, A3 → FINE-P2 → R2).

Decision. Raw and R2 multi-candidate VOC screens selected S0 in 5/5 folds; S0+R2 stays at 0.8242 VOC / 0.6621 COCO. S8+R2 (F/C16 min) had the best full VOC (0.8258) and S38+R2 (three-way min) the best forced OOF (0.8256, +0.0028), both below the promotion gate. S9+R2 (F x C16 product) reached 0.7054 frozen COCO but lost VOC (forced OOF 0.8123; VOC threshold 0.300 at the grid edge). Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand.

Table 78. **SG27: every cell** (84 cells). Screen references (`S0`, `S0+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S0	0.450	0.8114	+0.00	0.8097	0.6599	+0.00
S0+R2	0.425	0.8242	+0.00	0.8228	0.6621	+0.00
S1	0.525	0.8032	-0.81	0.8020	0.6869	+2.70
S1+R2	0.500	0.8183	-0.60	0.8170	0.6975	+3.54
S2	0.475	0.8106	-0.08	0.8089	0.6695	+0.96
S2+R2	0.450	0.8228	-0.14	0.8217	0.6717	+0.96
S3	0.500	0.8103	-0.11	0.8091	0.6821	+2.22
S3+R2	0.475	0.8241	-0.01	0.8241	0.6881	+2.60
S4	0.500	0.8106	-0.07	0.8088	0.6820	+2.21
S4+R2	0.475	0.8245	+0.02	0.8233	0.6879	+2.58
S5	0.500	0.8108	-0.06	0.8091	0.6823	+2.24
S5+R2	0.475	0.8247	+0.04	0.8236	0.6882	+2.61
S6	0.500	0.8099	-0.14	0.8088	0.6820	+2.21
S6+R2	0.475	0.8236	-0.06	0.8236	0.6881	+2.60
S7	0.550	0.8050	-0.64	0.8037	0.6881	+2.82
S7+R2	0.500	0.8189	-0.53	0.8162	0.6885	+2.64
S8	0.450	0.8125	+0.11	0.8125	0.6706	+1.06
S8+R2	0.425	0.8258	+0.16	0.8248	0.6743	+1.22
S9	0.300	0.8022	-0.92	0.8022	0.6927	+3.28
S9+R2	0.300	0.8123	-1.20	0.8123	0.7054	+4.33
S10	0.700	0.8096	-0.18	0.8096	0.6622	+0.23
S10+R2	0.700	0.8239	-0.03	0.8239	0.6786	+1.65
S13	0.425	0.8091	-0.23	0.8091	0.6525	-0.74
S13+R2	0.400	0.8211	-0.32	0.8211	0.6538	-0.83
S14	0.450	0.8119	+0.06	0.8119	0.6694	+0.95
S14+R2	0.425	0.8253	+0.10	0.8242	0.6730	+1.09
S16	0.475	0.8116	+0.02	0.8099	0.6705	+1.06
S16+R2	0.450	0.8246	+0.04	0.8246	0.6732	+1.11
S17	0.475	0.8116	+0.03	0.8100	0.6703	+1.04
S17+R2	0.450	0.8247	+0.04	0.8247	0.6730	+1.09
S18	0.475	0.8117	+0.03	0.8101	0.6705	+1.06
S18+R2	0.450	0.8247	+0.05	0.8247	0.6732	+1.11
S19	0.475	0.8116	+0.02	0.8099	0.6704	+1.05
S19+R2	0.450	0.8246	+0.03	0.8246	0.6731	+1.10
S20	0.475	0.8109	-0.05	0.8094	0.6641	+0.42
S20+R2	0.475	0.8235	-0.08	0.8220	0.6772	+1.51
S21	0.450	0.8117	+0.03	0.8117	0.6655	+0.56
S21+R2	0.425	0.8246	+0.04	0.8237	0.6682	+0.61
S22	0.300	0.7967	-1.46	0.7967	0.6875	+2.76
S22+R2	0.300	0.8037	-2.05	0.8037	0.6971	+3.50
S23	0.700	0.8143	+0.29	0.8143	0.6626	+0.27
S23+R2	0.700	0.8257	+0.14	0.8245	0.6750	+1.29
S25	0.450	0.8107	-0.07	0.8107	0.6626	+0.27
S25+R2	0.425	0.8233	-0.10	0.8233	0.6657	+0.36
S26	0.450	0.8115	+0.02	0.8106	0.6650	+0.51
S26+R2	0.425	0.8247	+0.04	0.8236	0.6673	+0.52
S27	0.500	0.8089	-0.24	0.8074	0.6810	+2.11
S27+R2	0.475	0.8225	-0.17	0.8211	0.6869	+2.48
S28	0.500	0.8092	-0.22	0.8079	0.6812	+2.13
S28+R2	0.475	0.8228	-0.15	0.8213	0.6871	+2.50
S29	0.500	0.8094	-0.20	0.8081	0.6814	+2.15
S29+R2	0.475	0.8229	-0.13	0.8199	0.6872	+2.51
S30	0.550	0.8057	-0.56	0.8048	0.6893	+2.94
S30+R2	0.525	0.8194	-0.48	0.8170	0.6986	+3.65
S31	0.475	0.8108	-0.06	0.8071	0.6783	+1.84
S31+R2	0.450	0.8243	+0.00	0.8243	0.6832	+2.11
S32	0.700	0.8070	-0.44	0.8070	0.6591	-0.08
S32+R2	0.700	0.8222	-0.20	0.8222	0.6764	+1.43
S33	0.500	0.8108	-0.06	0.8090	0.6817	+2.18
S33+R2	0.475	0.8241	-0.02	0.8225	0.6871	+2.50
S34	0.500	0.8110	-0.03	0.8093	0.6817	+2.18
S34+R2	0.475	0.8243	+0.00	0.8228	0.6871	+2.50
S35	0.475	0.8112	-0.02	0.8094	0.6720	+1.21

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S35+R2	0.475	0.8244	+0.02	0.8230	0.6872	+2.51
S36	0.500	0.8106	-0.08	0.8094	0.6815	+2.16
S36+R2	0.475	0.8238	-0.04	0.8228	0.6869	+2.48
S37	0.550	0.8060	-0.54	0.8050	0.6867	+2.68
S37+R2	0.525	0.8195	-0.47	0.8171	0.6951	+3.30
S38	0.450	0.8124	+0.11	0.8112	0.6718	+1.19
S38+R2	0.425	0.8256	+0.14	0.8256	0.6760	+1.39
S39	0.700	0.7672	-4.41	0.7672	0.5749	-8.51
S39+R2	0.700	0.7883	-3.59	0.7883	0.5937	-6.84
S40	0.475	0.8110	-0.04	0.8094	0.6718	+1.19
S40+R2	0.450	0.8239	-0.04	0.8230	0.6746	+1.25
S41	0.525	0.8090	-0.24	0.8074	0.6852	+2.53
S41+R2	0.500	0.8224	-0.18	0.8211	0.6913	+2.92
S42	0.475	0.8119	+0.05	0.8103	0.6773	+1.74
S42+R2	0.450	0.8247	+0.04	0.8229	0.6808	+1.87
S43	0.425	0.8095	-0.18	0.8080	0.6510	-0.89
S43+R2	0.425	0.8222	-0.21	0.8209	0.6641	+0.20
S44	0.425	0.8098	-0.15	0.8083	0.6520	-0.79
S44+R2	0.425	0.8223	-0.20	0.8207	0.6651	+0.30
S45	0.500	0.8089	-0.24	0.8074	0.6746	+1.47
S45+R2	0.475	0.8225	-0.18	0.8209	0.6789	+1.68

SG28: SG28 PEARL-inspired components (Procrustes alignment, Laplacian smoothing) (Sec. 110, 112). Tests two independent, never-combined components adapted from PEARL on top of the working default A3 → FINE-P2. (a) Recompute module-15 SA after per-head Q/K centering and/or Procrustes alignment of keys to queries. (b) Smooth the final normalized score on a 64x64 grid graph, optionally with image-boundary edge weights and/or cross-scale confidence. Raw and after R2.

Cell names. P0 = exact historical A3 → FINE-P2 map (SA module 15, calls 10-20, 8 heads, two observation-wise passes). Caution: here P1-P3 are Procrustes variants, not propagation depths. PA branch: P1 = weighted query-norm centering then SVD Procrustes rotation K → Q (rotation applied to uncentered K); P2 = weighted centering only; P3 = uniform centering then SVD Procrustes. Laplacian branch, solving (D_rho + L)F = D_rho f on the 4-neighbour grid with f = normalized P0 score: B1 = unit edges, rho = 1; B2 = image-boundary edge weights exp(-5|I_i - I_j|) from the grayscale generated image, rho = 1; C1 / C2 / C3 = unit edges with confidence rho = 1-|f-c16|, 1-|f-c32|, or 1 minus the average three-scale pairwise disagreement (c16/c32 = the coarse-path maps); D1 / D2 / D3 = the same rho with B2's boundary edges. +R2 = fixed DenseCRF.

Reference. P0 / P0+R2, the working default.

Decision. Every selector kept P0 (raw) and P0+R2 (final) in 5/5 VOC folds; nothing promoted. Baseline after R2 0.8228 forced OOF / 0.8242 full VOC / 0.6621 COCO. Best component B2: raw forced OOF 0.8140 (+0.0043, below the +0.005 floor), raw COCO 0.6654; after R2 0.8230 (+0.0001) and COCO 0.6661. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Independent audit (protocol Sec. 130): all 24 cells recomputed from the per-image shards match the report; P0 and P0+R2 equal the historical SG27 S0 counts on all 490 VOC and 707 COCO images. The +0.005 floor applies to each fold's training gain, which for B2 is only +0.0017 to +0.0030. Raw B2 on COCO gains +0.0068 [+0.0050, +0.0086] in paired mean image IoU; after R2 the interval includes zero. Centering K cancels in the softmax, so P2 is effectively query centering only; the cross-scale trust saturates because all three maps share the final fine-scale pass from the same seed.

Table 79. **SG28: every cell** (24 cells). Screen references (P0, P0+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
P0	0.450	0.8114	+0.00	0.8097	0.6599	+0.00
P0+R2	0.425	0.8242	+0.00	0.8228	0.6621	+0.00
P1	0.425	0.8129	+0.15	0.8129	0.6443	-1.56
P1+R2	0.425	0.8234	-0.08	0.8220	0.6543	-0.78
P2	0.475	0.7973	-1.41	0.7961	0.6555	-0.44
P2+R2	0.450	0.8167	-0.76	0.8155	0.6640	+0.19
P3	0.425	0.8129	+0.15	0.8129	0.6444	-1.55
P3+R2	0.425	0.8234	-0.08	0.8220	0.6543	-0.78
B1	0.475	0.8031	-0.83	0.8031	0.6537	-0.62
B1+R2	0.450	0.8155	-0.88	0.8138	0.6585	-0.36

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
B2	0.475	0.8140	+0.27	0.8140	0.6654	+0.55
B2+R2	0.450	0.8230	-0.13	0.8230	0.6661	+0.40
C1	0.475	0.8021	-0.93	0.8021	0.6530	-0.69
C1+R2	0.450	0.8146	-0.97	0.8129	0.6580	-0.41
C2	0.475	0.8027	-0.86	0.8027	0.6534	-0.65
C2+R2	0.450	0.8152	-0.91	0.8135	0.6583	-0.38
C3	0.475	0.8024	-0.90	0.8024	0.6532	-0.67
C3+R2	0.450	0.8149	-0.94	0.8132	0.6581	-0.40
D1	0.475	0.8139	+0.25	0.8139	0.6652	+0.53
D1+R2	0.450	0.8227	-0.16	0.8227	0.6658	+0.37
D2	0.475	0.8140	+0.26	0.8140	0.6653	+0.54
D2+R2	0.450	0.8228	-0.14	0.8228	0.6659	+0.38
D3	0.475	0.8140	+0.26	0.8140	0.6653	+0.54
D3+R2	0.450	0.8228	-0.15	0.8228	0.6659	+0.38

12.4.6. Paper-inspired propagation, hierarchy and combined studies

SG29: SG29 NERVE-inspired head weighting and stopped random walk (Sec. 113, 115). Adapts three ideas from NERVE to the working default A3 → FINE-P2: class-conditioned head weights from the spatial entropy of each head's one-step output, a geometrically stopped random walk instead of exactly two passes, and a local Q/K 8-neighbour transition fused with the global SA. Seven main cells, raw and after R2, plus three walk-length diagnostics. Single-class spatial entropy replaces NERVE's multiclass entropy, so this is an adaptation, not a reproduction.

Cell names. All cells: historical A3 CA seed, SA module 15 at 64x64, calls 10-20, 8 heads; global transitions are the original SD softmax SA. N0 = exact historical observation-wise two-pass map (FINE-P2; = SG28 P0). N1 = N0 with class-conditioned head weights softmax(-3H) within each call, H = spatial entropy of the head's one-step output divided by log(4096). N2 = geometrically stopped walk sum_d=0..10 0.9^d A^d c / sum_d 0.9^d per head, heads averaged. N3 = the same walk on a local transition only (8-neighbour (cos(Q,K)+1)/2 + 1e-6, plus 0.01 self weight, row-normalized). N4 = walk on the fused transition 0.8 x global + 0.2 x local. N5 = N1 head weights + N2 walk. N6 = N4 walk + entropy head weights computed from the fused one-step output. N2-L1 / N2-L2 / N2-L5 = N2 truncated at walk length 1 / 2 / 5 (diagnostic only, not selectable). +R2 = fixed DenseCRF.

Reference. N0 / N0+R2, the exact historical working default (checked per image against same-generation FINE-P2; SG28 counts were a replay audit only).

Decision. Every selector kept N0 (raw) and N0+R2 (final) in 5/5 VOC folds; nothing promoted. N0+R2: 0.8229 forced OOF / 0.8243 full VOC / 0.6623 COCO. N1+R2 won 5/5 pairwise folds but gained only +0.0001 forced OOF; entropy weights are near uniform (7.998/8 effective heads) and do not rank heads within a call. Stopped walks trade VOC for COCO: N2+R2 0.8135 / 0.8147 / 0.6696, N4+R2 0.8103 / 0.8110 / 0.6704, N6+R2 0.8104 / 0.8110 / 0.6705; N4+R2's COCO gain loses 0.0356 person IoU and its paired mean-image interval crosses zero. Local-only N3+R2 fails (0.7278 / 0.7278 / 0.6118). Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 80. **SG29: every cell** (20 cells). Screen references (N0, N0+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
N0	0.450	0.8114	+0.00	0.8098	0.6598	+0.00
N0+R2	0.425	0.8243	+0.00	0.8229	0.6623	+0.00
N1	0.450	0.8114	+0.00	0.8098	0.6602	+0.04
N1+R2	0.425	0.8243	+0.01	0.8229	0.6627	+0.04
N2	0.450	0.7994	-1.20	0.7994	0.6610	+0.11
N2+R2	0.425	0.8147	-0.96	0.8135	0.6696	+0.73
N3	0.400	0.6999	-11.14	0.6988	0.5784	-8.14
N3+R2	0.425	0.7278	-9.65	0.7278	0.6118	-5.05
N4	0.450	0.7949	-1.64	0.7949	0.6600	+0.02
N4+R2	0.425	0.8110	-1.32	0.8103	0.6704	+0.81
N5	0.450	0.7994	-1.20	0.7994	0.6611	+0.13
N5+R2	0.425	0.8147	-0.96	0.8136	0.6697	+0.74
N6	0.450	0.7950	-1.64	0.7950	0.6601	+0.03
N6+R2	0.425	0.8110	-1.32	0.8104	0.6705	+0.82
N2-L1	0.400	0.7537	-5.77	0.7525	0.6226	-3.73
N2-L1+R2	0.400	0.7794	-3.20	0.7794	0.6522	-0.77
N2-L2	0.425	0.7745	-3.69	0.7745	0.6425	-1.74

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
N2-L2+R2	0.400	0.7958	-1.56	0.7958	0.6623	+0.25
N2-L5	0.425	0.7934	-1.79	0.7922	0.6537	-0.61
N2-L5+R2	0.425	0.8108	-0.06	0.8108	0.6748	+1.49

SG30: SG30 SPARK-inspired nonlinear affinity sharpening (Sec. 116, 118). Adapts SPARK's expansion, inflation and pruning of a random-walk graph to the fine SA graph of A3 → FINE-P2 (module 15, calls 10-20, 8 heads, observation-wise). Varies where an elementwise power r = 2.6 is applied, whether A is first expanded to A^2, pruning at tau = 1e-7, and one feature-local/global fused update. Constants are fixed from the paper, no sweep; author code was unavailable, so the local graph is an explicit adaptation. Eight cells raw and after R2, plus two raw iteration diagnostics.

Cell names. Infl(X) = elementwise power X^2.6 then row normalization. S0 = historical observation-wise P2 mean_i(A_i^2 c) (same-generation FINE-P2). S1 = Infl(A) used for both passes. S2 = ordinary first pass, Infl(A) on the second pass. S3 = Infl(A^2) applied once to c. S4 = ordinary first pass Ac, then Infl(A^2). S5 = S3 with entries of the unnormalized power below tau = 1e-7 pruned and rows renormalized (an emptied row falls back to its pre-pruned row). S6 = S4 with the same pruning. S7 = one SPARK-like update (expand, inflate, prune) on 0.6 x global SA + 0.4 x a local graph from module-15 input features (after group norm, before Q/K/V; negative cosine clamped to 0, self weight 1, row-normalized), applied to c. S5-I2 / S5-I3 = the S5 update repeated two / three times (raw diagnostics only). +R2 = fixed DenseCRF. Caution: apart from S0 = FINE-P2, S cells here are unrelated to SG31's or SG32's same-numbered cells; SG30 S2 reappears as SG32 S1.

Reference. S0 / S0+R2 = same-generation FINE-P2, the working default.

Decision. Every selector kept S0 (raw) and S0+R2 (final) in 5/5 VOC folds; nothing promoted. S0+R2: 0.8231 forced OOF / 0.8244 full VOC / 0.6623 COCO. Second-pass inflation S2+R2 reached 0.6855 COCO but 0.8125 forced OOF (1/5 pairwise wins) / 0.8156 full VOC; S6+R2 0.8144 / 0.8144 / 0.6836; S4+R2 0.8187 / 0.8200 / 0.6732. S1, S5 and feature-local S7 (S7+R2 0.7356 / 0.7356 / 0.6206) fail. Inflation concentrates mass (top-quartile edge mass 0.825 → 0.976), yet graph purity does not predict masks; S6 pruning removes about 99% of entries. S2 trades recall for precision. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 81. **SG30: every cell** (18 cells). Screen references (S0, S0+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S0	0.450	0.8115	+0.00	0.8098	0.6600	+0.00
S0+R2	0.425	0.8244	+0.00	0.8231	0.6623	+0.00
S1	0.400	0.7634	-4.80	0.7634	0.6378	-2.22
S1+R2	0.400	0.7841	-4.03	0.7841	0.6623	+0.00
S2	0.450	0.7997	-1.17	0.7997	0.6674	+0.74
S2+R2	0.450	0.8156	-0.89	0.8125	0.6855	+2.32
S3	0.425	0.7892	-2.23	0.7880	0.6587	-0.13
S3+R2	0.400	0.8069	-1.75	0.8056	0.6697	+0.74
S4	0.450	0.8048	-0.67	0.8032	0.6664	+0.64
S4+R2	0.425	0.8200	-0.45	0.8187	0.6732	+1.09
S5	0.425	0.7632	-4.82	0.7620	0.6402	-1.98
S5+R2	0.425	0.7806	-4.38	0.7795	0.6619	-0.04
S6	0.450	0.7978	-1.37	0.7964	0.6638	+0.37
S6+R2	0.450	0.8144	-1.00	0.8144	0.6836	+2.12
S7	0.400	0.7069	-10.46	0.7069	0.5866	-7.34
S7+R2	0.425	0.7356	-8.89	0.7356	0.6206	-4.17
S5-I2	0.475	0.7235	-8.80	0.7216	0.6248	-3.53
S5-I3	0.475	0.7031	-10.84	0.7031	0.6046	-5.54

SG31: SG31 SeeDiff-inspired re-seeding and background propagation (Sec. 117, 119, 120). Adapts SeeDiff's coarse-to-fine re-seeding and explicit background expansion to A3 with the SG25 16/32/64 SA modules (same call, calls 10-20, 8 heads). Hard 0.5 seed sets from SeeDiff are expanded by averaging selected SA rows, then re-thresholded scale by scale; separate cells gate the fine map by propagated background support. Includes historical controls and a score-squaring control. Eleven cells raw and after R2. A shard stopped on a too-tight float tolerance in the S1 control and was resumed after relaxing it.

Cell names. F = normalized fine map; selected-row expansion E(S) = mean over p in S of head-mean A[p,:] (SeeDiff orientation, not A @ c); every scale step is spatially min-max normalized. S0 = same-generation FINE-P2. S1 = historical

SG25 16→64 continuous path. S2 = continuous 16→32→64 (three same-call SA applications). S3 = hard selected-row re-seeding 16→32→64 (seed = normalized map $>$ 0.5 at each scale; empty sets fall back to top-1). S4 = the same hard re-seeding 32→64 only. S5 = soft control, rows weighted by the nonnegative score, 16→32→64. Background seeds F $\leq$ 0.5; B = their normalized support; gated maps are minmax(F(1-B)). S6 = S0 gated by one-step selected-row B. S7 = S0 gated by observation-wise A64ˆ2 background. S8 / S9 = S3 / S4 gated by selected-row B. S10 = normalized Fˆ2 (monotone sharpening control). +R2 = fixed DenseCRF. Caution: apart from S0 = FINE-P2, S cells are unrelated to SG30's or SG32's same-numbered cells; SG31 S2 is the ordinary hierarchy replayed as SG33 H0, SG-combined-v1 C1 and v2 V1.

Reference. S0 / S0+R2 = same-generation FINE-P2, the working default.

Decision. Every selector kept S0 (raw) and S0+R2 (final) in 5/5 VOC folds; nothing promoted. S0+R2: 0.8228 forced OOF / 0.8243 full VOC / 0.6622 COCO. Historical S1+R2: 0.8170 / 0.8183 / 0.6977; continuous S2+R2: 0.8182 / 0.8193 / 0.6923. Hard re-seeding collapses recall (S3+R2 0.2122 / 0.2122 / 0.2606; seed fraction 30.9% at 16x16 → 6.4% at 64x64 on VOC). Background S6+R2 0.8199 / 0.8209 / 0.6757; score squaring S10+R2 0.7968 / 0.7968 / 0.6855, matching S7's COCO gain and weakening the case that propagated background is distinctive. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 82. **SG31: every cell** (22 cells). Screen references (`S0`, `S0+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S0	0.450	0.8114	+0.00	0.8097	0.6600	+0.00
S0+R2	0.425	0.8243	+0.00	0.8228	0.6622	+0.00
S1	0.525	0.8033	-0.81	0.8020	0.6870	+2.70
S1+R2	0.500	0.8183	-0.60	0.8170	0.6977	+3.54
S2	0.525	0.8075	-0.39	0.8060	0.6782	+1.82
S2+R2	0.525	0.8193	-0.50	0.8182	0.6923	+3.01
S3	0.300	0.2465	-56.49	0.2465	0.2846	-37.54
S3+R2	0.300	0.2122	-61.20	0.2122	0.2606	-40.16
S4	0.300	0.3907	-42.06	0.3907	0.4425	-21.75
S4+R2	0.300	0.3695	-45.48	0.3695	0.4303	-23.20
S5	0.300	0.4920	-31.93	0.4920	0.4601	-19.99
S5+R2	0.300	0.4863	-33.80	0.4863	0.4658	-19.64
S6	0.350	0.8083	-0.31	0.8075	0.6729	+1.29
S6+R2	0.300	0.8209	-0.34	0.8199	0.6757	+1.35
S7	0.300	0.8078	-0.35	0.8078	0.6785	+1.85
S7+R2	0.300	0.8132	-1.10	0.8132	0.6847	+2.25
S8	0.300	0.2130	-59.84	0.2130	0.2479	-41.21
S8+R2	0.300	0.1785	-64.58	0.1785	0.2224	-43.99
S9	0.300	0.3405	-47.09	0.3405	0.3978	-26.22
S9+R2	0.300	0.3126	-51.16	0.3126	0.3801	-28.22
S10	0.300	0.7907	-2.06	0.7907	0.6768	+1.68
S10+R2	0.300	0.7968	-2.75	0.7968	0.6855	+2.33

SG32: SG32 reliability-adaptive second SA pass (Sec. 121). An original hypothesis from SG22/SG27/SG30/SG31: keep the first fine pass u = A c and sharpen the second pass only where a per-pixel reliability signal is low. Reliability comes from cross-scale agreement of first-pass maps (16/32/64) and/or selected-row background support, per call and shared across the 8 heads. Two forms: convex blend of ordinary and sharpened second passes, or a row-wise exponent. Twelve cells raw and after R2; no sweep.

Cell names. Aˆsharp = row-wise power 2.6, row-normalized (SG30). Per call, f, m, k = separately min-max normalized head-mean first-pass maps at 64, 32→64 and 16→64. Reliability R (clamped to [0,1]): c16 = 1-|f-k|; pair = 1 - mean of the three pairwise |differences|; range = 1 - (max - min) of the three; bg = 1 - B, with B = normalized mean SA row over first-pass background seeds f $\leq$ 0.5; pair x bg; range x bg. S0 = ordinary P2, A u = A(Ac) (= FINE-P2). S1 = SG30-S2 fixed sharpening, Aˆsharp u. Convex cells R x A u + (1-R) x Aˆsharp u: S2 c16, S3 pair, S4 bg, S5 pair x bg, S6 range x bg. Row-temperature cells, per-row exponent r = 1 + 1.6(1-R) then row normalization: S7 c16, S8 pair, S9 bg, S10 pair x bg, S11 range x bg. Low reliability means more sharpening. +R2 = fixed DenseCRF. Caution: apart from S0 = FINE-P2, S cells are unrelated to SG30's or SG31's same-numbered cells.

Reference. S0 / S0+R2 = same-generation FINE-P2, the working default.

Decision. Every selector kept S0 (raw) and S0+R2 (final) in 5/5 VOC folds; nothing promoted. S0+R2: 0.8229 forced OOF / 0.8243 full VOC / 0.6622 COCO. Best new cell S7+R2 (c16 temperature): 0.8234 forced OOF (+0.0005, 2/5 pairwise

wins) / 0.8248 / 0.6773. The highest COCO cell is the historical sharpening control S1+R2 (0.8122 / 0.8153 / 0.6854), which loses 0.0107 forced OOF. Three-scale agreement saturates on distant background; background support is more spatially discriminative, but no adaptive combination clears the VOC rule. The same-generation control tolerance was set to atol 1e-5, rtol 1e-4. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 83. **SG32: every cell** (24 cells). Screen references (`S0`, `S0+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S0	0.450	0.8114	+0.00	0.8097	0.6598	+0.00
S0+R2	0.425	0.8243	+0.00	0.8229	0.6622	+0.00
S1	0.450	0.7996	-1.18	0.7996	0.6673	+0.75
S1+R2	0.450	0.8153	-0.90	0.8122	0.6854	+2.32
S2	0.475	0.8101	-0.13	0.8088	0.6693	+0.95
S2+R2	0.450	0.8239	-0.04	0.8223	0.6750	+1.28
S3	0.450	0.8104	-0.10	0.8090	0.6598	+0.00
S3+R2	0.450	0.8238	-0.05	0.8223	0.6747	+1.25
S4	0.475	0.8101	-0.13	0.8091	0.6639	+0.41
S4+R2	0.475	0.8232	-0.10	0.8220	0.6788	+1.66
S5	0.475	0.8092	-0.22	0.8081	0.6641	+0.43
S5+R2	0.475	0.8227	-0.15	0.8214	0.6796	+1.74
S6	0.475	0.8087	-0.27	0.8076	0.6642	+0.44
S6+R2	0.475	0.8226	-0.17	0.8212	0.6800	+1.78
S7	0.475	0.8104	-0.10	0.8094	0.6692	+0.94
S7+R2	0.450	0.8248	+0.05	0.8234	0.6773	+1.51
S8	0.475	0.8107	-0.07	0.8094	0.6700	+1.03
S8+R2	0.450	0.8246	+0.03	0.8231	0.6772	+1.50
S9	0.500	0.8081	-0.33	0.8067	0.6714	+1.17
S9+R2	0.475	0.8218	-0.24	0.8206	0.6788	+1.66
S10	0.475	0.8068	-0.46	0.8054	0.6641	+0.43
S10+R2	0.475	0.8216	-0.27	0.8202	0.6811	+1.90
S11	0.475	0.8065	-0.49	0.8051	0.6649	+0.51
S11+R2	0.475	0.8213	-0.29	0.8198	0.6820	+1.98

SG33: SG33 hierarchical residuals and coarse Markov reachability (Sec. 122). Adapts FA-Seg (retain intermediate SA resolutions as residual support) and M2N2 (Markov first-arrival maps, IPF balancing, temperature 0.65) to the sequential 16→32→64 SA hierarchy seeded by A3 (matched SG25 modules, calls 10-20, 8 heads). Residual cells fuse normalized stage maps; Markov cells replace only the 16x16 start with a class-seeded reachability map. Twelve cells raw and after R2; no sweep; not a reproduction of either paper.

Cell names. S0 = same-generation FINE-P2. Per call, r16, r32, r64 = min-max normalized stages of the ordinary head-mean 16→32→64 hierarchy, upsampled to 64. H0 = ordinary hierarchy (replays SG31 S2). H1 = (r16+r32+r64)/3; H2 = 0.2 r16 + 0.3 r32 + 0.5 r64; H3 = r16/9 + 5 r32/9 + 3 r64/9; H4 = max(r64, 0.5 r32, 0.25 r16). Markov start on the 16x16 head-mean SA transition from the normalized CA seed: a node's arrival is the first step (step 0 included) at which its walk probability exceeds 0.3 x the maximum, unreached nodes get the cap 256, support = 1 - normalized arrival; the support then runs the ordinary 16→32→64 path. M1 = ordinary transition; M2 = IPF-balanced transition (tolerance 1e-5, at most 100 iterations); M3 = rows sharpened at T = 0.65 before IPF. C1 / C2 / C3 = M1 / M2 / M3 plus the equal three-stage residual. +R2 = fixed DenseCRF. Caution: these C1-C3 are unrelated to SG-combined-v1's C1-C3, and M2 is unrelated to an earlier backlog cell named M2; SG-combined C10a replays SG33 M2.

Reference. S0 / S0+R2 = same-generation FINE-P2, the working default.

Decision. The raw selector froze M2 with 4/5 votes (a raw-stage promotion; selected OOF 0.8159 vs 0.8098). M2: 0.8216 forced OOF (5/5 pairwise wins) / 0.8226 full VOC / 0.6463 COCO vs S0 0.8098 / 0.8114 / 0.6599; its threshold 0.675 sits near the grid top and the gain does not transfer. The final stage stayed at the reference: S0+R2 5/5 (0.8229 / 0.8243 / 0.6623); M2+R2 had 0.8313 forced OOF / 0.8313 / 0.6528 but only 3/5 pairwise wins and no votes. Residual H3+R2 reached 0.6976 COCO but 0.8106 / 0.8126 VOC (0/5). Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 84. **SG33: every cell** (24 cells). Screen references (S0, S0+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S0	0.450	0.8114	+0.00	0.8098	0.6599	+0.00
S0+R2	0.425	0.8243	+0.00	0.8229	0.6623	+0.00
H0	0.525	0.8077	-0.37	0.8061	0.6782	+1.83
H0+R2	0.525	0.8196	-0.48	0.8184	0.6924	+3.01
H1	0.550	0.7892	-2.22	0.7879	0.6770	+1.70
H1+R2	0.525	0.8096	-1.47	0.8083	0.6961	+3.38
H2	0.550	0.7983	-1.31	0.7968	0.6840	+2.40
H2+R2	0.525	0.8153	-0.90	0.8141	0.6973	+3.50
H3	0.550	0.7941	-1.72	0.7926	0.6822	+2.22
H3+R2	0.525	0.8126	-1.18	0.8106	0.6976	+3.53
H4	0.525	0.8072	-0.42	0.8061	0.6774	+1.75
H4+R2	0.525	0.8191	-0.52	0.8180	0.6916	+2.93
M1	0.700	0.7967	-1.46	0.7961	0.5986	-6.13
M1+R2	0.675	0.8039	-2.05	0.8039	0.5994	-6.29
M2⋆	0.675	0.8226	+1.13	0.8216	0.6463	-1.36
M2+R2	0.675	0.8313	+0.70	0.8313	0.6528	-0.95
M3	0.700	0.8093	-0.20	0.8093	0.6191	-4.09
M3+R2	0.700	0.8167	-0.76	0.8167	0.6250	-3.73
C1	0.700	0.7567	-5.46	0.7567	0.5581	-10.18
C1+R2	0.700	0.7761	-4.82	0.7761	0.5725	-8.98
C2	0.700	0.7824	-2.89	0.7824	0.6186	-4.14
C2+R2	0.700	0.8038	-2.05	0.8038	0.6372	-2.51
C3	0.700	0.7642	-4.72	0.7642	0.5890	-7.10
C3+R2	0.700	0.7840	-4.04	0.7840	0.6056	-5.67

SG-combined-v1: SG-combined-v1 constrained transport and budgeted completion (Sec. 123). Our own unified mechanism, not a paper reproduction: coarse class evidence should be conserved through 16→32→64 refinement, and new evidence from a second fine SA pass admitted only under a background-aware budget. Each 2x2 child group is projected onto its coarse parent's mass with the closed-form KL solution (exact or 0.5-relaxed), then optional positive-residual completion. A3 seed, matched SA calls 10-20, 8 heads. Ten cells raw and after R2; no sweep.

Cell names. C0 = same-generation FINE-P2. C1 = ordinary 16→32→64 hierarchy, no conservation (replays SG31 S2; raw call maps averaged, then normalized). Transport cells work on per-call sum-normalized distributions; KL projection z_child = q_child x parent mass / sum of the group's q (uniform fallback when the group mass is near zero); min-max scoring only after transport. C2 = exact projection at 16→32 only. C3 = exact at 32→64 only. C4 = exact at both. C5 = 0.5-relaxed (half proposal, half exact) at both. Completion on a core: P1 = normalized core, P2 = normalized same-call head-mean fine SA applied to P1, residual [P2-P1]_+. C6 = C4 then unconditional second fine pass (P2). C7 = C4 + (1-B) x residual, B = SG32 selected-row background support. C8 = C4 + odds P1/(P1+B) x residual. C9 = C5 with the C8 rule. +R2 = fixed DenseCRF. Caution: these C1-C3 differ from SG33's C1-C3.

Reference. C0 / C0+R2 = same-generation FINE-P2, the working default.

Decision. Every selector kept C0 (raw) and C0+R2 (final) in 5/5 VOC folds; nothing promoted. C0: 0.8097 forced OOF / 0.8114 full VOC / 0.6599 COCO; C0+R2: 0.8228 / 0.8242 / 0.6622. Exact both-scale C4 fell to 0.7592 / 0.7592 / 0.6499; relaxed C5 recovered to 0.7926, unconditional completion C6 to 0.8035; budgeted C8 was 0.7774. After R2, C6+R2 0.8201 / 0.8221 / 0.7063 and C8+R2 0.8031 / 0.8046 / 0.7061: large COCO gains, not selected on VOC. KL mass/ratio checks passed; preserving high-scoring wrong coarse parents plausibly drives C4's loss, and the fixed budget discards useful TP additions. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 85. **SG-combined-v1: every cell** (20 cells). Screen references (C0, C0+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
C0	0.450	0.8114	+0.00	0.8097	0.6599	+0.00
C0+R2	0.425	0.8242	+0.00	0.8228	0.6622	+0.00
C1	0.525	0.8075	-0.39	0.8059	0.6782	+1.83
C1+R2	0.525	0.8192	-0.50	0.8181	0.6923	+3.01
C2	0.525	0.8049	-0.64	0.8034	0.6921	+3.22
C2+R2	0.500	0.8208	-0.35	0.8185	0.7038	+4.16
C3	0.525	0.7911	-2.03	0.7894	0.6843	+2.44

continued

Cell	t^*	VOC	Δ	OOF	COCO	Δ
C3+R2	0.500	0.8147	-0.96	0.8135	0.7022	+4.00
C4	0.475	0.7592	-5.22	0.7592	0.6499	-1.00
C4+R2	0.475	0.7917	-3.25	0.7917	0.6851	+2.29
C5	0.525	0.7940	-1.74	0.7926	0.6861	+2.62
C5+R2	0.500	0.8161	-0.81	0.8149	0.7059	+4.37
C6	0.525	0.8044	-0.70	0.8035	0.6933	+3.34
C6+R2	0.500	0.8221	-0.21	0.8201	0.7063	+4.41
C7	0.550	0.7785	-3.29	0.7785	0.6825	+2.26
C7+R2	0.525	0.8057	-1.86	0.8047	0.7054	+4.32
C8	0.550	0.7774	-3.40	0.7774	0.6823	+2.24
C8+R2	0.525	0.8046	-1.96	0.8031	0.7061	+4.39
C9	0.550	0.7966	-1.48	0.7951	0.6887	+2.88
C9+R2	0.525	0.8164	-0.79	0.8140	0.7057	+4.35

SG-combined C10: SG-combined-v1 C10 add-on: SG33 M2 coarse source (Sec. 124, 127). Three cells added after SG33 completed, without changing SG-combined-v1's ten cells: they start the 16→32→64 hierarchy from SG33 M2's IPF-balanced 16x16 first-arrival source and test ordinary propagation, exact two-transition KL projection, and background-odds completion. Only C10a/b/c are candidates; a same-generation S0 is the reference. Raw and after R2; VOC-only selection, frozen COCO transfer; no new parameter.

Cell names. S0 = same-generation FINE-P2 (reference only, not a candidate). C10a = SG33 M2 source through ordinary same-call 16→32→64 SA (verified against same-generation SG33 M2). C10b = the same source with exact parentwise KL projection at 16→32 and 32→64 (SG-combined-v1 C4 operator). C10c = C10b plus the SG32 background-row / foreground-odds positive completion (SG-combined-v1 C8 rule). A second fine pass on C10b is a completion-efficiency diagnostic only. +R2 = fixed DenseCRF.

Reference. S0 / S0+R2 = same-generation FINE-P2, the working default.

Decision. The raw selector froze C10a with 4/5 votes (a raw-stage promotion; selected OOF gain +0.0061). C10a: 0.8216 forced OOF (5/5 pairwise wins) / 0.8226 full VOC / 0.6465 COCO vs S0 0.8098 / 0.8114 / 0.6599; its threshold 0.675 sits near the grid top and COCO is worse. The final stage stayed at the reference: S0+R2 5/5 (0.8231 / 0.8244 / 0.6623); C10a+R2 0.8313 / 0.8313 / 0.6529, 3/5 pairwise wins. C10b collapses (0.6444 / 0.6473 / 0.5150; 64x64 stage AUC 0.975 → 0.888); C10c recovers partly (0.7622). M2 stays excluded from v2. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 86. **SG-combined C10: every cell** (8 cells). Screen references (S0, S0+R2) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
S0	0.450	0.8114	+0.00	0.8098	0.6599	+0.00
S0+R2	0.425	0.8244	+0.00	0.8231	0.6623	+0.00
C10a⋆	0.675	0.8226	+1.12	0.8216	0.6465	-1.34
C10a+R2	0.675	0.8313	+0.69	0.8313	0.6529	-0.94
C10b	0.425	0.6473	-16.41	0.6444	0.5150	-14.49
C10b+R2	0.425	0.6665	-15.79	0.6632	0.5343	-12.80
C10c	0.700	0.7622	-4.92	0.7622	0.6177	-4.22
C10c+R2	0.700	0.7852	-3.92	0.7852	0.6334	-2.89

SG-combined-v2: SG-combined-v2 one-sided coarse caps and route diagnostic (Sec. 126, 127). Following the calibration audit, replaces the exact two-sided KL projection with a one-sided cap: children may not exceed their coarse parent's mass, clipped mass is discarded, and raw call weights are kept, so the cap is isolated against the ordinary hierarchy. Tests the cap at 16→32 and at both transitions, two completion variants, and a non-selectable per-image route diagnostic. A3 seed, matched SA calls 10-20, 8 heads. Seven cells raw and after R2.

Cell names. Cap on 2x2 groups: z_j = q_j x min(1, p_g / (mean_k in g q_k + 1e-12)), the generalized-KL projection onto upper-mass constraints, applied to the raw (unrenormalized) hierarchy maps. V0 = same-generation FINE-P2 (reference). V1 = ordinary raw 16→32→64 hierarchy (SG31 S2, v1 C1). V2 = v1 C6 replay (exact KL at both transitions + unconditional completion). V3 = cap at 16→32 only. V4 = cap at 16→32 and 32→64 (capped 32x32 map as parent). V5 = V3 core + unconditional completion (C6 convention). V6 = V3 core + whole positive residual admitted only where P1 ≥ B (B = SG32 background support). Route score R = sum B_i [F_i - H_i]_+ / (sum [F_i - H_i]_+ + 1e-8), primary pair F = V0, H = V2; E =

fine-exclusive mean (secondary). +R2 = fixed DenseCRF. Caution: SG-A3G reuses V0 and V3 with the same meaning.

Reference. V0 / V0+R2 = same-generation FINE-P2, the working default.

Decision. Every selector kept V0 (raw) and V0+R2 (final) in 5/5 VOC folds; nothing promoted. V0+R2: 0.8231 forced OOF / 0.8244 full VOC / 0.6622 COCO. V3+R2: 0.8304 forced OOF (+0.0073, 5/5 pairwise wins, paired interval [+0.0017, +0.0125]) / 0.8304 / 0.6966. V3 was ineligible: the worst class (sheep, -0.092 after R2) failed the -0.050 guard in every fold. The cap binds in 58% of groups but removes only 7.6% of mass, changing mask extent rather than ranking. V4 adds nothing; completion re-adds false positives (V5+R2 0.8209 / 0.8230 / 0.6647). Route AUC 0.554 VOC / 0.594 COCO: no-go. Orientation note: A3's SA similarity uses a transposed affinity orientation relative to GoCA (protocol Sec. 107); the paired SG-A3G rerun (Sec. 131) found the corrected seed nearly identical (Pearson 0.99999) and the control inside the tie band, so the historical results stand. Owner-reported; not yet independently audited.

Table 87. **SG-combined-v2: every cell** (14 cells). Screen references (`V0`, `V0+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
V0	0.450	0.8115	+0.00	0.8098	0.6598	+0.00
V0+R2	0.425	0.8244	+0.00	0.8231	0.6622	+0.00
V1	0.525	0.8077	-0.38	0.8061	0.6783	+1.85
V1+R2	0.525	0.8195	-0.49	0.8184	0.6925	+3.03
V2	0.525	0.8046	-0.68	0.8037	0.6932	+3.34
V2+R2	0.500	0.8224	-0.20	0.8204	0.7063	+4.41
V3	0.550	0.8175	+0.60	0.8157	0.6821	+2.23
V3+R2	0.550	0.8304	+0.60	0.8304	0.6966	+3.44
V4	0.650	0.8168	+0.53	0.8151	0.6890	+2.92
V4+R2	0.625	0.8307	+0.63	0.8290	0.6964	+3.42
V5	0.550	0.8145	+0.30	0.8131	0.6652	+0.54
V5+R2	0.525	0.8230	-0.14	0.8209	0.6647	+0.25
V6	0.575	0.8156	+0.41	0.8143	0.6765	+1.67
V6+R2	0.550	0.8269	+0.25	0.8256	0.6792	+1.69

SG-A3G: SG-A3G GoCA-oriented A3 seed and paired rerun (Sec. 128, 131). Tests the open A3 orientation caveat directly. One diffusion run accumulates both the historical A3 seed and A3G, whose SA similarity uses GoCA's orientation; the two differ only in per-call CA module weights of resized modules, giving an exactly paired comparison per image. Each seed feeds the working default and the v2 V3 capped hierarchy. Pre-fixed questions: does G0+R2 - V0+R2 exceed the +/-0.002 tie band, and does V3's gain survive the corrected seed?

Cell names. V0 = historical A3 → FINE-P2 (reference; same as SG-combined-v2 V0). V3 = historical A3 → v2 V3 (one-sided cap at 16→32). G0 = A3G → FINE-P2. G3 = A3G → v2 V3. A3G = A3 with GoCA's affinity orientation <M,P> instead of the historical <MˆT,P>. +R2 = fixed DenseCRF. Candidates V3, G0, G3.

Reference. V0 / V0+R2 = historical A3 → FINE-P2.

Decision. Control inside the tie band: G0+R2 - V0+R2 = -0.0000 VOC forced OOF (paired CI [-0.0001, +0.0001]); frozen COCO -0.0006. The seeds are nearly identical (mean Pearson 0.99999, mean |difference| 0.0008, call-mean |weight difference| 0.0010). V3 holds on the corrected seed: G3+R2 0.8304 vs G0+R2 0.8228 (+0.0075, 5/5, CI [+0.0020, +0.0128]); G3+R2 - V3+R2 = +0.0012 (4/5), below any promotion band. Raw selector V0 (4/5), R2 selector V0+R2 (5/5). Decision: the A3 transpose is closed as immaterial; the historical A3 seed stays the default.

Table 88. **SG-A3G: every cell** (8 cells). Screen references (`V0`, `V0+R2`) in bold; promoted cell marked ⋆. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
V0	0.450	0.8113	+0.00	0.8097	0.6600	+0.00
V0+R2	0.425	0.8242	+0.00	0.8228	0.6623	+0.00
V3	0.550	0.8173	+0.60	0.8155	0.6821	+2.21
V3+R2	0.550	0.8302	+0.60	0.8292	0.6967	+3.44
G0	0.450	0.8114	+0.01	0.8098	0.6598	-0.02
G0+R2	0.425	0.8242	-0.00	0.8228	0.6617	-0.06
G3	0.550	0.8174	+0.62	0.8156	0.6815	+2.16
G3+R2	0.550	0.8304	+0.61	0.8304	0.6961	+3.38

SG-combined-v3: SG-combined-v3 single changes to the V3 capped hierarchy (Sec. 129, 132). Three owner-approved hypotheses, each changing exactly one thing relative to v2 V3: a per-head cap, a native coarse CA seed for the 16x16 start,

and a temporal cap across SA calls. Seed is historical A3; SA calls 10-20; cap clipped mass discarded. Five cells raw and after R2, a mechanism table against V3, fixed decision rules, GT cap-loss diagnostics and an owner-approved dense-map sidecar aimed at the frame-filling recall loss.

Cell names. R0 = FINE-P2 (= v2 V0), the registered selection reference. R1 = v2 V3 replay: r16 = head-mean 16x16 SA x c16, z32 = cap of the 32x32 head-mean pass against r16, then head-mean 64x64 pass; local mechanism reference. W1 = per-head cap: z32 = mean over heads of cap(r16, A^32_h U r16) with the head-mean parent (removes at least as much mass as R1). W2 = native coarse seed: the R1 chain starts from c16* = minmax of the A3 CA contributions at $\leq$ 16x16 modules (unchanged A3 weights), so 64-px CA no longer enters the hierarchy. W3 = temporal cap: each call's R1 map is capped by the area-downsampled mean of earlier calls' maps (first call uncapped). +R2 = fixed DenseCRF. Caution: R0/R1 are new names for v2 V0/V3.

Reference. R0 / R0+R2 (selection reference, = v2 V0); R1 is the mechanism reference for W1-W3.

Decision. No cell promoted: raw selector R0 (4/5), R2 selector R0+R2 (5/5). R1+R2 (= v2 V3) 0.8293 forced OOF / 0.6964 COCO. Against R1: W1+R2 +0.0001 (4/5, CI spans 0; head-wise cap removes 7.7% vs 7.6% of mass), W2+R2 -0.0012 (2/5; c16* correlates 0.996 with the downsampled seed), W3+R2 -0.0032 (1/5, worst on small objects). GT cap loss at 16$\rightarrow$32: the cap removes 13% of background proposal mass but 0.5% of object mass (sheep 1.1%). Dense-count diagnosis: the sheep loss comes from the shared global threshold (R1 moves the optimum from 0.45 to 0.55; sheep prefers 0.35, where R1 leaks), not from cap-removed mass. W1-W3 closed. Owner override (Sec. 135): A3 $\rightarrow$ V3 $\rightarrow$ R2 at 0.55 is adopted as the working default and paper method, labelled as not a registered promotion.

Table 89. **SG-combined-v3: every cell** (10 cells). Screen references (`R0`, `R0+R2`) in bold; promoted cell marked $\star$. Δ in mIoU points against the reference of the screen that lists the cell.

Cell	t^*	VOC	Δ	OOF	COCO	Δ
R0	0.450	0.8114	+0.00	0.8098	0.6599	+0.00
R0+R2	0.425	0.8244	+0.00	0.8230	0.6623	+0.00
R1	0.550	0.8174	+0.61	0.8157	0.6820	+2.21
R1+R2	0.550	0.8303	+0.60	0.8293	0.6964	+3.41
W1	0.550	0.8176	+0.63	0.8161	0.6827	+2.28
W1+R2	0.550	0.8305	+0.62	0.8294	0.6972	+3.49
W2	0.550	0.8168	+0.54	0.8155	0.6833	+2.34
W2+R2	0.550	0.8296	+0.53	0.8281	0.6975	+3.52
W3	0.575	0.8134	+0.20	0.8116	0.6792	+1.93
W3+R2	0.550	0.8261	+0.17	0.8261	0.6828	+2.05

SG-threshold: SG-threshold label-free per-image thresholds (CPU, on SG-combined-v3 counts) (Sec. 133, 134). Tests whether a per-image threshold computed from each image's own map can replace the global threshold, using the SG-combined-v3 counts and dense maps (no generation). Each rule's threshold is snapped to the 17-point grid, so the existing raw and R2 counts score it exactly; one-parameter rules expose their parameter on the threshold axis for fold selection.

Cell names. Maps R0 (FINE-P2) and R1 (V3). T0 = global threshold (reference rule). T1 = Otsu. T2 = two-Gaussian crossing. T3 = mean + k std (k fold-selected). T4 = most stable threshold (minimum area change). T5 = R1 ranking with the area of R0 $\geq$ t (t fold-selected; R1 only). T6 = Otsu + offset (fold-selected). +R2 = fixed DenseCRF read at the same threshold.

Reference. R0.T0 / R0.T0+R2 (= SG-combined-v3 R0 / R0+R2).

Decision. No promotion: raw selector R0.T0 (5/5), R2 selector R0.T0+R2 (4/5). No rule helps R1, raw or with R2; only R0.T6 raw beats its own T0 (+0.0057, 5/5) and it loses on COCO (0.635 vs 0.660). T5 removes the sheep loss (-0.007 vs -0.092) but gives back most of V3's gain (-0.0047 vs R1.T0+R2, 0/5). Per-image best-threshold ceiling: R0+R2 0.896 vs R1+R2 0.895 on VOC, so V3's gain is cross-image calibration, not within-image ranking. Per-image thresholding closed.

Cells are tabulated in the evidence tables (Sec. 12.3).

12.5. Transfer and published baselines

Track	Stage	t	mIoU	caption has class	caption lacks class	mean image IoU
class word (Sec. 54)	`CA/seed`	0.450	0.5489	0.5531	0.5419	0.5576
	`SA/REF`	0.450	0.6262	0.6264	0.6192	0.6487
	`P3/P1`	0.450	0.6244	0.6207	0.6257	0.6507
	`P3/R2`	0.450	0.6420	0.6406	0.6373	0.6800
	`P3/P1+R2`	0.450	0.6370	0.6317	0.6399	0.6762
few-shot token (Sec. 56)	`W/seed`	0.450	0.5489	0.5531	0.5419	0.5576
	`W/REF`	0.450	0.6262	0.6264	0.6192	0.6487
	`T/seed`	0.475	0.6383	0.6382	0.6442	0.6307
	`T/REF`	0.475	0.6683	0.6626	0.6833	0.6845
	`T/P1`	0.475	0.6608	0.6532	0.6805	0.6833
	`T/R2`	0.475	0.6852	0.6795	0.7010	0.7154
	`T/P1+R2`	0.475	0.6746	0.6664	0.6948	0.7078

Table 90. **Frozen COCO-Cap transfer by pipeline stage** (707 images; the caption names the class in 465 and not in 242). Thresholds frozen on VOC-Sim; nothing is chosen on COCO-Cap. `CA/seed`: cross-attention seed; `REF`: one SA pass; `P1`: two passes; `R2`: DenseCRF. `W/`: class word; `T/`: OVAM's released token.

DAAM variant	Data	OVAM t	mIoU at OVAM t	our t	mIoU at our t	paired Δ image IoU [95% CI]
training free	voc-sim	0.40	0.6616	folds: 0.45, 0.45, 0.45, 0.45, 0.45	0.6735 (OOF)	+0.0169 [+0.0095, +0.0244]
training free	coco-cap	0.40	0.4842	VOC 0.450	0.5267 (frozen)	–
token optimized	voc-sim	0.80	0.7970	folds: 0.78, 0.78, 0.78, 0.80, 0.78	0.7972 (OOF)	-0.0038 [-0.0075, -0.0002]
token optimized	coco-cap	0.80	0.6608	VOC 0.775	0.6417 (frozen)	–

Table 91. **DAAM with its threshold chosen in our folds** (Sec. 47; grid 0.30-0.90 (R3 grid extended to include OVAM's 0.8)). On COCO-Cap the threshold is the one chosen on VOC-Sim.

Method	VOC-Sim word (images)	VOC-Sim token	COCO-Cap word (images)	COCO-Cap token
OVAM	0.7036 (490)	0.8247 (490)	0.5823 (707)	—
DAAM	0.6616 (490)	0.7970 (490)	0.4842 (707)	0.6608 (707)
DAAM, threshold chosen in our folds (COCO-Cap: VOC-chosen threshold)	0.6735 (490, OOF)	0.7972 (OOF)	0.5267 (707)	0.6417
Attn2Mask	0.6028 (485)	0.8108 (485)	0.5844 (668)	0.6757 (668)
Grounded Diffusion (trains a module)	0.6213 (496)	0.8657 (496)	0.5015 (588)	0.7323 (588)
DatasetDM (trains a decoder)	0.8027 (500)	0.8057 (500)	0.5928 (728)	0.6048 (728)

Table 92. Released baseline masks scored with OVAM's evaluation-notebook procedures on each method's own released images and annotations

12.6. Diagnostics

These analyses select nothing; they motivated or explained the registered packages.

Pre-SG16 / Pre-SG17 region and second-pass diagnostics (Sec. 94). Diagnostic only (no method selected). Pre-SG16 asks whether any GT-free region statistic can tell which SPEC32 regions are really foreground, especially regions where the P1 prior is ambiguous (occupancy 0.2-0.8), scored by region ROC-AUC against a GT-majority label. Pre-SG17 asks whether observation agreement predicts whether a pixel change made by the second SA pass is useful (FN→TP, FP→TN) or harmful, at fixed threshold 0.45. Run on all 490 VOC / 707 COCO images.

Ambiguous-P1 regions, best occupancy ROC-AUC, VOC	0.686
Ambiguous-P1 regions, best occupancy ROC-AUC, COCO	0.673
Ambiguous-P1 regions, best observation statistic ROC-AUC VOC / COCO	0.597 (obs_top50_median) / 0.631 (obs_all_median)
All regions, p1_occupancy ROC-AUC VOC / COCO	0.951 / 0.907
Second-pass useful vs harmful ROC-AUC, second_q_direction, VOC / COCO	0.617 / 0.634
Preregistered SG17 go bar (ROC-AUC)	0.65
COCO top VOC-frozen reliability decile: useful fraction vs overall	72.2% vs 39.0%
Regions evaluated VOC / COCO	15,680 / 22,624

Finding. On ambiguous P1 regions the best simple prior-occupancy ROC-AUC is 0.686 VOC / 0.673 COCO vs 0.597 / 0.631 for the best observation statistic, so no SG16 recovery signal. Best second-pass signal (second_q_direction) gives useful/harmful ROC-AUC 0.617 VOC / 0.634 COCO, below the preregistered 0.65 SG17 bar.

Pre-SG18 / Pre-SG21 temporal and head-role diagnostics (Sec. 96). Diagnostic only. For every one of the 248 call/head observations of SA module 15 (one-pass A_i c from the class-word seed), and for fixed early/middle/late call stages, measures how many CA-relative true and false positive pixels it adds at fixed threshold 0.45, plus temporal features of each pixel's trajectory and structural descriptors of each head (entropy, local mass, attention distance, reciprocity). Motivated the SG18 window test and the SG21 head weighting.

Stage TP/FP ratio early / middle / late, VOC	1.354 / 3.494 / 3.228
Stage TP/FP ratio early / middle / late, COCO	0.509 / 0.941 / 0.935
Added FP >16 px fraction early / middle / late, VOC	0.643 / 0.456 / 0.438
Added FP >16 px fraction early / middle / late, COCO	0.804 / 0.776 / 0.774
Best temporal feature ROC-AUC (middle_up_fraction) VOC / COCO	0.617 / 0.617
VOC-to-COCO rank stability of 248 head/call cells, tp_fp_ratio (Spearman rho)	0.902
Strongest structural correlation, COCO (local_mass_r1 vs fp_separate_fraction)	0.949
Images VOC / COCO	490 / 707

Finding. Middle and late stages add far more TP per FP than early calls (VOC TP/FP 1.354 / 3.494 / 3.228; COCO 0.509 / 0.941 / 0.935). Head/call behaviour is highly stable across datasets (Spearman 0.902 for TP/FP ratio over 248 cells), and structural descriptors (local mass, entropy) correlate strongly (|rho| about 0.94) with head FP behaviour on COCO. The best temporal pixel feature, middle_up_fraction, reaches ROC-AUC only 0.617 on both datasets.

Pre-SG19 late-call support of middle-stage additions (Sec. 99). Diagnostic only. Takes the pixels that the middle-call map (calls 10-20) adds over the CA seed and asks whether late calls (21-30) can tell the true additions from false ones, which would justify a late-stage veto (SG19). Uses a GPU replay to get all 80 late observations per pixel; feature direction and decile edges are fixed on VOC and transferred to COCO.

Candidate middle-added pixels VOC / COCO	6,837,692 / 12,671,265
late_score TP vs FP ROC-AUC VOC / COCO (threshold 0.425)	0.586 / 0.675
late_minus_ca ROC-AUC VOC / COCO	0.535 / 0.564
late_up_consensus ROC-AUC VOC / COCO	0.556 / 0.530
late_score decile TP/FP ratio, lowest to highest bin, VOC	2.71 to 9.57
Same VOC decile edges on COCO, TP/FP ratio	0.55 to 7.68
VOC person best feature (late_minus_middle) ROC-AUC VOC → COCO	0.695 → 0.481
Fraction of candidates with L > c, VOC / COCO	96.8% / 96.1%

Finding. Raw late score separates middle-added TP from FP with ROC-AUC 0.586 VOC / 0.675 COCO, but the specifically temporal features are weak (L - c 0.535 / 0.564) and VOC-person's best signal does not transfer (0.695 → 0.481). Verdict: mixed, not a clear go; no SG19 mask was tested.

Pre-SG19 raw-score control (Sec. 100). Diagnostic only, CPU, from saved SG18 maps. Checks whether the late-score signal in Pre-SG19 is really late-specific, by scoring the same middle-added candidate pixels with five raw scores: the CA seed, the middle map, the late map, and their min and mean.

TP vs FP ROC-AUC, CA c, VOC / COCO	0.5590 / 0.6198
TP vs FP ROC-AUC, middle M, VOC / COCO	0.6081 / 0.6861
TP vs FP ROC-AUC, late L, VOC / COCO	0.5835 / 0.6733
TP vs FP ROC-AUC, min(M,L), VOC / COCO	0.6006 / 0.6784
TP vs FP ROC-AUC, (M+L)/2, VOC / COCO	0.5973 / 0.6814
TP vs FP average precision, middle vs late, VOC	0.8417 vs 0.8262
TP vs FP average precision, middle vs late, COCO	0.7099 vs 0.6931
Candidate pixels reconstructed VOC / COCO	6,816,320 / 12,665,100

Finding. The middle score beats the late score on both datasets and every contrast (TP/FP ROC-AUC 0.6081 / 0.6861 vs 0.5835 / 0.6733); min and mean do not beat middle. The apparent late signal is explained by general score strength, so the SG19 late veto is not justified.

SG15 region labels vs GT-majority oracle on 20 VOC images (Sec. 93). Visual diagnostic on 20 VOC-Sim images (the first annotated image of each class, fixed before scoring). For three frozen priors, compares the current SG15 region-relabelling rules with an oracle that labels each SPEC32 region by its GT majority, to see how much headroom hard region decisions have. Not an inference method, no selection, no COCO.

Mean IoU over 20 images, W/P1: prior / A050 / B / oracle (computed)	0.8031 / 0.7879 / 0.7946 / 0.8690
Mean IoU, W/SG11/trim20/rho0p80: prior / A050 / B / oracle (computed)	0.7864 / 0.7659 / 0.7838 / 0.8690
Mean IoU, W/SG12/top50: prior / A050 / B / oracle (computed)	0.7905 / 0.7761 / 0.7875 / 0.8690
Images where oracle IoU $>$ prior IoU, P1 / SG11 / SG12 (computed)	15 / 17 / 15 of 20
Images	20 (one per VOC class)

Finding. Averaged over the 20 images, the GT-majority oracle (0.869 mean IoU) is above every prior and both current rules, while A050 and B both score slightly below the unrefined prior for all three priors. Protocol Section 93 says the oracle values must be presented only as diagnostic upper bounds for hard region decisions on these 20 images.

SG18 middle vs late call-window diagnostic (Sec. 98). Diagnostic only, CPU, from the cached SG18 maps. Compares one-pass maps from the middle calls (10-20), the late calls (21-30) and both (10-30) at fixed thresholds 0.425 (primary, T1's frozen threshold) and 0.45, and counts which pixels flip when the late calls are added to the middle window. Asks why adding late calls did not help.

mIoU @0.425 middle / late / middle+late, VOC	0.7959 / 0.7890 / 0.7936
mIoU @0.425 middle / late / middle+late, COCO	0.6285 / 0.6191 / 0.6240
VOC middle $\rightarrow$ middle+late @0.425: TP gained / lost	261,001 / 255,053
VOC middle $\rightarrow$ middle+late @0.425: FP gained / removed	399,322 / 226,040
COCO middle $\rightarrow$ middle+late @0.425: TP gained / lost	178,332 / 581,249
COCO middle $\rightarrow$ middle+late @0.425: FP gained / removed	674,271 / 929,652
mIoU @0.45 middle / late / middle+late, VOC	0.7941 / 0.7868 / 0.7916
mIoU @0.45 middle / late / middle+late, COCO	0.6343 / 0.6251 / 0.6307

Finding. At 0.425, mIoU middle / late / middle+late is 0.7959 / 0.7890 / 0.7936 on VOC and 0.6285 / 0.6191 / 0.6240 on COCO. Adding late calls nets more false positives on VOC but mostly loses true positives on COCO, so protocol Section 98 concludes the effect differs by dataset and cannot be reduced to a single leakage story.

Finalizer reference audit (SG11/SG12 and SG2+DenseCRF recovery) (Sec. 89). Audit, not an experiment. The SG11/SG12 and SG2+DenseCRF finalizers stopped because new V100 masks did not match historical A100 masks within 0.2% of pixels at every threshold. This package compares the REF (one-pass) and other control counts across packages and GPU types, and binds the evidence by SHA256, so that within-package V100 comparisons could be recovered without changing any measurement.

Max per-image REF difference, SG11 (V100) vs SG8 (A100), VOC / COCO	8,185 / 39,917 px
Images above the 0.2% gate, SG11 vs SG8 REF, VOC / COCO	65 / 168
Max per-image REF difference, two V100 packages (Dense vs SG11), VOC / COCO	1 / 2 px
Max per-image REF difference, A100 SG8 vs SG13, VOC / COCO	4 / 4 px
Max per-image REF difference, SG2+CRF package vs original SG2, VOC / COCO	26,282 / 18,171 px
REF mIoU @0.45, SG11 V100 vs SG8 A100, VOC	0.788440 vs 0.788656
REF mIoU @0.45, SG11 V100 vs SG8 A100, COCO	0.625872 vs 0.626588
Live V100 replay (job 234952) of 4 worst samples	SG11/SG12 and raw SG2 counts reproduced exactly

Finding. Within-package V100 comparisons were recovered; historical pixel-level reproduction was not. Same-hardware runs agree within a few pixels, while cross-hardware REF differences reach 8,185 px (VOC) / 39,917 px (COCO) per image, yet REF mIoU at 0.45 moves by under 0.001. Recovered outcomes (protocol Section 89): SG11 OOF 0.787280 $\rightarrow$ 0.798400, SG12 $\rightarrow$ 0.796658, SG2+CRF does not beat FTF (0.811095 vs 0.813209).

Calibration and ranking audit of SG25-SG-combined-v1 (diagnostic only) (Sec. 125). A CPU-only re-read of the stored per-image counts of all 204 cells (raw and R2) of SG25, SG27, SG30, SG31, SG32, SG33 and SG-combined-v1, asking whether COCO transfer gains reflect better spatial ranking or only threshold calibration. Each cell's frozen transfer is set beside a label-using best-threshold mIoU and a threshold-free raw-map AP. No image is regenerated; no result, threshold, selection or promotion changes.

Columns. Frozen = registered result (VOC-selected threshold transferred unchanged to COCO). Best-threshold = cell and reference each take their best of the 17 thresholds on that dataset's own labels (optimistic, never a result; VOC is in-sample). Raw AP = mean per-image average precision of the raw 64x64 map with 20k paired bootstrap; threshold free, undefined for binary R2 masks. Gap = reference's COCO best-threshold minus frozen mIoU. Every delta is against the same package's same-stage reference. Cells: SG25 FINE-P2 (reference), COARSE32-P2 (both passes at 32x32, upsampled), 16-to-64 and 32-to-64 (coarse first pass, fine second pass); SG27 S0 (reference = FINE-P2), S8 (fine/16 min), S9 (fine x 16 product), S38 (three-way min); SG30 S0, S2 (second-pass inflation), S6 (S4 + pruning); SG31 S0, S1 (= SG25 16-to-64), S2 (continuous 16→32→64), S6 / S7 (background-gated), S10 (score squared); SG32 S0, S1 (fixed sharpening), S7 (c16 temperature); SG33 S0, H0 (ordinary hierarchy), H2 / H3 (stage residuals), M2 (IPF Markov source); SG-combined-v1 C0 (reference), C1 (ordinary hierarchy), C2 (exact KL 16→32), C5 (relaxed both), C6 (exact both + completion), C8 (exact both + odds budget). +R2 = fixed DenseCRF. The same S ids mean different operators in different packages.

Table 93. **Calibration and ranking audit** (48 cells). Every Δ is against the same package's same-stage reference. Best-threshold columns use each dataset's labels and are diagnostics, never results; AP is undefined for binary R2 masks.

Package	Cell	VOC OOF Δ	COCO frozen Δ	COCO best-thr Δ (thr)	VOC best-thr Δ	COCO AP Δ [95%]	VOC AP Δ [95%]
sg25	COARSE32-P2	-0.0173	+0.0094	+0.0025 (0.550/0.525)	-0.0190	+0.0013 [-0.0020, +0.0046]	-0.0001 [-0.0016, +0.0016]
sg25	COARSE32-P2+R2	-0.0090	+0.0327	+0.0142 (0.525/0.525)	-0.0095		
sg25	16-to-64	-0.0077	+0.0270	+0.0168 (0.575/0.525)	-0.0081	+0.0073 [+0.0043, +0.0106]	+0.0010 [-0.0005, +0.0026]
sg25	16-to-64+R2	-0.0058	+0.0354	+0.0216 (0.575/0.525)	-0.0060		
sg25	32-to-64	-0.0008	+0.0096	+0.0108 (0.550/0.525)	-0.0008	+0.0028 [+0.0005, +0.0050]	+0.0018 [+0.0010, +0.0028]
sg25	32-to-64+R2	-0.0012	+0.0096	+0.0114 (0.550/0.525)	-0.0014		
sg27	S8	+0.0028	+0.0106	+0.0094 (0.525/0.525)	+0.0011	+0.0036 [+0.0021, +0.0054]	+0.0008 [+0.0002, +0.0016]
sg27	S8+R2	+0.0020	+0.0122	+0.0104 (0.525/0.525)	+0.0016		
sg27	S9	-0.0075	+0.0328	+0.0138 (0.325/0.525)	-0.0092	+0.0051 [+0.0034, +0.0069]	+0.0013 [+0.0006, +0.0022]
sg27	S9+R2	-0.0106	+0.0433	+0.0163 (0.300/0.525)	-0.0120		
sg27	S38	+0.0015	+0.0119	+0.0103 (0.525/0.525)	+0.0011	+0.0038 [+0.0021, +0.0056]	+0.0010 [+0.0003, +0.0019]
sg27	S38+R2	+0.0028	+0.0139	+0.0110 (0.525/0.525)	+0.0014		
sg30	S2	-0.0101	+0.0074	-0.0061 (0.500/0.525)	-0.0117	+0.0036 [+0.0011, +0.0064]	-0.0008 [-0.0019, +0.0003]
sg30	S2+R2	-0.0106	+0.0232	+0.0004 (0.475/0.525)	-0.0089		
sg30	S6	-0.0135	+0.0037	-0.0054 (0.500/0.525)	-0.0137	+0.0020 [+0.0000, +0.0041]	-0.0026 [-0.0036, -0.0016]
sg30	S6+R2	-0.0087	+0.0212	+0.0019 (0.500/0.525)	-0.0100		
sg31	S1	-0.0077	+0.0270	+0.0166 (0.575/0.525)	-0.0081	+0.0073 [+0.0043, +0.0106]	+0.0010 [-0.0005, +0.0026]
sg31	S1+R2	-0.0058	+0.0354	+0.0214 (0.575/0.525)	-0.0060		
sg31	S2	-0.0038	+0.0182	+0.0167 (0.600/0.525)	-0.0039	+0.0010 [-0.0039, +0.0057]	+0.0008 [-0.0011, +0.0028]
sg31	S2+R2	-0.0047	+0.0301	+0.0168 (0.600/0.525)	-0.0050		
sg31	S6	-0.0022	+0.0129	+0.0004 (0.425/0.525)	-0.0031	-0.0042 [-0.0058, -0.0027]	-0.0001 [-0.0008, +0.0006]
sg31	S6+R2	-0.0030	+0.0135	-0.0014 (0.400/0.525)	-0.0034		
sg31	S7	-0.0019	+0.0185	-0.0016 (0.325/0.525)	-0.0035	-0.0107 [-0.0145, -0.0072]	-0.0021 [-0.0043, -0.0003]
sg31	S7+R2	-0.0096	+0.0225	-0.0047 (0.300/0.525)	-0.0110		
sg31	S10	-0.0190	+0.0168	-0.0037 (0.300/0.525)	-0.0206	+0.0000 [-0.0000, +0.0000]	+0.0000 [-0.0000, +0.0000]
sg31	S10+R2	-0.0260	+0.0233	-0.0038 (0.300/0.525)	-0.0275		
sg32	S1	-0.0101	+0.0075	-0.0060 (0.500/0.525)	-0.0118	+0.0037 [+0.0012, +0.0065]	-0.0008 [-0.0018, +0.0004]
sg32	S1+R2	-0.0107	+0.0232	+0.0003 (0.475/0.525)	-0.0090		
sg32	S7	-0.0003	+0.0094	-0.0015 (0.525/0.525)	-0.0010	+0.0014 [+0.0005, +0.0023]	+0.0002 [-0.0001, +0.0007]
sg32	S7+R2	+0.0005	+0.0151	+0.0017 (0.525/0.525)	+0.0005		
sg33	H0	-0.0036	+0.0183	+0.0167 (0.600/0.525)	-0.0037	+0.0009 [-0.0039, +0.0057]	+0.0008 [-0.0012, +0.0027]
sg33	H0+R2	-0.0045	+0.0301	+0.0169 (0.600/0.525)	-0.0048		
sg33	H2	-0.0130	+0.0240	+0.0132 (0.600/0.525)	-0.0131	+0.0087 [+0.0049, +0.0127]	+0.0030 [+0.0013, +0.0047]
sg33	H2+R2	-0.0088	+0.0350	+0.0208 (0.600/0.525)	-0.0090		
sg33	H3	-0.0172	+0.0222	+0.0111 (0.600/0.525)	-0.0172	+0.0086 [+0.0046, +0.0126]	+0.0026 [+0.0009, +0.0043]
sg33	H3+R2	-0.0123	+0.0353	+0.0203 (0.600/0.525)	-0.0118		
sg33	M2	+0.0119	-0.0136	-0.0277 (0.700/0.525)	+0.0113	-0.0374 [-0.0473, -0.0280]	-0.0147 [-0.0227, -0.0076]
sg33	M2+R2	+0.0084	-0.0095	-0.0302 (0.700/0.525)	+0.0070		
sg_combined_v1	C1	-0.0038	+0.0183	+0.0167 (0.600/0.525)	-0.0039	+0.0012 [-0.0036, +0.0059]	+0.0008 [-0.0011, +0.0028]
sg_combined_v1	C1+R2	-0.0047	+0.0301	+0.0169 (0.600/0.525)	-0.0050		
sg_combined_v1	C2	-0.0064	+0.0322	+0.0194 (0.575/0.525)	-0.0064	+0.0043 [+0.0011, +0.0077]	-0.0028 [-0.0045, -0.0011]
sg_combined_v1	C2+R2	-0.0043	+0.0416	+0.0243 (0.575/0.525)	-0.0035		
sg_combined_v1	C5	-0.0172	+0.0262	+0.0134 (0.575/0.525)	-0.0174	-0.0120 [-0.0165, -0.0074]	-0.0171 [-0.0200, -0.0142]
sg_combined_v1	C5+R2	-0.0079	+0.0437	+0.0249 (0.575/0.525)	-0.0081		
sg_combined_v1	C6	-0.0063	+0.0334	+0.0203 (0.575/0.525)	-0.0070	-0.0061 [-0.0101, -0.0020]	-0.0123 [-0.0148, -0.0098]
sg_combined_v1	C6+R2	-0.0027	+0.0441	+0.0255 (0.550/0.525)	-0.0021		
sg_combined_v1	C8	-0.0323	+0.0224	+0.0052 (0.575/0.525)	-0.0340	-0.0164 [-0.0205, -0.0121]	-0.0241 [-0.0272, -0.0212]
sg_combined_v1	C8+R2	-0.0197	+0.0439	+0.0218 (0.575/0.525)	-0.0196		

Finding. Diagnostic only; every recomputed frozen mIoU matches the registered value to 1e-9. The fine reference is miscalibrated on COCO in every package: VOC threshold 0.425 (R2) / 0.45 (raw) vs COCO best 0.525, worth +0.027 / +0.020 mIoU. Hierarchy cells keep a COCO gain at matched thresholds (after R2: SG25 16→64 +0.022, SG33 H2/H3 +0.021/+0.020, C2 +0.024, C6 +0.026) and raw AP rises for 16→64, H2/H3 and C2. Sharpening wins (SG30 S2, SG32

S1) fall to about 0.000 and SG31 S6/S7/S10 to ≤ 0; S10's AP delta is exactly 0. C5/C6/C8 gain after R2 while raw AP falls; SG33 M2 ranks worse (COCO AP -0.037). Of 53 R2 cells with frozen COCO gain $\geq +0.015$, 34 keep it at matched thresholds. Owner-reported; not yet independently audited.

12.7. Exploratory record (B1–B17)

Before the fold protocol existed, every choice was made on the images it was reported on, so the values below are optimistic by construction. They motivated the controlled screens and are never used as evidence.

Dataset	Method	mIoU	Evidence
VOC-Sim	OVAM CA only (released heatmaps)	0.6616	B16
VOC-Sim	OVAM CA only (regenerated)	0.6540	B8
VOC-Sim	OVAM, implemented (CA×SA, DenseCRF)	0.6934	attention_analysis.md
VOC-Sim	OVAM, released masks	0.7036	B16
VOC-Sim	+ SA propagation at U3.2, 4 passes	0.7777	B8
VOC-Sim	+ top-4-head CA seed, 2 passes, t=0.45	0.7922	B9
VOC-Sim	+ heads {0,1,2,4,5,6}, calls 15–30, 4 passes, 1% cleanup	0.7997	B10
VOC-Sim	Revised: 1 pass, all 16 CA layers	0.7824	B13
VOC-Sim	OVAM token-optimized, released masks	0.8247	B16
VOC-Sim	Earlier final + optimized token + DenseCRF (1 pass)	0.8525	B17
COCO-Cap	OVAM, implemented (paper COCO parameters)	0.5717	B17
COCO-Cap	Final method, frozen from VOC	0.6080	B12
COCO-Cap	Revised: 1 pass, all 16 CA layers	0.6322	B13
COCO-Cap	Earlier final + optimized token + DenseCRF (1 pass)	0.6612	B17
COCO-Cap	OVAM token-optimized (paper Table 1)	0.6920	OVAM paper

Table 94. How the method developed, with the baselines it is compared against

Table 95. Exploratory VOC-Sim experiments: best configuration per family (in-sample, 490 images)

Exp.	Family	Configurations	Best configuration	Best mIoU
B8	baseline	1	ovam_unoptimized	0.6540
B8	layers	24	mid_up	0.6828
B8	resolution	15	r8_r16	0.7122
B8	self_attention/progressive_16-32	229	iter4__non_residual__seed_t0.30__foreground	0.7413
B8	self_attention/progressive_16-32-64	229	iter2__non_residual__seed_t0.30__foreground	0.7577
B8	self_attention/progressive_16-8	229	iter1__residual_a0.50__seed_t0.30__background_difference	0.6913
B8	self_attention/progressive_32-16	229	iter2__residual_a0.50__seed_t0.30__background_difference	0.7201
B8	self_attention/progressive_32-16-8	229	iter2__residual_a0.50__seed_t0.30__background_difference	0.6944
B8	self_attention/progressive_32-64	229	iter4__non_residual__seed_t0.30__foreground	0.7683
B8	self_attention/progressive_64-32	229	iter4__residual_a0.75__seed_t0.70__background_difference	0.7558
B8	self_attention/progressive_64-32-16	229	iter4__residual_a0.50__seed_t0.70__background_difference	0.7282
B8	self_attention/progressive_ 64-32-16-8	229	iter4__residual_a0.50__seed_t0.30__background_difference	0.7000
B8	self_attention/progressive_8-16	229	iter1__non_residual__seed_t0.80__background_ratio	0.7005
B8	self_attention/progressive_8-16-32	229	iter1__non_residual__seed_t0.80__background_ratio	0.7264
B8	self_attention/progressive_ 8-16-32-64	229	iter4__residual_a0.75__seed_t0.30__background_difference	0.7474
B8	self_attention/r16	778	all_modules_all_steps__iter1__seed_t0.30__foreground_ residual_a0.50	0.7161
B8	self_attention/r32	778	module__up_blocks.2.attentions.2.transformer_ blocks.0.attn1__iter4__co	0.7641
B8	self_attention/r64	778	module__up_blocks.3.attentions.2.transformer_ blocks.0.attn1__iter4__co	0.7777
B8	self_attention/r8	238	window__steps_26_30__iter1__continuous__residual_a0.75	0.7034
B8	stage1/fusion	118	resolution_weights__16-1_32-0.5_64-2_8-0.5	0.6770
B8	stage1/heads	212	per_layer_top_4_heads	0.6830
B8	stage1/postprocessing	18	fill_holes	0.6661
B8	stage1/time_layer	237	all16__cumulative_00_25	0.6844
B8	threshold_normalization	204	all_16_layers__percentile_01_99__t0.40	0.6975
B8	time_resolution	105	all_00_30__r8_r16	0.7122
B8	time_window	7	steps_15_20	0.6752
B9	focused/cross_attention_seeds	6	top4_heads_percentile	0.6831
B9	focused/output_calibration	936	top4_heads_percentile__r64__iter2__percentile_01_99__ t0.45	0.7922
B9	focused/postprocessing	90	best12_percentile__r64__iter4__percentile_01_99__t0.45__ remove_fractio	0.7780
B9	focused/sa_heads	108	iter3__leave_head_6	0.7797
B9	focused/sa_time	348	iter3__trailing_10_30	0.7801
B9	focused/seed_operator	144	top4_heads_percentile__r64__iter2	0.7922
B10	joint/joint_head_time_threshold	3900	iter3__leave_3_7__steps_15_20__t0.40	0.7991

Exp.	Family	Configurations	Best configuration	Best mIoU
B10	joint/joint_postprocessing	780	iter4__leave_3_7__trailing_15_30__t0.45__remove_fraction_0.01	0.7997
B15	heads	255	heads_0_2_3_4_5_6_7	0.6764
B15	layers_structured	39	down_middle	0.6818
B15	resolution	15	r8_r16_r64	0.6957
B15	time	79	cumulative_00_20	0.6752
B15	time_resolution	1185	cumulative_00_25__r8_r16	0.6968

Table 96. Remove-one-component ablations of the final and revised systems

System	Dataset	Change	mIoU	Δ vs full
final (4 passes, 12 layers)	VOC-Sim	all sa calls	0.7938	-0.0060
final (4 passes, 12 layers)	VOC-Sim	all sa heads	0.7940	-0.0059
final (4 passes, 12 layers)	VOC-Sim	full system	0.7999	0.0000
final (4 passes, 12 layers)	VOC-Sim	max normalization	0.7176	-0.0822
final (4 passes, 12 layers)	VOC-Sim	one sa pass	0.7801	-0.0198
final (4 passes, 12 layers)	VOC-Sim	without ca head filter	0.7832	-0.0167
final (4 passes, 12 layers)	VOC-Sim	without ca layer filter	0.7975	-0.0024
final (4 passes, 12 layers)	VOC-Sim	without component cleanup	0.7992	-0.0006
final (4 passes, 12 layers)	VOC-Sim	without sa propagation	0.6805	-0.1194
final (4 passes, 12 layers)	COCO-Cap	all sa calls	0.6054	-0.0024
final (4 passes, 12 layers)	COCO-Cap	all sa heads	0.5935	-0.0143
final (4 passes, 12 layers)	COCO-Cap	full system	0.6078	0.0000
final (4 passes, 12 layers)	COCO-Cap	max normalization	0.4898	-0.1181
final (4 passes, 12 layers)	COCO-Cap	one sa pass	0.6263	0.0185
final (4 passes, 12 layers)	COCO-Cap	without ca head filter	0.6026	-0.0052
final (4 passes, 12 layers)	COCO-Cap	without ca layer filter	0.6120	0.0042
final (4 passes, 12 layers)	COCO-Cap	without component cleanup	0.6051	-0.0027
final (4 passes, 12 layers)	COCO-Cap	without sa propagation	0.5739	-0.0339
revised (1 pass, 16 layers)	VOC-Sim	all sa calls	0.7802	-0.0022
revised (1 pass, 16 layers)	VOC-Sim	all sa heads	0.7837	0.0013
revised (1 pass, 16 layers)	VOC-Sim	max normalization	0.7804	-0.0021
revised (1 pass, 16 layers)	VOC-Sim	revised full system	0.7824	0.0000
revised (1 pass, 16 layers)	VOC-Sim	without ca head filter	0.7625	-0.0199
revised (1 pass, 16 layers)	VOC-Sim	without component cleanup	0.7824	-0.0001
revised (1 pass, 16 layers)	VOC-Sim	without sa propagation	0.6969	-0.0855
revised (1 pass, 16 layers)	COCO-Cap	all sa calls	0.6240	-0.0083
revised (1 pass, 16 layers)	COCO-Cap	all sa heads	0.6274	-0.0049
revised (1 pass, 16 layers)	COCO-Cap	max normalization	0.6147	-0.0176
revised (1 pass, 16 layers)	COCO-Cap	revised full system	0.6322	0.0000
revised (1 pass, 16 layers)	COCO-Cap	without ca head filter	0.6253	-0.0069
revised (1 pass, 16 layers)	COCO-Cap	without component cleanup	0.6288	-0.0034
revised (1 pass, 16 layers)	COCO-Cap	without sa propagation	0.5862	-0.0460

Dataset	Configuration	mIoU
VOC-Sim	ovam_implemented	0.6945
VOC-Sim	plain_top4/sa_iter1	0.7802
VOC-Sim	plain_top4/sa_iter1+crf	0.7957
VOC-Sim	opt_top4/sa_iter1	0.8347
VOC-Sim	opt_top4/sa_iter1+crf	0.8525
VOC-Sim	opt_all16/sa_iter1+crf	0.8553
VOC-Sim	plain_top4/sa_iter4	0.8000
VOC-Sim	opt_top4/sa_iter4+crf	0.8521
COCO-Cap	ovam_implemented	0.5717
COCO-Cap	plain_top4/sa_iter1	0.6264
COCO-Cap	plain_top4/sa_iter1+crf	0.6432
COCO-Cap	opt_top4/sa_iter1	0.6419
COCO-Cap	opt_top4/sa_iter1+crf	0.6612
COCO-Cap	opt_all16/sa_iter1+crf	0.6552
COCO-Cap	plain_top4/sa_iter4	0.6077
COCO-Cap	opt_top4/sa_iter4+crf	0.6369
VOC-Sim	released OVAM token-optimized masks	0.8247
COCO-Cap	OVAM token-optimized (paper)	0.6920

Table 97. OVAM optimized token and DenseCRF on the earlier final method (exploratory)

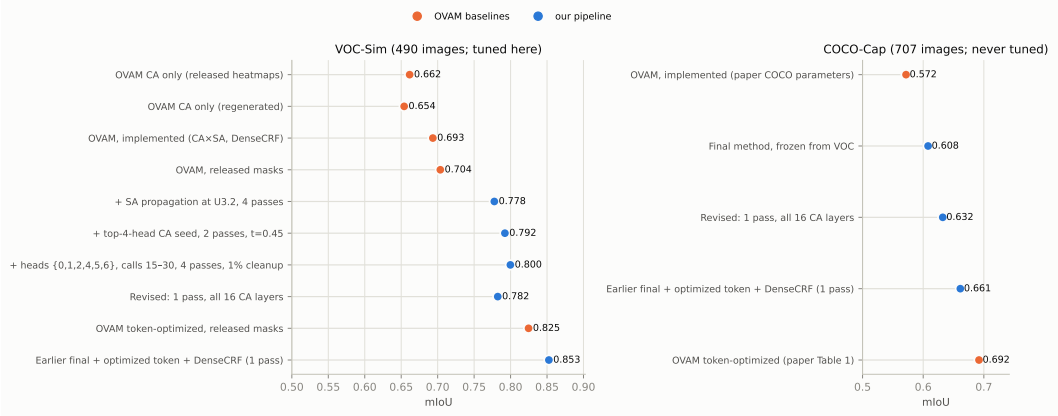


Figure 39. Each step of the earlier method and the baselines. VOC-Sim values are in-sample; COCO-Cap was never used to choose the earlier method.

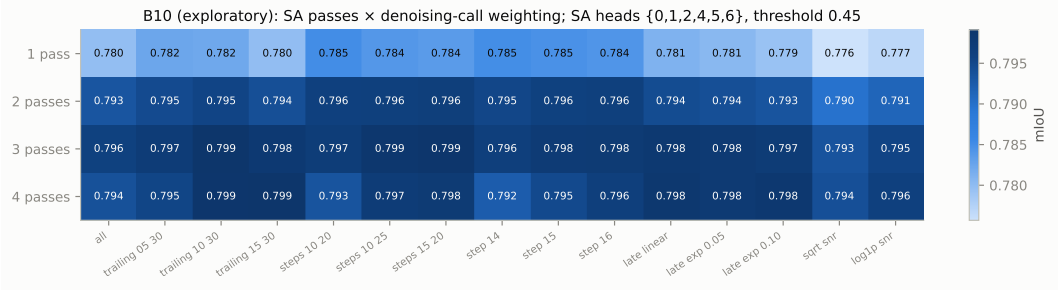


Figure 40. Exploratory B10: SA passes by denoising-call weighting (in-sample).

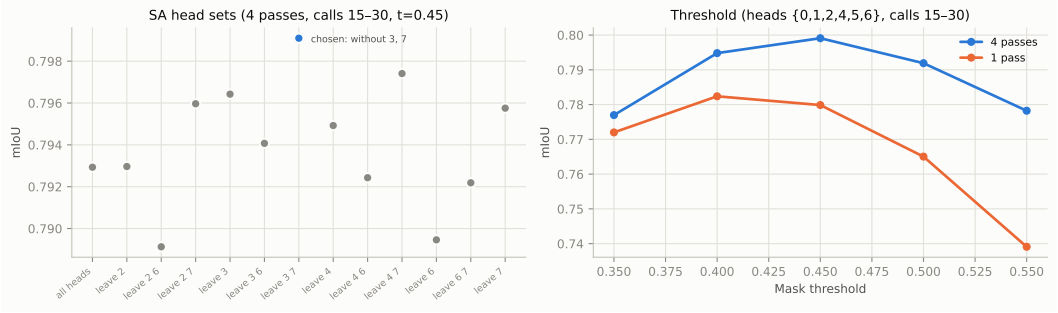


Figure 41. Exploratory B10: SA head subsets and the mask threshold (in-sample).

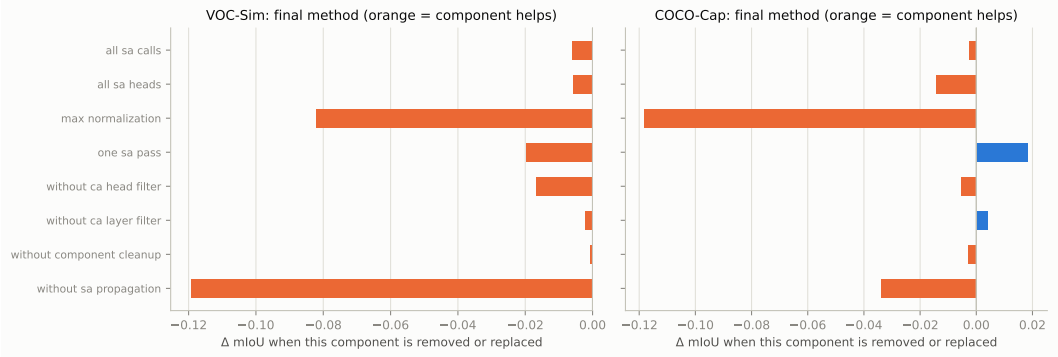


Figure 42. Removing one component of the earlier method, on VOC-Sim and on the untuned COCO-Cap.

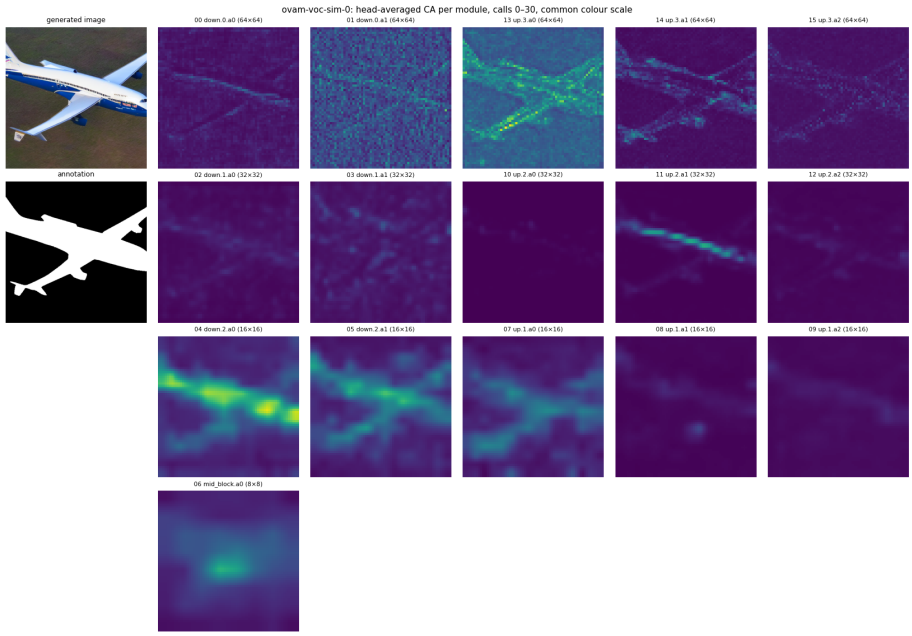


Figure 43. Head-averaged cross-attention of every module for one VOC-Sim image (calls 0–30, common colour scale, one row per resolution). Further images in the evidence package.

References

[1] Huy Che and Vinh-Tiep Nguyen. FA-Seg: A fast and accurate diffusion-based method for open-vocabulary segmentation, 2025. 15

[2] Markus Karmann and Onay Urfalioglu. Repurposing stable diffusion attention for training-free unsupervised interactive segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 15

[3] Kunal Mahatha, Jose Dolz, and Christian Desrosiers. SPARK: Stochastic propagation via affinity-guided random walk for training-free unsupervised segmentation, 2026. 15

[4] Kunal Mahatha B, Jose Dolz, and Christian Desrosiers. NERVE: Neighbourhood and entropy-guided random-walk for training free open-vocabulary segmentation, 2025. 14

[5] Benyuan Meng, Qianqian Xu, Zitai Wang, Xiaochun Cao, Longtao Huang, and Qingming Huang. Making training-free diffusion segmentors scale with the generative power. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 35861–35871, 2026. 1, 16

[6] Joon Hyun Park, Kumju Jo, and Sungyong Baik. SeeDiff: Off-the-shelf seeded mask generation from diffusion models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6406–6415, 2025. 15

[7] Gensheng Pei, Xiruo Jiang, Xinhao Cai, Tao Chen, Yazhou Yao, and Byeungwoo Jeon. PEARL: Geometry aligns semantics for training-free open-vocabulary semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17927–17937, 2026. 14